Chapter 32

Numerical Analysis

INTRODUCTION

Using Mathematical Modeling, most of the problems in Engineering and Physical and Economical sciences can be formulated in terms of systems of linear or non-linear equations, ordinay or partial differential equations or integral equations. In majority of the cases, the solutions to these problems in analytical form are non-existent or difficult or not amenable for direct interpretation. In all such problems, Numerical Analysis provides approximate solutions, practical and amenable for analysis. Numerical analysis does not strive for exactness. Instead, it yields approximations with specified degree of accuracy. The early disadvantage of the several number of computations involved has been removed through high speed computation using computers, giving results which are accurate, reliable and fast. Numerical analysis is not only a science but also an 'art' because the choice of 'appropriate' procedure which 'best' suits to a given problem yields 'good' solutions.

In this chapter we consider the bisection, regula falsi and Newton-Raphson's methods of obtaining solutions of transcendental equations. For an unknown function $f(x)$ given at a set of tabulated values, interpolation is the process of obtaining a simpler function $\phi(x)$. We study Newton-Gregory finite differences formulae. Sterling and Bessel's central differences for equally spaced value of x. Lagrange's interpolation and inverse interpolation is used for arbitrarily spaced x. For tabulated functions, the derivatives can be calculated using numerical differentiation. Numerical integration is useful even for integrals such as $\int \frac{\sin x}{x} dx$, $\int \sqrt{1+x^4} dx$, $\int_2^3 e^{-x^2} dx$ which can not be expressed in terms of elementary functions. We study trapezoidal, Simpson's $\frac{1}{3}$, $\frac{3}{8}$ rule, Weddle's and Boole's rules of integration. Quadratic and cubic splines which are powerful tools in piecewise-polynomial approximation are also considerd.

32.1 ROOTS OF TRANSCENDENTAL EQUATIONS

An algebraic equation of degree n is

$$P(x) \equiv a_0 x^n + a_1 x^{n-1} + \cdots + a_n = 0 \qquad (1)$$

where the coefficients $a_0, a_1, \ldots, a_n$ are real numbers and $a_0 \neq 0$. Here $n \geq 1$. Transcendental equations are non-algebraic equations involving transcendental functions such as exponential, logarithmic, trigonometric or hyperbolic functions. A general form of an algebraic or transcendental equation is

$$f(x) = 0 \qquad (2)$$

where the function $f(x)$ is defined and continuous on an interval $a < x < b$.

Root

Any value ξ for which $f(\xi) = 0$ is known as the root or solution of the equation (2) or ξ is called the zero of the function $f(x)$. In the case of the algebraic Equation (1), the roots can be determined in analytical (literal) form when $n = 1, 2, 3, 4$ (i.e., for linear, quadratic, cubic and biquadratic equations). For $n \geq 5$ no such results exists for the roots of (1). Also from the fundamental theorem of algebra, (1) has exactly n roots, real or complex whereas the number and analytical form of roots of a transcendental equation are not known at all.

Example: $\sin x = 2$ has no real roots, $\sin x = \frac{1}{2}$ has infinite number of roots, while $\sin x = \frac{x}{2}$ has three real roots (see Fig. 32.9 on page 32.6).

Therefore only approximate solutions (roots) of an algebraic or transcendental Equation (2) are to be found by numerical methods consisting of (a) isolating the roots (b) and then improving the value of the approximate roots. Here it is assumed that the roots of (2) are isolated i.e., for any root ξ of (2) there exists an interval containing no other root except ξ.

Geometrically, the root of Equation (2) is the point where the graph (or curve) of $y = f(x)$ crosses the x-axis (i.e., $y = f(x) = 0$). Although the roots can be isolated by drawing the graphs of the curve, these graphical methods are cumbersome. The following theorem is very useful in isolating the roots of (2).

Theorem 1: *If a continuous function $f(x)$ assumes values of opposite sign at the end points of an interval $[\alpha, \beta]$ i.e., $f(\alpha)f(\beta) < 0$, then the interval will contain at least one root of the equation $f(x) = 0$ i.e., there exists $\xi \in (\alpha, \beta)$ such that $f(\xi) = 0$.*

Note 1: Root ξ will be unique (only one) in the interval (α, β) if $f'(x)$ has the same sign in the interval (α, β) (i.e., $f' > 0$ or $f' < 0$ in $\alpha < x < \beta$).

Note 2: For an nth degree algebraic Equation (1), we get $(n + 1)$ sign changes.

Note 3: Descarte's rule of sign: The number of positive roots of $f(x) = 0$ can not exceed the number of changes of sign in $f(x)$. Also the number of negative roots of $f(x)$ can not exceed the number of changes of sign in $f(-x)$.

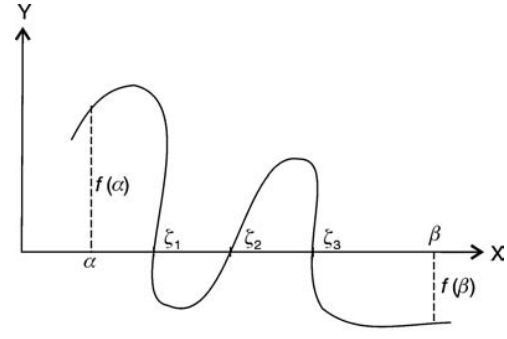


Fig. 32.1

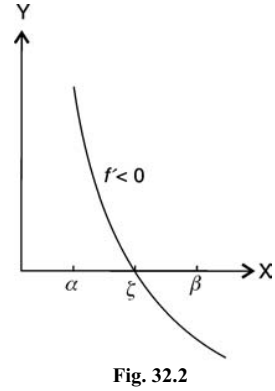


Fig. 32.2

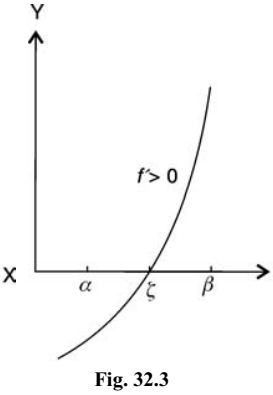


Fig. 32.3

Example: $f(x) = x^5 - 6x^2 - 4x + 5 = 0$.
$f(x) : + - - + :$ 2 changes of sign; no more than 2 positive roots.
$f(-x) : - - + +$: one change of sign; no more than one negative root.

Bisection Method (or Halving Method or Bolzano Method)

Consider the equation

$$f(x) = 0 \tag{2}$$

in $[a, b]$. Assume that $f(a)f(b) < 0$. The bisection method isolates the root in $[a, b]$ by halving process, approximately dividing the given interval $[a, b]$ into two, four, eight etc. equal parts. Thus in order to find a root of (2) lying in the interval $[a, b]$, divide

the interval in half. If $f\left(\frac{a+b}{2}\right) = 0$, then $\xi = \frac{a+b}{2}$ is the required root. If $f\left(\frac{a+b}{2}\right) \neq 0$, then choose that half $\left[a, \frac{a+b}{2}\right]$ or $\left[\frac{a+b}{2}, b\right]$, at the end points of which $f(x)$ has opposite signs. The newly reduced interval $[a, b]$ is again bisected and the above process is repeated. Thus we get a sequence of nested intervals $[a_1, b_1], [a_2, b_2], \ldots, [a_n, b_n]$ such that

$$f(a_n) f(b_n) < 0 \quad \text{for} \quad n = 1, 2, \ldots$$

and

$$b_n - a_n = \frac{1}{2^n}(b - a).$$

The required root $\xi = \frac{1}{2}(a_n + b_n)$.

Rugula-Falsi Method (or Method of False Position or Linear Interpolation or Method of Chords or Method of Proportional Parts)

This method is probably the oldest, faster method more generally applicable.

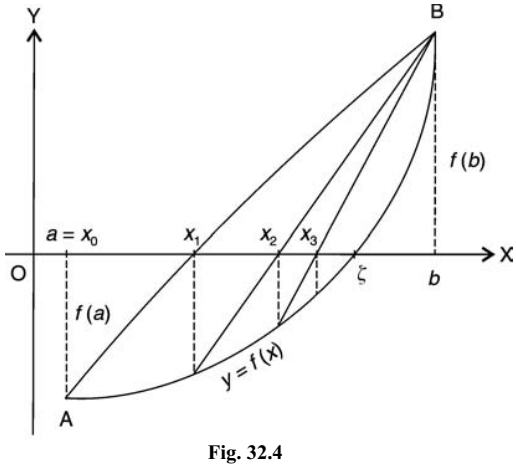


Fig. 32.4

To find root ξ of the equation $f(x) = 0$ in the interval $[a, b]$ assume that $f(a) < 0$ and $f(b) > 0$ so that $f(a) \cdot f(b) < 0$.

Geometrically, this method is equivalent to replacing the curve $y = f(x)$ by a chord that passes through the points $A(a, f(a))$ and $B(b, f(b))$. The equation of the chord AB is

$$\frac{x - a}{b - a} = \frac{y - f(a)}{f(b) - f(a)}$$

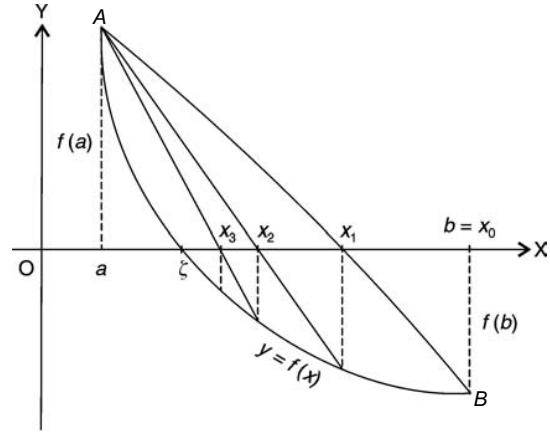


Fig. 32.5

If chord AB meets x-axis at $x = x_1$, then $y = 0$.

$$\frac{x_1 - a}{b - a} = \frac{0 - f(a)}{f(b) - f(a)} \quad \text{or}$$

$$x_1 = a - \frac{f(a)}{f(b) - f(a)}(b - a)$$

If $f(a) < 0$, then the end point b is fixed and the successive approximations

$$x_0 = a$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f(b) - f(x_n)}(b - x_n),$$

(for $n = 0, 1, 2, 3, \ldots$) form a bounded increasing monotonic sequence and

$$x_0 < x_1 < x_2 \cdots < x_n < x_{n+1} < \cdots < \xi < b.$$

If $f(a) > 0$, end point a is fixed and successive approximations are

$$x_0 = b$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f(x_n) - f(a)}(x_n - a)$$

and

$$a < \xi < \cdots < x_{n+1} < x_n < \cdots < x_1 < x_0.$$

Note 1: Fix the end point for which sign of f and f'' is same.

Note 2: Successive approximations x_n lie on the side of root ξ where sign of f is opposite to the sign of f''.

Newton-Raphson Method (or Method of Tangents)

Newton's method is used to obtain a better (refined) approximation of a root using the earlier approximations obtained by bisection method or Regula-Falsi method. Geometrically, Newton's method is equivalent to replacing a small arc of the curve $y = f(x)$ by a tangent line drawn at a point of the curve. Draw a tangent to the curve at B_0 which meets x-axis at x_1. Then draw a tangent at B_1 which meets x-axis at x_2. Continuing this process, the root ξ is obtained.

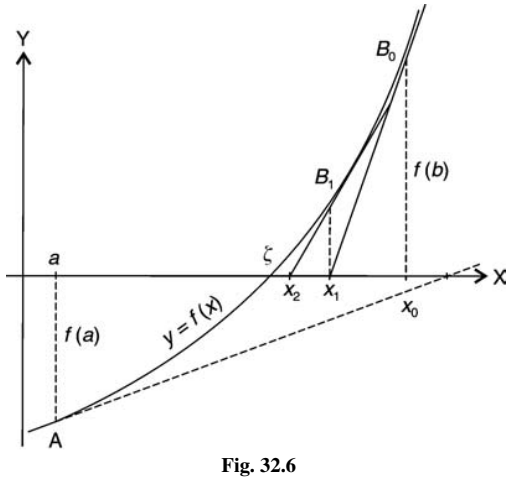


Fig. 32.6

Suppose $\xi = x + h$ where h is a small quantity. Then applying Taylor's formula

$$0 = f(x+h) \approx f(x) + hf'(x)$$

or
$$h = -\frac{f(x)}{f'(x)}$$

Thus
$$\xi = x + h = x - \frac{f(x)}{f'(x)}.$$

In general,

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}, \quad n = 0, 1, 2, \ldots.$$

Note 1: A root ξ of equation $f(x) = 0$ can be computed to any degree of accuracy if a 'good' initial approximation x_0 is chosen for which

$$f(x_0)f''(x_0) > 0$$

i.e., choose the end point of the interval at which f and f'' have the same sign.

Note 2: Newton's method converge slow if f' is small (fails when $f' = 0$ because tangent in this case is parallel to x-axis and will never meet it).

Worked Out Examples

Example 1: Isolate the roots of the equation

$$x^3 - 4x + 1 = 0$$

Find all the roots using bisection method.

Solution: Here $f(x) = x^3 - 4x + 1$.

x:	$-\infty$	-3	-2	-1	0	1	2	3	4	∞
sign of $f(x)$:	$-$	-14	1	4	1	-2	1	16	49	∞

Since equation is cubic (degree three) and since there are 3 changes of signs, a unique (one) root lies in the three intervals $(-3, -2)$, $(0, 1)$, $(1, 2)$. Now consider the interval $(0, 1)$ and apply bisection method.

x_i	$f(x_i)$	new interval (with signs)
$x_1 = \frac{1+0}{2} = \frac{1}{2}$	$f(x_1) = -8.75$	$+ \quad -$ $(0,\ 0.5)$
$x_2 = \frac{0+0.5}{2} = \frac{1}{4}$	$f(x_2) = f(0.25) = 0.015625$	$+ \quad -$ $(0.25,\ 0.5)$
$x_3 = \frac{.25+.5}{2} = .35$	$f(.35) = -0.357$	$+ \quad -$ $(0.25,\ 0.35)$
$x_4 = \frac{.25+.35}{2} = .3$	$f(.3) = -0.173$	$+ \quad -$ $(.25,\ .3)$
$x_5 = \frac{.25+.3}{2} = \frac{.55}{2}$	$f(.275) = -0.0792$	$+ \quad -$ $(.25,\ .275)$
$x_6 = \frac{.25+.275}{2} = .2625$	$f(.2625) = -0.0319$	$+ \quad -$ $(.25,\ .2625)$
$x_7 = \frac{.25+.2625}{2} = 0.25625$	$f(.25625) = -0.00817$	$+ \quad -$ $(0.25,\ .25625)$
$x_8 = \frac{.25+.25625}{2} = 0.253125$	$f(.253125) = 0.0100$	$+ \quad -$ $(0.253125,\ .25625)$
$x_9 = \frac{.253125+.25625}{2} = 0.2546875$	$f(.2546875) = -0.00222$	$+ \quad -$ $(.253125,\ .2546875)$
$x_{10} = \frac{0.253125+0.2546875}{2} = 0.25390625$	$f(0.25390625) = 0.000743$	$+ \quad -$ $(.25390625,\ .2546875)$

The approximate root is

$$\xi = \frac{0.25390625 + 0.2546875}{2} = 0.254296875 \approx 0.2540$$

$$f(0.254296875) = -0.00074290925$$

By synthetic division

0.254	1	0	−4	1
		0.254	0.064516	−0.99612
	1	0.254	−3.93546	0.000387 remainder

Roots of the quadratic equation

$$x^2 + 0.254x - 3.93546 = 0$$

are $x = -2.11475 \approx -2.115$, $x = 1.86075 \approx 1.8608$.

Example 2: Using Regula-Falsi method, compute the real root of the equation $x^3 - 4x - 9 = 0$.

Solution: Here $f(x) = x^3 - 4x - 9$.

x:	0	1	2	3
$f(x)$:	−9	−12	−9	6

By bisection method, $f\left(\frac{2+3}{2}\right) = f(2.5) = -3.375$ so a root of $f(x)$ lies in the interval $(2.5, 3)$. Here $f' = 3x^2 - 4$, $f'' = 6x$, $f''(3) = 18 > 0$, $f''(2.5) = 15 > 0$. Since sign of f and f'' is same at $x = 3$, fix the point $b = 3$ and since $f(a) = f(2.5) = -3.375 < 0$, use

$$x_{n+1} = x_n - \frac{f(x_n)}{f(b) - f(x_n)}(b - x_n)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{6 - f(x_n)}(3 - x_n)$$

Then $x_0 = 2.5$, $f(x_0) = -3.375$, $n = 0$, then

$$x_1 = x_0 - \frac{f(x_0)(3 - x_0)}{6 - f(x_0)} = 2.5 + \frac{3.375}{6 + 3.375}(3 - 2.5)$$

$$x_1 = 2.5 + 0.18 = 2.68$$

At $x_1 = 2.68$, $f(2.68) = -0.471168$.

$$x_2 = 2.68 + \frac{0.471168}{6 + 0.471168}(3 - 2.68)$$

$$= 2.68 + 0.0232993$$

$$x_2 = 2.7033$$

At x_2, $f(x_2) = f(2.7033) = -0.05794$

$$x_3 = 2.7033 + \frac{0.05794}{6 + 0.05794}(3 - 2.7033)$$

$$x_3 = 2.7033 + 0.0028377 = 2.706$$

$$f(2.706) = -0.009488$$

$$x_4 = 2.706 + \frac{0.009488}{6.009488}(3 - 2.706) = 2.70646$$

$$x_4 = 2.70650.$$

Example 3: Determine the root of $xe^x - 2 = 0$ by method of false position.

Solution: Here $f(x) = xe^x - 2$.

x:	0	0.5	0.8	0.9	1.0
$f(x)$:	−2	−1.1756	−0.2196	0.2136	0.718

A root lies between 0.8 and 0.9. Now

$$f'(x) = e^x + xe^x, \qquad f''(x) = 2e^x + xe^x$$

$$f''(0.8) = 6.2315, \qquad f''(0.9) = 7.1328.$$

Since f and f'' have the same (positive) sign at $x = 0.9$, fix this point $b = 0.9$. Note that $f(0.8) < 0$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f(0.9) - f(x_n)}(0.9 - x_n)$$

Take $n = 0$, $x_0 = 0.8$, $f(x_0) = f(0.8) = -0.2196$

$$x_1 = 0.8 + \frac{0.2196}{0.2136 + 0.2196}(0.9 - 0.8)$$

$$x_1 = 0.8 + 0.05069252 = 0.851$$

$$f(x_1) = f(0.851) = -0.00697$$

$$x_2 = 0.851 + \frac{0.00697}{0.2136 + 0.00697}(0.9 - 0.851)$$

$$x_2 = 0.851 + 0.0015484 = 0.85256$$

$$f(x_2) = -0.0001977$$

$$x_3 = 0.85256 + \frac{0.0001977}{0.2136 + 0.0001977}(0.9 - 0.85256)$$

$$x_3 = 0.85256 + 0.000043868 = 0.8526$$

$$f(x_3) = -0.0000239$$

$$x_4 = 0.8526 + \frac{0.0000239}{0.2136 + 0.0000239}(0.9 - 0.8526)$$

$$x_4 = 0.8526 + 0.0000053 = 0.8526$$

$\therefore$ Approximate root is 0.8526.

Example 4: Find the approximate value of the real root of the equation $2x - \log_{10}^{x} - 7 = 0$ by using Newton-Raphson method.

Solution: Here $f(x) = 2x - \log_{10}^{x} - 7$. Since log function is involved, x must be positive

x:	1	2	3	4
$f(x)$:	-5	-3.301	-1.4771	0.3979

Roots lies in between 3 and 4. Now

x:	3.5	3.7	3.8
$f(x)$:	-0.5441	-0.1682	0.0202

Here $f'(x) = 2 - \frac{\log_{10}^{e}}{x}$ and $f''(x) = \frac{\log_{10}^{e}}{x^2} = \frac{0.4343}{x^2}$. Since f and f'' have the same sign at $x = 3.8$, choose $x_0 = 3.8$. Then $f(x_0) = 0.0202$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'x_n} = x_n - \frac{(2x_n - \log_{10} x_n - 7)}{\left(2 - \frac{\log_{10}^{e}}{x_n}\right)}$$

Take $n = 0$, $x_1 = 3.8 - \frac{0.0202}{1.88571} = 3.8 - 0.010712$

$x_1 = 3.7893$, $f(x_1) = 0.000041$, then

$$x_2 = 3.7893 - \frac{0.000041}{1.88538} = 3.7893 - 0.00002175$$

$x_2 = 3.789278 \approx 3.7893$ is the required root.

Example 5: Find a positive root of $x^4 - x = 10$ using Newton-Raphson's method.

Solution: Here $f(x) = x^4 - x - 10$.

x :	0	1	2
$f(x)$:	-10	-10	4

A root lies between 1 and 2

x :	1	1.5	1.75	1.8	1.9	2
$f(x)$:	-10	-6.4375	-2.3711	-1.3024	1.1321	4

Root lies between 1.8 and 1.9. Now

$$f'(x) = 4x^3 - 1. \quad \text{So}$$

$$x_{n+1} = x_n - \frac{x_n^4 - x_n - 10}{4x_n^3 - 1}.$$

Since both f and f'' have the same sign at $x = 1.9$, choose $x_0 = 1.9$ as the starting point. Now

$$x_1 = x_0 - \frac{x_0^4 - x_0 - 10}{4x_0^3 - 1}$$

$$= 1.9 - \frac{f(1.9)}{f'(1.9)} = 1.9 - \frac{1.1321}{26.436}$$

$x_1 = 1.9 - 0.042824 = 1.8572$

$f(x_1) = 0.03972$

$$x_2 = 1.8572 - \frac{0.03972}{24.623} = 1.8572 - 0.0016131$$

$x_2 = 1.855586897$

$f(x_2) = +0.000058169$.

So $x_2 = 1.855587$ is the required root.

EXERCISE

1. Using graphical method find an approximate roots of the transcendental equations:

   (a) $3x - \cos x - 1 = 0$; (b) $e^x - 3x = 0$; (c) $\sin x = 2$; (d) $\sin x = \frac{1}{2}$; (e) $\sin x = \frac{1}{2}x$.

   Hint:

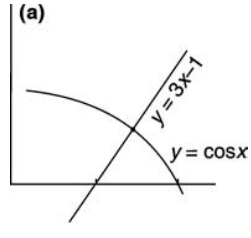


Fig. 32.7

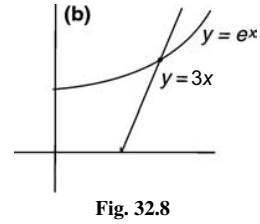


Fig. 32.8

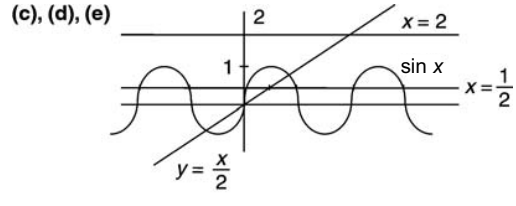


Fig. 32.9

Ans. (a) 0.6; (b) 0.619; (c) no root; (d) infinite number of roots; (e) $x = 30^\circ$

2. Obtain an approximate root using bisection method, for the following equations:

   (a) $x^4 - x - 10 = 0$; (b) $x^3 - x - 1 = 0$; (c) $x^4 - 4x - 9 = 0$; (d) $x^4 + 2x^3 - x - 1 = 0$.

Ans. (a) (1.85546875, 1.857421855) at 9th iteration; (b) $x_5 = 1.38125$; (c) $x_4 = 2.6875$; (d) x_6: 0.867

3. Isolate the roots of the equations:

   (a) $x^3 - 6x + 2 = 0$; (b) $x + e^x = 0$.

Ans. **a.** $(-3, -1), (0, 1), (1, 3)$
b. $(-\infty, 1), (1, +\infty)$

4. Using Regula-Falsi method find an approximate root of the following equations:

   (a) $x^3 - 4x + 1 = 0$; (b) $x^3 - 0.2x^2 - 0.2x - 1.2 = 0$; (c) $x^3 - 2x - 5 = 0$; (d) $xe^x - 3 = 0$; (e) $x \log_{10}^x - 2 = 0$; (f) $xe^x - \cos x = 0$.

Ans. (a) 0.2541; (b) 1.198; (c) 2.094548; (d) 1.050; (e) 2.74065; (f) 0.5177

5. Using Newton-Raphson method, find an approximate root of the following equations:

   (a) $(x - 1)\sin x - x = 1$; (b) $e^{-x} = \sin x$; (c) $x^3 - 25 = 0$; (d) $x^4 - x - 1 = 0$; (e) $x + e^x = 0$; (f) $3x^3 + 5x - 40 = 0$; (g) $x - e^{-x} = 0$; (h) $2\sin x = x$; (i) $x^2 + 4\sin x = 0$; (j) $x^3 - 3x - 5 = 0$; (k) $x^4 - 3x^2 + 75x - 10,000 = 0$.

Ans. (a) −0.42036; (b) 0.5885; (c) 2.924; (d) 1.22138; (e) −0.567; (f) 0.5635; (g) 0.5671; (h) 1.895494; (i) −1.9338; (j) 2.7984; (k) −10.261

6. Use Newton's method to find the smallest positive root of the equation $\tan x = x$.

   Hint: Roots lies in $\left(\pi, \frac{3\pi}{2}\right)$.

Ans. 4.49343

7. Find the real root of $x \log_{10}^x = 1.2$ using Newton's iterative method.

   Hint: Root lies in (2, 3).

Ans. 2.74065

8. Apply Newton-Raphson method to evaluate approximately $\sqrt{12}$.

   Hint: Solve $x^2 - 12 = 0$, root lies in (3, 4).

Ans. 3.4641.

32.2 FINITE DIFFERENCES

Let $y = f(x)$ be a function and $\Delta x = h$ denote the increment in the independent variable x. Assume that Δx, increment in the argument x (also known as *interval or spacing*) is fixed. i.e., $h =$ constant. Then the first finite difference of the function y is defined

$$\Delta y \equiv \Delta f(x) = f(x + \Delta x) - f(x).$$

Similarly, finite differences of higher orders are defined as follows

$$\begin{aligned}\Delta^2 y = \Delta(\Delta y) &= \Delta(f(x + \Delta x) - f(x)) \\ &= \Delta(f(x + \Delta x)) - \Delta f(x) \\ &= \Big[f(x + 2\Delta x) - f(x + \Delta x)\Big] \\ &\quad - \Big[f(x + \Delta x) - f(x)\Big] \\ \Delta^2 y &= f(x + 2\Delta x) - 2f(x + \Delta x) + f(x).\end{aligned}$$

In general,

$$\boxed{\Delta^n y = \Delta(\Delta^{n-1} y)}, \quad \text{for } n = 2, 3, 4, \ldots$$

Now consider the function $y = f(x)$ specified by tabular values $y_i = f(x_i)$ for a set of ***equidistant*** points x_i where $i = 0, 1, 2, \ldots$ and $\Delta x_i = x_{i+1} - x_i = h =$ constant. Thus the tabulated function consists of ordered pairs $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k), \ldots$. Here y_k's are known as entries.

Forward differences

The first forward difference is denoted by Δy_k and defined as

$$\Delta y_k = y_{k+1} - y_k \tag{1}$$

Here the symbol Δ is the forward differences operator, having the following properties:

(i) $\Delta c = 0$ (differences of a constant function are zero)

(ii) $\Delta(cy_k) = c(\Delta y_k)$ where c is a constant

(iii) $\Delta(u_k + v_k) = \Delta u_k + \Delta v_k$

(iv) $\Delta(u_k v_k) = v_{k+1}\Delta u_k + u_k \Delta v_k$

(v) $\Delta^m(\Delta^n y_k) = \Delta^{m+n} y_k$

where m and n are non-negative integers and $\Delta^0 y_k = y_k$ (by definition). Note that because of (ii) and (iii) Δ is a linear operator.

The higher order forward differences are similarly defined: The second order forward difference of y_k is

$$\Delta^2 y_k = \Delta(\Delta y_k) = \Delta y_{k+1} - \Delta y_k$$

– – – – – – – – – – – – – – –

In general,

$$\Delta^n y_k = \Delta(\Delta^{n-1} y_k) = \Delta^{n-1} y_{k+1} - \Delta^{n-1} y_k \quad (2)$$

defines the nth order forward differences. For example, $\Delta^3 y_k = \Delta^2 y_{k+1} - \Delta^2 y_k = (\Delta y_{k+2} - \Delta y_{k+1}) - (\Delta y_{k+1} - \Delta y_k) = y_{k+3} - 3y_{k+2} + 3y_{k+1} - y_k$.

From (1) $y_{k+1} = y_k + \Delta y_k = (1 + \Delta) y_k$

Similarly, $y_{k+2} = y_{k+1} + \Delta y_{k+1} = (1 + \Delta) y_{k+1}$

or $y_{k+2} = (1 + \Delta)(1 + \Delta) y_k = (1 + \Delta)^2 y_k$

Thus $y_{k+3} = (1 + \Delta)^3 y_k$

– – – – – – – – – – – – – – – – – –

$$y_{k+n} = (1 + \Delta)^n y_k.$$

Expanding $(1 + \Delta)^n$ by binomial theorem

$$y_{k+n} = y_k + n_{c_1} \Delta y_k + n_{c_2} \Delta^2 y_k + \cdots + \Delta^n y_k.$$

Conversely, we have

$$\Delta^n y_k = \Big[(1 + \Delta) - 1\Big]^n y_k$$

$$= (1 + \Delta)^n y_k - n_{c_1} (1 + \Delta)^{n-1} y_k +$$

$$+ n_{c_2} (1 + \Delta)^{n-2} y_k - \cdots + \cdots + (-1)^n y_i$$

or $$\Delta^n y_k = y_{n+k} - n_{c_1} y_{n+k-1}$$

$$+ n_{c_2} y_{n+k-2} + \cdots + (-1)^n y_k \quad (3)$$

Thus any higher order forward differences can be expressed in terms of the successive values y_k's of the function. For example,

$$\Delta^2 y_k = y_{k+2} - 2y_{k+1} + y_k$$

$$\Delta^3 y_k = y_{k+3} - 3y_{k+2} + 3y_{k+1} - y_k \quad \text{etc.}$$

Finite differences of various orders are conveniently arranged in the form of a forward (diagonal) differences table (Table 32.1).

In general, y_0, the first entry is known as the leading terms, and Δy_0, $\Delta^2 y_0$, $\Delta^3 y_0$, $\Delta^4 y_0$, $\Delta^5 y_0$, $\Delta^6 y_0$, $\Delta^7 y_0$ are known as the leading differences.

Table 32.1: Forward differences table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$	$\Delta^7 y$
x_0	y_0	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$	$\Delta^5 y_0$	$\Delta^6 y_0$	$\Delta^7 y_0$
x_1	y_1	Δy_1	$\Delta^2 y_1$	$\Delta^3 y_1$	$\Delta^4 y_1$	$\Delta^5 y_1$		
x_2	y_2	Δy_2	$\Delta^2 y_2$	$\Delta^3 y_2$	$\Delta^4 y_2$	$\Delta^5 y_2$		
x_3	y_3	Δy_3	$\Delta^2 y_3$	$\Delta^3 y_3$	$\Delta^4 y_3$			
x_4	y_4	Δy_4	$\Delta^2 y_4$	$\Delta^3 y_4$				
x_5	y_5	Δy_5	$\Delta^2 y_5$					
x_6	y_6	Δy_6						
x_7	y_7							

Differences of a Polynomial

Book Work: If $P_n(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_n$ is an nth degree polynomial, then prove that

$$\Delta^n P_n(x) = n! a_0 h^n = \text{constant} \quad (4)$$

where $\Delta x = h$.

Proof: Consider $\Delta P_n(x) = P_n(x + h) - P_n(x)$

$$= [a_0(x + h)^n + a_1(x + h)^{n-1} + \cdots + a_n] -$$

$$-[a_0 x^n + a_1 x^{n-1} + \cdots + a_n]$$

$$= a_0[(x + h)^n - x^n] + a_1[(x + h)^{n-1} - x^{n-1}] +$$

$$+ \cdots + a_{n-1}[x + h - x].$$

Expanding by binomial theorem

$$(x + h)^n - x^n = x^n + nhx^{n-1} + n_{c_2} h^2 x^{n-2} + \cdots - x^n$$

Similar terms in the other brackets. Thus

$$Q_{n-1}(x) = \Delta P_n(x) = b_0 x^{n-1} + b_1 x^{n-2} + \cdots + b_{n-1}$$

which is a polynomial of $(n - 1)$th degree. Here $b_0 = nha_0$. Now

$$\Delta^2 P_n(x) = \Delta(\Delta P_n(x)) = Q_{n-1}(x + h) - Q_{n-1}(x).$$

Expanding by binomial theorem, observe that $\Delta^2 P_n(x)$ is a polynomial of degree $n-2$:

$$\Delta^2 P_n(x) = c_0 x^{n-2} + c_1 x^{n-3} + \cdots + c_{n-2} \quad \text{where}$$
$$c_0 = (n-1)hb_0 = n(n-1)h^2 a_0.$$

Continuing in this manner successively, we get

$$\Delta^n P_n(x) = n! a_0 h^n = \text{constant}$$

Thus the nth order differences of a polynomial of nth degree are constant.

Corollary:

$$\Delta^S P_n(x) = 0 \qquad \text{for } s > n \tag{5}$$

i.e., $(n+1)$th order differences of a polynomial of nth degree are zero.

Converse: If the nth differences of a tabulated function are constant when the values of the independent variable (argument) are taken in arithmetic progression (i.e., at equal intervals apart), then the function is a polynomial of degree n.

Backward Differences

The first (order) backward difference is denoted by ∇ and defined as

$$\nabla y_k = y_k - y_{k-1} \tag{6}$$

Second (order) backward difference

$$\nabla^2 y_k = \nabla(\nabla y_k) = \nabla y_k - \nabla y_{k-1}$$

In general,

$$\nabla^n y_k = \nabla(\nabla^{n-1} y_k) = \nabla^{n-1} y_k - \nabla^{n-1} y_{k-1}.$$

Now
$$\begin{aligned}\nabla^3 y_k &= \nabla^2 y_k - \nabla^2 y_{k-1} \\ &= \nabla y_k - \nabla y_{k-1} - \nabla y_{k-1} + \nabla y_{k-2} \\ &= y_k - y_{k-1} - 2y_{k-1} + 2y_{k-2} + y_{k-2} - y_{k-3} \\ &= y_k - 3y_{k-1} + 3y_{k-2} - y_{k-3}\end{aligned}$$

In general,

$$\nabla^n y_k = \sum_{i=0}^{n} (-1)^i n_{c_i} y_{k-i} \tag{7}$$

Here y_4, the last entry, is the leading term and $\nabla y_4, \nabla^2 y_4, \nabla^3 y_4, \nabla^4 y_4$ are known as leading backward differences.

Table 32.2: Backward differences table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
		∇y_1			
x_1	y_1		$\nabla^2 y_2$		
		∇y_2		$\nabla^3 y_3$	
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$
		∇y_3		$\nabla^3 y_4$	
x_3	y_3		$\nabla^2 y_4$		
		∇y_4			
x_4	y_4				

Generalized Power or Factorial

The generalized nth power of a number x, denoted by $x^{[n]}$ or $[x]^n$, is defined as the product of n consecutive factors, the first of which is equal to x and each subsequent factor is h less than the preceding:

$$x^{[n]} = [x]^n = x \cdot (x-h) \cdot (x-2h) \cdots (x-(n-1)h) \tag{8}$$

Here h is some fixed constant $[x]^n$ is also known as a "factorial function". Here $[x]^0 = x^{[0]} = 1$.

Corollary 1: For $h = 0$, the generalized power coincides with the ordinary power i.e., $x^{[n]} = [x]^n = x^n$.

Corollary 2: For $h = 1$, $[x]^n = x(x-1)(x-2)\cdots(x-n+1)$.

Differences of a Generalized Power

The first difference of the factorial function is

$$\begin{aligned}\Delta[x]^n &= [x+h]^n - [x]^n \\ &= \Big\{(x+h)(x)(x-h)\cdots(x-(n-2)h)\Big\} - \\ &\quad -\Big\{x(x-h)\cdots(x-(n-1)h)\Big\} \\ &= \Big\{x(x-h)\cdots(x-(n-2)h)\Big\} \times \\ &\quad \times \Big\{x+h-(x-(n-1)h)\Big\} \\ &= \Big\{x(x-h)\cdots(x-(n-2)h)\Big\} nh = nh[x]^{n-1}.\end{aligned}$$

Thus $\Delta[x]^n = nh[x]^{n-1}$.

Now the second difference

$$\Delta^2[x]^n = \Delta(\Delta[x]^n) = \Delta(nh[x]^{n-1})$$
$$= nh\Delta[x]^{n-1} = nh \cdot (n-1) \cdot h \cdot [x]^{n-2}$$
$$\Delta^2[x]^n = n(n-1)h^2[x]^{n-2}.$$

By mathematical induction

$$\Delta^k[x]^n = n(n-1)\cdots(n-(k-1))h^k[x]^{n-k} \quad (9)$$

where $k = 1, 2, 3, \ldots, n$.

Corollary 1: For $k = n$, from (9), we have

$$\Delta^n[x]^n = n(n-1)(n-2)\cdots 2 \cdot 1 \cdot h^n[x]^0$$
$$\Delta^n[x]^n = n!h^n = \text{constant} \quad (10)$$

since $[x]^0 = 1$.

Corollary 2: For $k = n+1$, from (9),

$$\Delta^{n+1}[x]^n = \Delta(\Delta^n[x]^n) = n!h^n - n!h^n = 0.$$

Thus

$$\Delta^s[x]^n = 0 \quad \text{when } s > n \quad (11)$$

Corollary 3: When $h = 1$, from (10), we have

$$\Delta^n[x]^n = n! \quad (12)$$

i.e., the result of *differencing* $[x]^n$ is analogous to that of *differentiating* x^n. In other words, the operator Δ is equivalent the differential operator $D = \frac{d}{dx}$ i.e., $\Delta \equiv D$.

Corollary 4: Every polynomial of degree n can be expressed as a factorial polynomial of the same degree and vice versa.

WORKED OUT EXAMPLES

Example 1: Show that

$$\Delta(u_k v_k) = v_{k+1}\Delta u_k + u_k \Delta v_k.$$

Solution: $\Delta(u_k v_k) = u_{k+1}v_{k+1} - u_k v_k$. We know that $\Delta u_k = u_{k+1} - u_k$, $\Delta v_k = v_{k+1} - v_k$

$$\therefore\ u_{k+1} \cdot v_{k+1} = (u_k + \Delta u_k)(v_k + \Delta v_k)$$
$$= u_k v_k + u_k \Delta v_k + v_k \Delta u_k + \Delta u_k \Delta v_k.$$

Thus

$$\Delta(u_k v_k) = u_{k+1} \cdot v_{k+1} - u_k v_k$$
$$= u_k \Delta v_k + \Delta u_k\{v_k + v_{k+1} - v_k\}$$
$$= u_k \Delta v_k + v_{k+1}\Delta u_k.$$

Note: $\Delta(u_k v_k) = v_k \Delta u_k + u_{k+1}\Delta v_k$ is true.

Example 2: Evaluate $\Delta^n(e^{3x+5})$.

Solution: $\Delta(e^{3x+5}) = e^{3(x+1)+5} - e^{3x+5}$
$= e^{3x+5}\,(e^3 - 1)$. Now

$$\Delta^2(e^{3x+5}) = \Delta(\Delta e^{3x+5}) = \Delta\left\{(e^3-1)(e^{3x+5})\right\}$$
$$= (e^3-1)\Delta(e^{3x+5}), \text{ using the first result}$$
$$= (e^3-1)(e^3-1)e^{3x+5} = (e^3-1)^2 e^{3x+5}.$$

By induction $\Delta^n(e^{3x+5}) = (e^3-1)^n e^{3x+5}$.

Example 3: Evaluate $\Delta^2\left\{\frac{4x^2-25x+31}{(x-1)(x-2)(x-3)}\right\}$.

Solution: Resolving into partial fractions

$$\Delta^2\left(\frac{4x^2-25x+31}{(x-1)(x-2)(x-3)}\right)$$
$$= \Delta^2\left\{\frac{5}{x-1} + \frac{3}{x-2} - \frac{4}{x-3}\right\}$$
$$= \Delta\left[\Delta\left(\frac{5}{x-1}\right) + \Delta\left(\frac{3}{x-2}\right) + \Delta\left(\frac{-4}{x-3}\right)\right]$$
$$= \Delta\Bigg[5\left(\frac{1}{x} - \frac{1}{x-1}\right) + 3\left(\frac{1}{x-1} - \frac{1}{x-2}\right)$$
$$-4\left(\frac{1}{x-2} - \frac{1}{x-3}\right)\Bigg]$$
$$= 5\left\{\left(\frac{1}{x+1} - \frac{1}{x}\right) - \left(\frac{1}{x} - \frac{1}{x-1}\right)\right\}$$
$$+3\left\{\left(\frac{1}{x} - \frac{1}{x-1}\right) - \left(\frac{1}{x-1} - \frac{1}{x-2}\right)\right\}$$
$$-4\left\{\left(\frac{1}{x-1} - \frac{1}{x-2}\right) - \left(\frac{1}{x-2} - \frac{1}{x-3}\right)\right\}$$
$$= \frac{5}{x+1} - \frac{7}{x} - \frac{5}{x-1} + \frac{11}{x-2} - \frac{4}{x-3}.$$

Example 4: If $f(x) = x^3 + 5x - 7$, then form the table of backward differences for $x =$

$-1, 0, 1, 2, 3, 4, 5$. Continue the table to obtain $f(6)$.

Solution: For $x = -1, 0, 1, 2, 3, 4, 5$, the values of $f(x)$ are respectively $-13, -7, -1, 11, 35, 77, 143$. The backward differences table is

Table 32.3:

i	x	$y = f(x)$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$	$\nabla^6 y$
0	−1	−13						
			6					
1	0	−7		0				
			6		6			
2	1	−1		6		0		
			12		6		0	
3	2	11		12		0		0
			24		6		0	
4	3	35		18		0		[0]
			42		6		[0]	
5	4	77		24		[0]		
			66		[6]			
6	5	143		[30]				
			[96]					
7	[6]	[239]						

Values obtained by continuation at $f(6)$ are squared boxes.

Example 5: Construct the missing values in the following table

x :	0	5	10	15	20	25
y :	6	10	—	17	—	31

Solution: Name the missing values as y_2 and y_4. Then the differences table is Table 32.4
Since only four entries y_0, y_1, y_3, y_5 are given, y can be represented by third degree polynomial. Consequently its fourth order differences are zero. Thus

$$\Delta^4 y_0 = y_4 + 6y_2 - 102 = 0 \quad \text{and}$$
$$\Delta^4 y_1 = 143 - 4y_4 - 4y_2 = 0$$

Solving $y_2 = 13.25$, $y_4 = 22.5$.

Generalized power (factorial function)

Example 6: Express $x^3 - 2x^2 + x - 1$ in generalized powers (or into factorial polynomial). Hence

Table 32.4:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	6				
		4			
5	10		$y_2 - 14$		
		$y_2 - 10$		$41 - 3y_2$	$y_4 + 6y_2 - 102$
10	y_2		$27 - 2y_2$		
		$17 - y_2$		$y_4 + 3y_2 - 61$	
15	17		$y_4 - 34 + y_2$		$143 - 4y_4 - 4y_2$
		$y_4 - 17$		$82 - 3y_4 - y_2$	
20	y_4		$48 - 2y_4$		
		$31 - y_4$			
25	31				

show that $\Delta^4 f(x) = 0$.

Solution: Assume that

$$f(x) = x^3 - 2x^2 + x - 1 = [x]^3 + B[x]^2 + C[x] + D$$
$$= x(x-1)(x-2) + Bx(x-1) + cx + D.$$

Put $x = 0$, then $-1 = D$
Put $x = 1$, then $C + D = -1 \quad \therefore C = 0$
Put $x = 2$, then $2B + C + D = 1 \quad \therefore B = 1$
Thus $f(x) = [x]^3 + [x]^2 - 1$
Since differencing of power function $[x]^r$ amounts differentiation of x^r, we have

$$\Delta f(x) = \Delta[x]^3 + \Delta[x]^2 - \Delta[x]^0$$
$$= \frac{d}{dx}x^3 + \frac{d}{dx}x^2 - \frac{d}{dx}1 = 3[x]^2 + 2[x]$$
$$\Delta^2 f = 6[x] + 2$$
$$\Delta^3 f = 6 \quad \text{and} \quad \Delta^4 f(x) = 0.$$

Example 7: Obtain the function whose first difference is $2x^3 + 3x^2 - 5x + 4$.

Solution: Let $\Delta f(x) = 2x^3 + 3x^2 - 5x + 4$.
Now like the above problem

$$2x^3 + 3x^2 - 5x + 4 = 2[x]^3 + B[x]^2 + C[x] + D$$

Solving $D = 4$, $C = 0$, $B = 9$. Thus

$$\Delta f(x) = 2[x]^3 + 9[x]^2 + 0 + 4$$

Integrating, $f(x) = \frac{2[x]^4}{4} + 9\frac{[x]^3}{3} + 4[x] + C$

$$f(x) = \frac{1}{2}x(x-1)(x-2)(x-3) + \\ +3(x)(x-1)(x-2) + 4x + C$$

where C is constant of integration.

Example 8: Evaluate $\Delta^3[(1-x)(1-2x)(1-3x)]$.

Solution:

$$\Delta^3[(1-x)(1-2x)(1-3x)] \\ = \Delta^3[(-1)(-2)(-3)x^3 + (11)x^2 + (-6)x + 1] \\ = -6\Delta^3 x^3 + 0 + 0 + 0 = -6 \cdot 3! = -6 \cdot 6 = -36 \\ \because\ \Delta^n f(x) = a \cdot n! h^n.$$

EXERCISE

1. Evaluate (a) $\Delta \tan^{-1}\left(\frac{n-1}{n}\right)$; (b) $\Delta^2 \cos 2x$; (c) $\Delta^n(e^{2x+3})$.

Ans. (a) $\tan^{-1}\frac{1}{2n^2}$; (b) $-4\sin^2 h\cos(2x+2h)$; (c) $(e^2-1)e^{2x+3}$

2. Evaluate

   a. $\Delta^2\left(\frac{5x+12}{x^2+5x+16}\right)$

   b. $\Delta^2\left(\frac{1}{(x^2)+5x+6}\right)$

   with internal spacing $h = 1$.

Ans. (a) $\frac{2(5x+16)}{(x+2)(x+3)(x+4)(x+5)}$; (b) $\frac{-2}{(x+2)(x+3)(x+4)}$

3. Evaluate (a) $\Delta^{10}[(1-x)(1-2x^2)(1-3x^3)\times \times(1-4x^4)]$ if the interval of spacing is 2.

   Hint: L.H.S.$=\Delta^{10}[(-1)(-2)(-3)(-4)x^{10} +$ terms containing powers of x less than 10] $= 24\Delta^{10}[x^{10}] + 0 + \cdots + 0 = 24 \cdot 10! 2^{10}$ since $\Delta^n f(x) = n! h^n$.

Ans. $24 \times 2^{10} \times 10!$

4. Show that $\Delta\left(\frac{u_k}{v_k}\right) = \frac{v_k \Delta u_k - u_k \Delta v_k}{v_k v_{k+1}}$.

   Hint: $\Delta\left(\frac{u_k}{v_k}\right) = \frac{u_{k+1}}{v_{k+1}} - \frac{u_k}{v_k} = \frac{u_{k+1}v_k - u_k v_{k+1}}{v_k v_{k+1}}$

   $= \frac{(u_{k+1}v_k - u_k v_k + u_k v_k - u_k v_{k+1})}{(u_k v_{k+1})}$

5. Express $f(x) = 2x^3 - 3x^2 + 3x - 10$ in generalized power (in factorial notation) and hence find $\Delta^3 y$.

Ans. $f(x) = 2[x]^3 + 3[x]^2 + 2[x] - 10$, $\Delta^3 y = 12$

6. Compute the missing values in the following table:

x :	45	50	55	60	65
y :	3.0	—	2.0	—	−2.4

   Hint: Solve $\Delta^3 y_0 = 3y_1 + y_3 - 9 = 0$, $\Delta^3 y_1 = y_1 + 3y_2 - 3.6 = 0$

Ans. $y_1 = 2.925$, $y_3 = 0.225$

7. Represent $f(x) = x^4 - 12x^3 + 42x^2 - 30x + 9$ and its successive differences in generalized power.

Ans. $f(x) = [x]^4 - 6[x]^3 + 13[x]^2 + [x]^1 + 9$, $\Delta y = 4[x]^3 - 18[x]^2 + 26[x] + 1$, $\Delta^4 y = 24$.

8. Obtain the function whose first difference is $x^3 + 3x^2 + 5x + 12$.

Ans. $\frac{1}{4}[x]^4 + 2[x]^3 + \frac{9}{2}[x]^2 + 125[x]^1 + C$.

32.3 INTERPOLATION

To fix the height of dam across a river, an engineer utilizes a set of data of the form (x_i, y_i) where x_i denotes the year and y_i, the peak flood level across the river; to estimate the highest possible flood level in future. From the recorded data of (t_i, A_i), time and altitude of a rocket, a physicist would like to estimate what was the altitude at a particular time $t_0 (\neq t_i)$. Thus most of the experimental or observed data is in the form a set of say $(n+1)$ ordered pairs $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ which is the tabular form of an unknown function $y = f(x)$. The process of determining the value of y for an $x \in [x_0, x_n]$ is known as interpolation. Here $x_0, x_1, \ldots, x_n$ are

called interpolation (or mesh) points. Note that x differs from the interpolation points. Thus interpolation is the "art of reading *between* the lines of a table." In extrapolation value of y is determined for an $x \notin [x_0, x_n]$ i.e., for x outside the interval $[x_0, x_n]$. But generally interpoaltion includes extrapolation also.

The problem of interpolation is to construct a new (interpolating) function $F(x)$ which collocates (coincides) with the unknown function $f(x)$ at the tabulated $(n+1)$ interpolation points

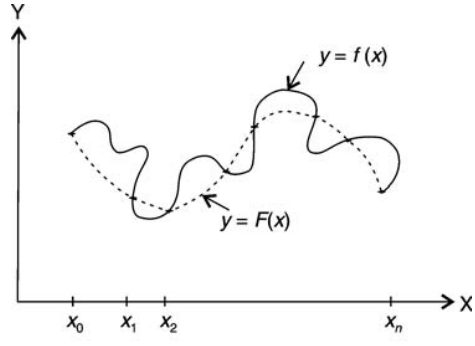


Fig. 32.10

i.e., $y_0 \equiv f(x_0) = F(x_0), \quad y_1 \equiv f(x_1) = F(x_1), \ldots$

$y_n \equiv f(x_n) = F(x_n).$

Geometrically, this means that graphs of $y = f(x)$ and $y = F(x)$ coincide at the $n+1$ points. Since a unique straight line passes through two given points, a unique parabola through three given points, a unique polynomial $F(x)$ of degree n can be determined (passing) satisfying the given set of $(n+1)$ points. In this case it is called a polynomial interpolation. Now from $y = F(x)$, y can be computed for a given x. This interpolating polynomial will be expressed in terms of finite differences in several forms leading to the Newton-Gregory forward, backward and central differences formulae of Gauss, Stirling, Bessel, Everett etc. Lagrange's interpolation and Newton's divided differences formulae are applicable for unequally spaced interpolation points x_i.

32.4 NEWTON-GREGORY FORWARD INTERPOLATION FORMULA

Suppose the values of $y_i = f(x_i)$ are given for equally spaced values of the independent variable (argument) $x_i = x_0 + ih$ for $i = 0, 1, 2, \ldots, n$. Here h, known as the size of the interval or spacing, is constant. Assume that the nth degree interpolating polynomial is given by

$$F(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0)(x - x_1)\cdots(x - x_{n-1}) \quad (1)$$

Using the $n+1$ conditions, $y_i = F(x_i)$ for $i = 0, 1, 2, \ldots, n$, we determine the $(n+1)$ unknown coefficients $a_0, a_1, a_2, \ldots, a_n$ in (1).

Putting $x = x_0$ in (1), we get

$$y_0 = F(x_0) = a_0 + 0 + \cdots + 0 \quad \therefore \quad a_0 = y_0$$

Putting $x = x_1$ in (1), we have

$$y_1 = F(x_1) = a_0 + a_1(x_1 - x_0)$$

But $\quad a_0 = y_0 \quad$ and $\quad x_1 - x_0 = h$

$$\therefore \quad a_1 = \frac{y_1 - a_0}{x_1 - x_0} = \frac{y_1 - y_0}{h} = \frac{1}{h}\Delta y_0$$

Now with $x = x_2$ in (1), we get

$$y_2 = F(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$$
$$= a_0 + a_1 \cdot 2h + a_2 \cdot 2h \cdot h$$

$$\text{so} \quad a_2 = \frac{y_2 - a_0 - 2h \cdot a_1}{2h^2} = \frac{y_2 - y_0 - 2h\frac{1}{h}\Delta y_0}{2h^2}$$

$$a_2 = \frac{y_2 - y_0 - 2(y_1 - y_0)}{2h}$$

$$= \frac{y_2 - 2y_1 + y_0}{2h} = \frac{1}{2h}\Delta^2 y_0.$$

Similarly, at $x = x_3$, we have

$$y_8 = F(x_3) = a_0 + a_1(x_3 - x_0) + a_2(x_3 - x_0)(x_3 - x_1) + a_3(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)$$
$$= a_0 + a_1 \cdot 3h + a_2 \cdot 3h \cdot 2h + a_3 \cdot 3h \cdot 2h \cdot h$$

Solving $\quad a_3 = \dfrac{y_3 - 3y_2 + 3y_1 - y_0}{3!h^2} = \dfrac{1}{3!h^2}\Delta^3 y_0.$

This way, we get

$$a_4 = \frac{1}{4!h^4}\Delta^4 y_0, \quad a_5 = \frac{1}{5!h^5}\Delta^5 y_0 \quad \text{etc. and}$$

$$\boxed{a_n = \frac{1}{n!h^n}\Delta^n y_0} \tag{2}$$

Substituting these values of $a_0, a_1, \ldots, a_n$ in (1), we get the Newton-Gregory forward interpolation formula (also known as *Newton's first interpolation formula*) as

$$y = F(x) = y_0 + \frac{\Delta y_0}{h}(x - x_0) + \frac{\Delta^2 y_0}{2!h^2}(x - x_0)(x - x_1) + \\ + \frac{\Delta^3 y_0}{3!h^3}(x - x_0)(x - x_1)(x - x_2) + \cdots \\ + \frac{\Delta^n y_0}{n!h^n}(x - x_0)(x - x_1)(x - x_2)\cdots(x - x_{n-1}) \tag{3}$$

Introducing $q = \frac{x - x_0}{h}$, the above formula (3) can be written in more convenient way.

Now $$\frac{x - x_1}{h} = \frac{x - (x_0 + h)}{h} = \frac{x - x_0}{h} - 1 = q - 1,$$

$$\frac{x - x_2}{h} = \frac{x - (x_0 + 2h)}{h} = \frac{x - x_0}{h} - 2 = q - 2 \text{ etc.}$$

$$\frac{x - x_{n-1}}{h} = \frac{x - (x_0 + (n-1)h)}{h} \\ = q - (n-1) = q - n + 1.$$

Substituting these values, we have

$$F(x) = F(x_0 + hq) = g(q) = y_0 + \Delta y_0 \cdot q + \\ + \frac{\Delta^2 y_0}{2!}q(q-1) + \frac{\Delta^3 y_0}{3!}q(q-1)(q-2) + \cdots \\ + \frac{q(q-1)\cdots(q-n+1)}{n!}\Delta^n y_0 \tag{4}$$

Note that the coefficients of Δ's are binomial coefficients. Since (4) involves only the "forward differences" $\Delta y_0, \Delta^2 y_0, \ldots, \Delta^n y_0$, Newton-Gregory forward interpolation formula given by (4) is most often used to interpolate (and extrapolate) for values of y at the ***begining*** of a set of tabular data. For $n = 1$ in (4), we get linear interpolation

$$P_1(x) = y_0 + q\Delta y_0.$$

For $n = 2$ in (4), we have parabolic interpolation

$$P_2(x) = y_0 + q\Delta y_0 + \frac{q(q-1)}{2}\Delta^2 y_0.$$

Newton-Gregory Backward Interpolation Gormula

It is mainly useful to interpolate near the ***end*** of the table. Assume the polynomial as

$$y = F(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + \\ + a_3(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \cdots + \\ + a_n(x - x_n)(x - x_{n-1})\cdots(x - x_1) \tag{5}$$

We use $y_i = F(x_1)$ to determine $a_0, a_1, \ldots, a_n$. Put $x = x_n$ in (5). Then

$$y_n = F(x_n) = a_0 + 0 \cdots + 0 \quad \therefore a_0 = y_n.$$

When $x = x_{n-1}$ in (5), we get

$$y_{n-1} = F(x_{n-1}) = a_0 + a_1(x_{n-1} - x_n)$$

or $$a_1 = \frac{y_{n-1} - a_0}{x_{n-1} - x_n} = \frac{y_n - y_{n-1}}{x_n - x_{n-1}} = \frac{1}{h}\nabla y_n$$

For $x = x_{n-2}$ in (5), we have

$$y_{n-2} = F(x_{n-2}) = a_0 + a_1(x_{n-2} - x_n) + \\ + a_2(x_{n-2} - x_n)(x_{n-2} - x_{n-1})$$

$$a_2 = \frac{y_{n-2} - 2y_{n-1} + y_n}{2h^2} = \frac{1}{2h^2}\nabla^2 y_n.$$

Similarly, we get

$$\boxed{a_n = \frac{1}{n!h^n}\nabla^n y_n}$$

Substituting these values of $a_0, a_1, \ldots, a_n$ in (5), we get the Newton-Gregory backward interpolation formula (also known as *Newton's second interpolation formula*) as

$$y = F(x) = y_n + \frac{(x - x_n)}{h}\nabla y_n + \\ + \frac{(x - x_n)(x - x_{n-1})}{2!h^2}\nabla^2 y_n + \cdots \\ + \frac{(x - x_n)(x - x_{n-1})\cdots(x - x_1)}{n!h^n}\nabla^n y_n \tag{6}$$

Introducing $q = \frac{x - x_n}{h}$ and noting that

$$\frac{x - x_{n-1}}{h} = q + 1, \quad \frac{x - x_{n-2}}{h} = q + 2, \ldots$$

$$\frac{x - x_n}{h} = \frac{x - (x_n - \overline{(n-1)}h)}{h} = q + n - 1,$$

$$y = F(x) = F(x_n + hq) = y_n + q\nabla y_n + \frac{q(q+1)}{2!}\nabla^2 y_n + \\ + \frac{q(q+1)(q+2)}{3!}\nabla^3 y_n + \cdots \\ + q(q+1)(q+2)\cdots(q+n-1)\cdot\frac{1}{n!}\nabla^n y_n \qquad (7)$$

Generally (4) is used for forward interpolation and backward extrapolation and (7) is used for backward interpolation and forward extrapolation.

WORKED OUT EXAMPLES

Example 1: Compute (a) $y(9)$; (b) $y(7)$; (c) $y(17)$ and (d) $y(19)$ from the following data:

x :	8	10	12	14	16	18
y :	10	19	32.5	54	89.5	15.4

Solution: Since $x = 9$ and $x = 7$ are at the begining of the table (data), use Newton-Gregory Forward Interpolation Formula (NGFIF) for interpolation at $x = 9$ and extrapolation at $x = 7$. Similarly, since $x = 17$ and $x = 19$ occur at the end of the data, use Newton-Gregory Backward Interpolation Formula (NGBIF) for interpolation at $x = 17$ and extrapolation at $x = 19$. Here interval size is $h = 2$ (points x_i are equidistant).

The finte differences table (from which both forward and backward differences can be read) is obtained below:

Table 32.5:

i	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	8	10					
			9				
1	10	19		4.5			
					3.5		
			13.5			2.5	
2	12	32.5		8			6.5
			21.5		6		
3	14	54		14		9	
			35.5		15		
4	16	89.5		29			
			64.5				
5	18	154					

Newton-Gregory forward interpolation formula is

$$y = y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_0 + \\ + \frac{q(q-1)(q-2)}{3!}\Delta^3 y_0 + \frac{q(q-1)(q-2)(q-3)}{4!}\Delta^4 y_0 + \\ + \frac{q(q-1)(q-2)(q-3)(q-4)}{5!}\Delta^5 y_0.$$

From the difference table, $y_0 = 10$, $\Delta y_0 = 9$, $\Delta^2 y_0 = 4.5$, $\Delta^3 y_0 = 3.5$, $\Delta^4 y_0 = 2.5$ and $\Delta^5 y_0 = 6.5$.

a. To interpolate at $x = 9$: $q = \frac{x - x_0}{h}$ so $q = \frac{9-8}{2} = \frac{1}{2}$.

Substituting the above data in the formula

$$y(9) = 10 + \frac{1}{2}(9) + \frac{1}{2}\left(\frac{1}{2} - 1\right)\frac{1}{2!}(4.5) \\ + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right)}{3!}(3.5) \\ + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right)\left(\frac{1}{2} - 3\right)}{4!}(2.5) \\ + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right)\left(\frac{1}{2} - 3\right)\left(\frac{1}{2} - 4\right)}{5!}(6.5)$$

$y(9) = 14.455.$

b. To extrapolate at $x = 7$: $q = \frac{7-8}{2} = -\frac{1}{2}$.

$$y(7) = 10 - \frac{9}{2} + \left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\frac{1}{2!}(4.5) \\ - \frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\frac{1}{3!}(3.5) \\ + \frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\frac{1}{4!}(2.5) \\ - \frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\left(\frac{9}{2}\right)\frac{1}{5!}(6.5)$$

$y(7) = 5.1777.$

Newton-Gregory backward interpolation formula is

$$y = y_n + q\nabla y_n + \frac{q(q+1)}{2!}\nabla^2 y_n + \\ + \frac{q(q+1)(q+2)}{3!}\nabla^3 y_n +$$

$$+\frac{q(q+1)(q+2)(q+3)}{4!}\nabla^4 y_n + \\ +\frac{q(q+1)(q+2)(q+3)(q+4)}{5!}\nabla^5 y_n.$$

From the difference table, $x_n = 18$, $y_n = 154$, $\nabla y_n = 64.5$, $\nabla^2 y_n = 29$, $\nabla^3 y_n = 15$, $\nabla^4 y_n = 9$, $\nabla^5 y_n = 6.5$.

c. To interpolate at $x = 17$, we have

$$q = \frac{x - x_n}{h} = \frac{17 - 18}{2} = -\frac{1}{2}$$

Substituting these values in the formula

$$y(17) = 154 + \left(-\frac{1}{2}\right)(64.5) \\ +\left(-\frac{1}{2}\right)\left(-\frac{1}{2}+1\right)\cdot\frac{1}{2!}(29) \\ +\left(-\frac{1}{2}\right)\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)\frac{1}{3!}15 \\ +\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)\left(-\frac{1}{2}+3\right)(9)}{4!} \\ +\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)\left(-\frac{1}{2}+3\right)\left(-\frac{1}{2}+4\right)(6.5)}{5!}$$

$y(17) = 126.841$.

d. To extrapolate at $x = 19$, take $q = \frac{19-18}{2} = \frac{1}{2}$. Then

$$y(19) = 154 + \frac{1}{2}(64.5) + \frac{1}{2}\left(\frac{3}{2}\right)\frac{1}{2!}(29) \\ +\frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\frac{1}{3!}(15) \\ +\frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\frac{1}{4!}(9) \\ +\frac{1}{2}\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\left(\frac{9}{2}\right)\frac{1}{5!}(6.5)$$

$y(19) = 219.208$.

Example 2: Fit a polynomial of degree three which takes the following values:

x :	3	4	5	6
y :	6	24	60	120

Solution: The Newton-Gregory forward interpolating polynomial collocates (takes or geometrically passes through) the given set of points (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3) and is given by

$$y = y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_0 + \frac{q(q-1)(q-2)}{3!}\Delta^3 y_0$$

Here $q = \frac{x-x_0}{h}$ and $h = 1$ (given).
The finite differences table is

Table 32.6:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
3	6	18		
4	24		18	
		36		6
5	60		24	
		60		
6	120			

From the table, $x_0 = 3$, $y_0 = 6$, $\Delta y_0 = 18$, $\Delta^2 y_0 = 18$, $\Delta^3 y_0 = 6$, $q = \frac{x-x_0}{h} = \frac{x-3}{1} = x - 3$
Substituting these values

$$y = 6 + (x-3)(18) + (x-3)\frac{(x-3-1)}{2!}(18) \\ +(x-3)(x-3-1)\frac{(x-3-2)}{3!}(6) \\ = 6 + 18(x-3) + 9(x-3)(x-4) \\ +(x-3)(x-4)(x-5)$$

$$y(x) = x^3 - 3x^2 + 2x$$

is the required cubic polynomial (which takes the given data for example, $y(3) = 3^3 - 3(3^2) + 2(3) = 6$ etc.).

Note: Using Newton's backward interpolating polynomial (using backward differences), we get the same cubic polynomial

$$y = y_n + q\nabla y_n + \frac{q(q+1)}{2!}\nabla^2 y_n \\ +\frac{q(q+1)(q+2)}{3!}\nabla^3 y_n \\ = 120 + (x-6)60 + \frac{(x-6)(x-5)}{2!}(24) \\ +\frac{(x-6)(x-5)(x-4)}{3!}6$$

$$y(x) = x^3 - 3x^2 + 2x.$$

EXERCISE

1. Using Newton's forward formula compute the pressure of the steam at temperature 142° from the following steam table:

Temperature:	140	150	160	170	180
Pressure:	3.685	4.854	6.302	8.076	10.225

Hint: Forward differences 1.169, 0.279, 0.047, 0.002.

Ans. 3.899

2. Using Newton's backward formula compute $f(43)$, $f(84)$ from the following table. Fit a polynomial

x :	40	50	60	70	80	90
$f(x)$:	184	204	226	250	276	304

Hint: Backward differences: 28, 2, 0, 0, 0.

Ans. $f(43) = 190$, $f(84) = 287$; $f(x) = 84 + 2.9x - 0.01x^2$

3. Fit a cubic polynomial which takes the following values. Hence find $y(4)$.

x :	0	1	2	3
y :	1	0	1	10

Ans. $x^3 - 2x^2 + 1$, $y(4) = 33$, forward differences: $-1, 2, 6$

4. Estimate the population in 1895 and 1925 from the following statistics:

Year x :	1891	1901	1911	1921	1931
Population y :	46	66	81	93	101

Hint: Forward differences: 20, −5, 2, −3;

Backward differences: 8, −4, −1, −3.

Ans. 54.85, 96.84

5. Compute the first and tenth term of the following series assuming that values of y are consecutive.

x :	3	4	5	6	7	8	9
y :	2.7	6.4	12.5	21.6	34.3	51.2	72.9

Hint: Forward differences: 3.7, 2.4, 0.6, 0;

Backward differences: 21.7, 4.8, 0.6, 0.

Ans. $y(1) = 0.1$, $y(10) = 100$

6. Construct an empirical formula for the function y specified in the following table. Hence find $y(-1)$, $y(0.5)$,

x :	0	1	2	3	4	5
y :	5.2	8.0	10.4	12.4	14.0	15.2

Ans. $y = 5.2 + 3x - 0.2x^2$, $y(-1) = 2$, $y(0.5) = 6.65$

7. Determine $\log_{10} 1044$ using backward formula from the data below:

x:	1000	1010	1020
$\log_{10} x$:	3	3.0043214	3.0086002
x:	1030	1040	1050
$\log_{10} x$:	3.0128372	3.017033	3.601193

Hint: Backward differences: 0.004156, 0.0000401, 0.0000008.

Ans. 3.0187005

Extrapolation

8. Compute $\sin 14°$ and $\sin 56°$ from the data below:

x:	15°	20°	25°	30°	35°
$\sin x$:	0.2588	0.342	0.4226	0.5	0.5736
x:	40°	45°	50°	55°	
$\sin x$:	0.6428	0.7071	0.706	0.8192	

Hint: Forward differences: 0.0832, −0.0026, −0.0006

Backward differences: 0.0532, −0.0057, −0.0003

Ans. $\sin 14° = 0.24192$, $\sin 56° = 0.82904$

9. Find the polynomial of degree four which takes the following values:

x :	2	4	6	8	10
y :	0	0	1	0	0

Hint: Differences: 0, 1, −3, 6.

Ans. $\frac{1}{64}[x^4 - 24x^3 + 196x^2 - 624x + 640]$

10. Estimate the number of students who secured marks between 40 and 45 from the following table:

Marks x :	30 to 40	40 to 50	50 to 60
No. Students y :	31	42	51
Marks x :	60 to 70	70 to 80	
No. Students y :	35	31	

Ans. $y(45) = 47.87 \approx 48$, number of students between 40 and 45 is $48 - 31 = 17$

11. Compute $y(1.5)$ and $y(8.5)$ from the following data:

x :	1	2	3	4	5	6	7	8
y :	1	8	27	64	125	216	343	512

Hint: F.D.: 7, 12, 6, 0, 0, B.D.: 169, 42, 6, 0, 0.

Ans. $y(1.5) = 3.375$, $y(8.5) = 614.125$

12. If $F(\phi) = \int_0^{\phi} \frac{dt}{\sqrt{1-\frac{1}{2}\sin^2 t}}$ (elliptic integral), compute $F(23.5)$ by both N-G forward and backward formulae.

ϕ :	21	22	23	24	25	26
$F(\phi)$:	0.3706	0.3887	0.4068	0.425	0.4433	0.4616

Hint: F.D.: $\phi = 22$, $F(22) = 0.3887$, 0.0181, 0.0001

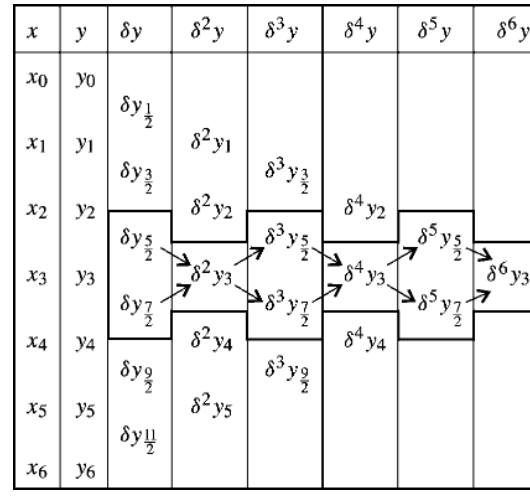
B.D.: $\phi = 25$, $F(25) = 0.4433$, 0.0183, 0, −0.0001.

Ans. Forward: $F(23.5) = 0.4159$, Backward: 0.41588.

32.5 CENTRAL DIFFERENCES

Observe that the Newton-Gregory forward difference formula involves only the leading differences Δy_0, $\Delta^2 y_0$, $\Delta^3 y_0$ etc. corresponding to the leading term (initial value) x_0. Thus only those values of function were used which lie on one side of the chosen initial value. Similarly, the Newton-Gregory backward difference formula used one-sided values ∇y_n, $\nabla^2 y_n$, $\nabla^3 y_n$ which lie on one side of y_n. Interpolation formulae that contain both preceding and succeeding values of the funtion w.r.t., the initial value, give very useful results. For this purpose, the differences located in a horizontal row (line) corresponding to the initial values x_0 and y_0 in a diagonal difference table are used. These are known as central differences and are used in the Gauss, Stirling and Bessel interpolation formulae, **for interpolation near the middle of the table**, taken as the initial value. Central differences are defined by the operator $\boldsymbol{\delta}$ as follows:

Table 32.7: Central differences table



$$\delta y_{\frac{1}{2}} = y_1 - y_0, \delta y_{\frac{3}{2}} = y_2 - y_1, \ldots, \delta y_{n-\frac{1}{2}} = y_n - y_{n-1}$$

Higher order central differences are

$$\delta^2 y_1 = \delta y_{\frac{3}{2}} - \delta y_{\frac{1}{2}}, \delta^2 y_2 = \delta y_{\frac{5}{2}} - \delta y_{\frac{3}{2}}, \quad \text{etc.}$$

Note 1: The subscript of δy for any difference is the average of the subscripts of the two members of the difference.

Note 2: Differences on the same horizontal line have the same suffix.

Note 3: Odd order differences are known only for half values of the suffix (having fractional suffix) while even order differences for only integral values of the suffix.

Note 4: Observe that $\Delta y_0 = y_1 - y_0 = \nabla y_1 = \delta y_{\frac{1}{2}}$. Also $\Delta^3 y_2 = \nabla^3 y_5 = \delta^3 y_{\frac{7}{2}}$ etc. i.e., same numbers occur in the same positions but identified in different ways.

In order to derive the central-differences interpolation formulas of Gauss, Stirling and Bessel, we rearrange the central differences table with the *initial values x_0 and y_0 placed at the middle of the table* as shown below. Here

$$x_i = x_0 + ih \quad \text{for } i = 0, \pm1, \pm2, \ldots$$

$$y_i = f(x_i) \quad \text{and}$$

$$\Delta y_i = y_{i+1} - y_i,\ \Delta^2 y_i = \Delta y_{i+1} - \Delta y_i \quad \text{etc.}$$

32.6 STIRLING AND BESSEL'S INTERPOLATION FORMULAE

Gaussian Interpolation Formulae

Consider $(2n + 1)$ equally spaced points

$$x_{-n}, x_{-(n-1)}, \ldots, x_{-1}, x_0, x_1, \ldots, x_{n-1}, x_n$$

where

$$\Delta x_i = x_{i+1} - x_i = h = \text{constant}$$

with $i = -n, -(n-1), \ldots, -1, 0, 1, \ldots, n-1, n$.

Let $P(x)$ be a polynomial of degree $2n$, given by

$$P(x) = a_0 + a_1(x - x_0) + \cdots + a_{2n}(x - x_{-(n-1)}) \cdots (x - x_{-1}) \times \\ \times (x - x_0)(x - x_1) \cdots (x - x_{n-1})(x - x_n).$$

If $P(x_i) = y_i = f(x_i)$ for $i = 0, \pm1, \pm2, \ldots, \pm n$ then $a_0 = y_0$, $a_1 = \frac{\Delta y_0}{h}$, $a_2 = \frac{\Delta^2 y_{-1}}{2!h^2}$, $\ldots$, $a_{2n} = \frac{\Delta^{2n} y_{-n}}{(2n)!h^{2n}}$.

Introducing $q = \frac{x - x_0}{h}$, we get Gauss first (GI) interpolation formula

$$P(x) = y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_{-1} \\ + \frac{(q+1)q(q-1)}{3!}\Delta^2 y_{-1} \\ + \frac{(q+1)q(q-1)(q-2)}{4!}\Delta^2 y_{-2} + \cdots \\ + \frac{(q+n-1)\cdots(q-n)}{(2n)!}\Delta^{2n} y_{-n} \qquad (1)$$

Similarly, Gaussian second (GII) interpolation formula is

$$P(x) = y_0 + q\Delta y_{-1} \\ + \frac{(q+1)q}{2!}\Delta^2 y_{-1} + \frac{(q+1)(q-1)}{3!}\Delta^3 y_{-2} \\ + \frac{(q+2)(q+1)q(q-1)}{4!}\Delta^4 y_{-2} + \cdots \\ + \frac{(q+n)(q+n-1)\cdots(q-n+1)}{(2n)!}\Delta^{2n} y_{-n} \qquad (2)$$

To shift the initial value from (x_0, y_0) to (x_1, y_1) replace q by $q - 1$ and increase the indices in R.H.S. of (2) by (1). This yields another Gaussian

Table 32.8 Central differences table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
x_{-4}	y_{-4}						
		Δy_{-4}					
x_{-3}	y_{-3}		$\Delta^2 y_{-4}$				
		Δy_{-3}		$\Delta^3 y_{-4}$			
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$		$\Delta^4 y_{-4}$		
		Δy_{-2}		$\Delta^3 y_{-3}$		$\Delta^5 y_{-4}$	
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$		$\Delta^6 y_{-4}$
		Δy_{-1}		$\Delta^3 y_{-2}$		$\Delta^5 y_{-3}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$		$\Delta^6 y_{-3}$
		Δy_0		$\Delta^3 y_{-1}$		$\Delta^5 y_{-2}$	
x_1	y_1		$\Delta^2 y_0$		$\Delta^4 y_{-1}$		$\Delta^6 y_{-2}$
		Δy_1		$\Delta^3 y_0$		$\Delta^5 y_{-1}$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$		
		Δy_2		$\Delta^3 y_1$			
x_3	y_3		$\Delta^2 y_2$				
		Δy_3					
x_4	y_4						

G_2 ----- G_1 —— G_3 .....

interpolation (GIII) formula

$$P(x) = y_1 + (q-1)\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_0 + \frac{q(q-1)(q-2)}{3!}\Delta^3 y_{-1} + \frac{(q+1)q(q-1)(q-2)}{4!}\Delta^4 y_{-1} + \cdots + \frac{(q+n-1)\cdots(q-n)}{(2n)!}\Delta^{2n} y_{-(n-1)} \quad (3)$$

Stirling's Interpolation Formula

is the arithmetic mean of GI and GII i.e., of (1) and (2)

$$P(x) = y_0 + q\frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{q^2}{2}\Delta^2 y_{-1} + \frac{q(q^2-1)}{3!}\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{q^2(q^2-1)}{4!}\Delta^4 y_{-2} + \frac{q(q^2-1)(q^2-2^2)}{5!}\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \cdots \quad (4)$$

Bessel's Interpolation Formula

is the arithmetic mean of GI and GIII i.e., of (1) and (3).

$$P(x) = y_0 + q\Delta y_0 + \frac{q(q-1)}{2}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{(q-\frac{1}{2})q(q-1)}{3!}\Delta^3 y_{-1} + \frac{q(q-1)(q+1)(q-2)}{4!}\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \cdots.$$

Note 1: It is advised to use Stirling's formula for $|q| \le 0.25$ and Bessel's formula for $0.25 \le q \le 0.75$.

Note 2: Stirling's formula consists of the even differences along the central line and mean of the odd differences just above and below the central line

$$-\ y_0 \ \underset{\Delta y_0}{\overset{\Delta y_{-1}}{---}}\ \Delta^2 y_{-1} \ \underset{\Delta^5 y_{-1}}{\overset{\Delta^3 y_{-2}}{---}}\ \Delta^4 y_{-2} \ \underset{\Delta^5 y_{-2}}{\overset{\Delta^5 y_{-3}}{---}}\ \Delta^6 y_{-3} - \text{central line.}$$

Note 3: Bessel's formula consists of means of even differences along and below the central line and odd differences below the central line

$$y_0 \text{ -- } \Delta^2 y_{-1} \text{ --- } \Delta^4 y_{-2} \text{ --- } \Delta^6 y_{-3} \text{ -- central line.}$$
$$\Delta y_0 \qquad \Delta^3 y_{-1} \qquad \Delta^5 y_{-2}$$
$$\Delta^2 y_0 \qquad \Delta^4 y_{-1} \qquad \Delta^6 y_{-2}.$$

WORKED OUT EXAMPLES

Example 1: Using *Stirling's* formula, compute $f(1.22)$ from the following data:

x :	1.0	1.1	1.2	1.3	1.4
$f(x)$:	0.841	0.891	0.932	0.963	0.985

Solution: Choose origin as $x_0 = 1.2$ given $h =$ interval size $= 0.1$ then $q = \frac{x - x_0}{h} = \frac{1.22 - 1.2}{0.1} = 0.2$
The central differences table is 32.9 below.
Since $q = 0 : 2$ lies in between $-\frac{1}{4}$ and $\frac{1}{4}$, Stirling's formula is applicable.

Table 32.9

x	q	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	−2	$y_{-2} = 0.841$				
			$\Delta y_{-2} = 0.05$			
1.1	−1	$y_{-1} = 0.891$		$\Delta^2 y_{-2} = -0.009$		
			$\Delta y_{-1} = 0.041$		$\Delta^3 y_{-2} = -0.001$	
1.2	0	$y_0 = 0.932$		$\Delta^2 y_{-1} = -0.01$		$\Delta^4 y_{-2} = 0.002$
			$\Delta y_0 = 0.031$		$\Delta^3 y_{-1} = +0.001$	
1.3	1	$y_1 = 0.963$		$\Delta^2 y_0 = -0.009$		
			$\Delta y_1 = 0.022$			
1.4	2	$y_2 = 0.985$				

Stirling's formula is

$$y = y_0 + q\left(\frac{\Delta y_{-1} + \Delta y_0}{2}\right) + \frac{q^2}{2!}\Delta^2 y_{-1} + \frac{q(q^2-1)}{3!}\left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2}\right) + \frac{q^2(q^2-1)}{4!}\Delta^4 y_{-2} + \cdots$$

Substituting the values from the table (underlined)

$$y(1.22) = 0.932 + 0.2\frac{(.041 + .031)}{2} + \frac{(0.2)^2}{2!}(-0.01) + \frac{(0.2)[(0.2)^2 - 1]}{6}\frac{(-0.001 + 0.001)}{2} + \frac{(0.2)^2[(0.2)^2 - 1]}{24}(0.002)$$

$$y(1.22) = 0.9389968.$$

Example 2: Use *Bessel's* formula to compute $f(1.95)$ from the following data:

x :	1.7	1.8	1.9	2.0	2.1	2.2	2.3
$f(x)$:	2.979	3.144	3.283	3.391	3.463	3.997	4.491

Solution: Choose the origin at $x_0 = 2.0$, given $h = 0.1$ so $q = \frac{x - x_0}{h} = \frac{1.95 - 2.0}{0.1} = -0.5$. Since $q = -0.5$ lies between $-\frac{1}{4}$ and $\frac{3}{4}$, Bessel's formula is applicable. The central difference table is:

Table 32.10:

x	q	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1.7	−3	2.979					
			0.165				
1.8	−2	3.144		−0.026			
			0.139		−0.005		
1.9	−1	3.283		−0.031		0	
			0.108		−0.005		0.503
2.0	0	3.391		−0.036		0.503	
			0.072		0.498		−1.503
2.1	1	3.463		0.462		−1.0	
			0.53		−0.502		
2.2	2	3.997		−0.04			
			0.494				
2.3	3	4.491					

The Bessel's formula is

$$y = y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right) + \frac{\left(q - \frac{1}{2}\right)q(q-1)}{3!}\Delta^3 y_{-1} + \frac{(q+1)(q)(q-1)(q-2)}{4!}\left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right)$$

Substituting the values from the table (underlined)

$$y(1.95) = 3.391 + (-0.5)(0.072) + \frac{(-0.5)(-0.5-1)}{2}\frac{(-0.036 + 0.462)}{2} + \frac{\left(-0.5 - \frac{1}{2}\right)(-0.5)(-0.5-1)}{6}(0.498) + \frac{(-0.5+1)(0.5)(-0.5-1)(-0.5-2)}{24}\frac{(0.503 - 1.0)}{2}$$

$$y(1.95) = 3.362917.$$

EXERCISE

1. Compute $\sinh 1.41710$ from the following data:

x :	1.0	1.1	1.2	1.3	1.4
$\sinh x$:	1.1752	1.33565	1.50946	1.69838	1.90430
x :	1.5	1.6	1.7	1.8	
$\sinh x$:	2.12928	2.37557	2.64563	2.94217	

Hint: Use Stirlings with $x_0 = 1.4$, $y_0 = 1.90430$, differences 0.20592, 0.22498, 0.01906, 0.00206, 0.00225.

Ans. 1.94136

2. Calculate $f(0.5437)$ for the probability integral $f(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$ with the following data:

x :	0.51	0.52	0.53	0.54
$f(x)$:	0.5292437	0.5378987	0.5464641	0.5549392
x :	0.55	0.56	0.57	
$f(x)$:	0.5633233	0.5716157	0.5798158	

Hint: Use Stirlings with $x_0 = 0.54$, $y_0 = 0.5549392$, differences 0.0084751, 0.0083841, −0.0000910, −0.0000007, 0.

Ans. 0.5580520

3. Compute e^x when $x = 0.644$ by (a) Stirling; (b) Bessel's formula from the following data. Also find $e^{0.638}$ by (a) Stirling; (b) Bessel's formulae.

x :	0.61	0.62	0.63	0.64
e^x :	1.840431	1.858928	1.877610	1.896481
x :	0.65	0.66	0.67	
e^x :	1.9515541	1.934792	1.954237	

Hint:

a, b. Differences: $x_0 = 0.64$, $y_0 = 1.896481$, 0.018871, 0.019060, 0.000189, 0, 0.000002, 0.000002.

c. $x_0 = 0.64$, $x_n = 0.638$, $q = -0.2$

d. $x_0 = 0.63$, $x_n = 0.638$, $q = 0.8$

Ans. **a.** 1.904082
b. 1.904082
c. 1.892692
d. 1.89262

4. Find $f(16)$ by Stirling's formula from the following data:

x :	0	5	10	15	20	25	30
$f(x)$:	0	0.0875	0.1763	0.2679	0.364	0.4663	0.5774

Hint: $x_0 = 15$, $y_0 = 0.2679$, Differences: 0.0916, 0.0961, 0.0045, 0.0017, 0.0017, 0, −0.0002, 0.0009.

Ans. 0.2867

5. Compute $y(x = 5)$ using Bessel's formula

x :	0	4	8	12
y :	143	158	177	199

Ans. 162.41

6. Find $y(12.2)$ using Stirling's formula

x :	10	11	12	13	14
y :	0.23967	0.28060	0.31788	0.35209	0.38368

Hint: $x_0 = 12$, $h = 1$, $p = 12.2 - 12 = 0.2$, $y_0 = 0.31788$, Differences: 0.03728, 0.03421, −0.00365, −0.00307, 0.00058, −0.00045, −0.00013.

Ans. 0.32497

7. Compute $y(25)$ using Bessel's formula

x :	20	24	28	32
y :	2854	3162	3544	3992

Hint: $x_0 = 24$, $h = 4$, $p = \frac{25-24}{4} = \frac{1}{4}$; $y_0 = 3162$, Differences: 382, 74, 66, −8.

Ans. 3250.875

32.7 LAGRANGE'S INTERPOLATION

The interpolation formulae derived so far, are applicable only when argument x is equally spaced (equidistant $x_{i+1} - x_i = h =$ constant). Lagrange's interpolation formula is a more general one and can be applied for arbitrarily specified points i.e., for unequally spaced argument (i.e., $x_{i+1} - x_i = h =$ variable). For example tables of empirical data contian variable intervals.

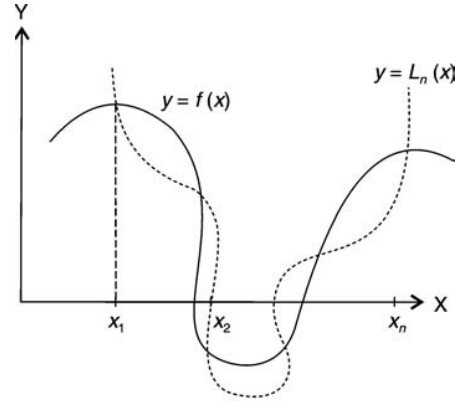


Fig. 32.11

In order to construct a polynomial $L_n(x)$ having (i) degree not exceeding n and (ii) satisfying the $(n + 1)$ set of points i.e., $L_n(x_i) = y_i$ for $i = 0, 1, 2, \ldots, n$ choose

$$L_n(x) = \sum_{i=0} P_i(x) \cdot y_i \qquad (1)$$

Geometrically, the second condition implies that the curve $y = L_n(x)$ meets the curve $y = f(x)$ at these $(n + 1)$ points $x_0, x_1, x_2, \ldots, x_n$. Since the desired polynomial $L_i(x)$ vanishes at n points

$x_0, x_1, \ldots, x_{i-1}, x_{i+1,\ldots,x_n}$ and $L_i(x_i) = y_i$ choose

$$P_i(x) = c_i(x - x_0)(x - x_1)\cdots(x - x_{i-1}) \times \times(x - x_{i+1})\cdots(x - x_n) \quad (2)$$

where c_i is a constant coefficient. To satisfy the first condition, at $x = x_j$

$$L_n(x_j) = \sum_{i=0}^{n} P_i(x_j)y_j = y_j$$

we must have

$$P_i(x_j) = \delta_{ij} = \begin{Bmatrix} 1, \text{ if } & j = i \\ 0, \text{ if } & j \neq i \end{Bmatrix}$$

This determines c_i, put $x = x_i$ in (2), then

$$1 = P_i(x_i) = c_i(x_i - x_0)(x_i - x_1)\cdots(x_i - x_{i-1}) \cdot(x_i - x_{i+1})\cdots(x_i - x_n)$$

or

$$c_i = \frac{1}{(x_i - x_0)(x_i - x_1)\cdots(x_i - x_{i-1})(x_i - x_{i+1})\cdots(x_i - x_n)}$$

Substituting c_i in (2), we get the Lagrangian coefficient functions

$$P_i(x) = \frac{(x-x_0)(x-x_1)\cdots(x-x_{i-1})(x-x_{i+1})\cdots(x-x_n)}{(x_i-x_0)(x_i-x_1)\cdots(x_i-x_{i-1})(x_i-x_{i+1})\cdots(x_i-x_n)} \quad (3)$$

The degree of the polynomial $P_i(x)$ is not higher than n (i.e., $\leq n$). Substituting (3) in (1), we get the required Lagrange's interpolation formula

$$y = L_n(x) = = \sum_{i=0}^{n} y_i \cdot \frac{(x-x_0)(x-x_1)\cdots(x-x_{i-1})(x-x_{i+1})\cdots(x-x_n)}{(x_i-x_0)(x_i-x_1)\cdots(x_i-x_{i-1})(x_i-x_{i+1})\cdots(x_i-x_n)}$$

or in the expanded form

$$y = L_n(x) = \frac{(x - x_1)(x - x_2)\cdots(x - x_n)}{(x_0 - x_1)(x_0 - x_2)\cdots(x_0 - x_n)}y_0 + \frac{(x - x_0)(x - x_2)\cdots(x - x_n)}{(x_1 - x_0)(x_1 - x_2)\cdots(x_1 - x_n)}y_1 + - + \frac{(x-x_0)(x-x_1)\cdots(x-x_{i-1})(x-x_{i+1})\cdots(x-x_n)}{(x_i-x_0)(x_i-x_1)\cdots(x_i-x_{i-1})(x_i-x_{i+1})\cdots(x_i-x_n)}y_i - + \frac{(x-x_0)(x-x_1)(x-x_2)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)(x_n-x_2)\cdots(x_n-x_{n-1})} \cdot y_n \quad (4)$$

Note: If the points are equally spaced, Lagrange's interpolation polynomial coincids with the corresponding Newton's interpolation polynomial.

To determine y at a point x^*, put $x = x^*$ in (4). The unique polynomial (4) containing y_i explicitly is applicable for both unequally spaced and equally spaced points, abscissa $x_0, x_1, \ldots, x_n$ need not be in order, but inconvenient to move from one interpolation polynomial to another of degree one greater.

Note: The given function $y = f(x)$ can be split into partial fractions by dividing (4) throughout by $(x - x_0)(x - x_1)\cdots(x - x_i)\cdots(x - x_n)$ having $n + 1$ factors. Then

$$\frac{f(x)}{(x - x_0)\cdots(x - x_i)\cdots(x - x_n)} = \frac{y_0}{(x_0 - x_1)(x_0 - x_2)\cdots(x_0 - x_n)} \cdot \frac{1}{(x - x_0)} + + \cdots \frac{y_1}{(x_1 - x_0)(x_1 - x_2)\cdots(x_1 - x_n)} \cdot \frac{1}{x - x_1} + - + \frac{y_n}{(x_n - x_0)(x_n - x_1)\cdots(x_n - x_{n-1})} \cdot \frac{1}{(x - x_n)} = \frac{A_0}{x - x_0} + \frac{A_1}{x - x_1} + \cdots + \frac{A_n}{x - x_n}$$

where $A_0, A_1, \ldots, A_n$ coefficients are completely determined by the points $(x_0, y_0), (x_1, y_1) \cdots (x_n, y_n)$.

32.8 INVERSE INTERPOLATION USING LAGRANGE'S INTERPOLATION FORMULA

In the inverse interpolation, for a given value of y, the corresponding value of x is to be determined. For example to find value of the root x for which the function $y = f(x)$ becomes zero is an inverse interpolation problem. Inverse interpolation problem is similar to direct interpolation since the roles of x and y are interchanged.

Now interchanging the roles of x and y, the Lagrange's interpolation formula for inverse inter-

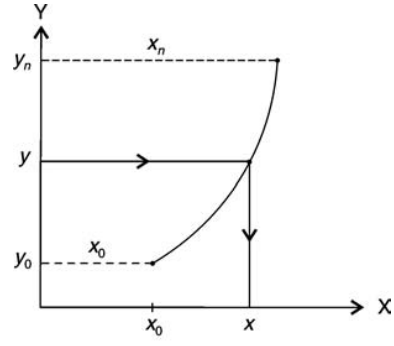


Fig. 32.12

polation takes the form given by

$$x=\sum_{i=0}^{n}\cdot\frac{(y-y_1)(y-y_2)\cdots(y-y_{i-1})(y-y_{i+1})\cdots(y-y_n)}{(y_i-y_1)(y_i-y_2)\cdots(y_i-y_{i-1})(y_i-y_{i+1})\cdots(y_i-y_n)}\cdot x_i$$

or in the expanded form

$$x=\frac{(y-y_1)(y-y_2)\cdots(y-y_n)}{(y_0-y_1)(y_0-y_2)\cdots(y_0-y_n)}x_0+$$
$$+\frac{(y-y_0)(y-y_2)\cdots(y-y_n)}{(y_1-y_0)(y_1-y_2)\cdots(y_1-y_n)}\cdot x_1+\cdots+$$
$$+\frac{(y-y_1)(y-y_2)\cdots(y-y_{i-1})(y-y_{i+1})\cdots(y-y_n)}{(y_i-y_1)(y_i-y_2)\cdots(y_i-y_{i-1})(y_i-y_{i+1})\cdots(y_i-y_n)}\cdot x_i+$$
$$+\cdots+\frac{(y-y_0)(y-y_1)\cdots(y-y_{n-1})}{(y_n-y_0)(y_n-y_1)\cdots(y_n-y_{n-1})}\cdot x_n \qquad (5)$$

Putting $y=y^*$ in (5), we get the value of x^*.

Worked Out Examples

Example 1: Use Lagrange's interpolation formula to fit a polynomial to the following data. Hence find $y(-2)$, $y(1)$ and $y(4)$.

x :	-1	0	2	3
y :	-8	3	1	2

Solution: Here $x_0=-1$, $x_1=0$, $x_2=2$, $x_3=3$ and $y_0=-8$, $y_1=3$, $y_2=1$, $y_3=2$. Using Lagrange's interpolation formula to this

$$y=f(x)=\frac{(x-0)(x-2)(x-3)}{(-1-0)(-1-2)(-1-3)}(-8)+$$
$$+\frac{(x+1)(x-2)(x-3)}{(0+1)(0-2)(0-3)}\cdot 3+\frac{(x+1)(x-0)(x-3)}{(2+1)(2-0)(2-3)}\cdot 1+$$
$$+\frac{(x+1)(x-0)(x-2)}{(3+1)(3-0)(3-2)}\cdot 2$$
$$=\frac{2}{3}x(x-2)(x-3)+\frac{1}{2}(x+1)(x-2)(x-3)-$$
$$-\frac{1}{6}x(x+1)(x-3)+\frac{1}{6}(x+1)x(x-2)$$
$$=(x-2)(x-3)\left(\frac{7x+3}{6}\right)+\frac{1}{6}x(x+1)$$
$$y=\frac{1}{6}[7x^3-31x^2+28x+18]$$

is the required 3rd degree polynomial.

$$y(-2)=\frac{1}{6}[-56-124-56+18]=-\frac{218}{6}=-36.33$$
$$y(1)=\frac{1}{6}[7-31+28+18]=\frac{22}{6}=3.666$$
$$y(4)=\frac{1}{6}[448-496+112+18]=\frac{82}{6}=13.666.$$

Partial fractions

Example 2: Express the function

$$\frac{x^2+6x-1}{(x^2-1)(x-4)(x-6)}$$

as a sum of partial fractions, using Lagrange's formula.

Solution: Consider $f(x)=x^2+6x-1$. Tabulate $f(x)$ at the roots of the denominator of the given expression i.e., at $x=-1,+1,4,6$.

x :	-1	1	4	6
$f(x)$:	-6	6	39	71

We fit a polynomial using Lagranges formula

$$f(x)=\frac{(x-1)(x-4)(x-6)}{(-1-1)(-1-4)(-1-6)}\cdot(-6)+$$
$$+\frac{(x+1)(x-4)(x-6)}{(1+1)(1-4)(1-6)}\cdot 6+\frac{(x+1)(x-1)(x-6)}{(4+1)(4-1)(4-6)}\cdot 39+$$
$$+\frac{(x+1)(x-1)(x-4)}{(6+1)(6-1)(6-4)}\cdot 71$$

$$f(x)=\frac{3}{35}(x-1)(x-4)(x-6)+\frac{1}{5}(x+1)(x-4)(x-6)-$$
$$-\frac{13}{10}(x+1)(x-1)(x-6)+\frac{71}{70}(x+1)(x-1)(x-4)$$

Dividing both sides by $(x^2-1)(x-4)(x-6)$,

$$\frac{f(x)}{(x^2-1)(x-4)(x-6)}$$
$$=\frac{x^2+6x-1}{(x^2-1)(x-4)(x-6)}=\frac{3}{35}\frac{1}{x+1}+\frac{1}{5}\frac{1}{x-1}-$$
$$-\frac{13}{10}\frac{1}{x-4}+\frac{71}{70}\frac{1}{x-6}.$$

Inverse interpolation

Example 3: Compute the value of x, when $y=8$ by inverse interpolation using Lagrange's formula

x :	-2	-1	1	2
y :	-7	2	0	11

Solution: Here $x_0=-2, x_1=1, x_2=1, x_3=2$ and $y_0=-7, y_1=2, y_2=0, y_3=11$. Now by Lagrange's formula, we have

$$x=\frac{(y-2)(y-0)(y-11)}{(-7-2)(-7-0)(-7-11)}(-2)+$$
$$+\frac{(y+7)(y-0)(y-11)}{(2+7)(2-0)(2-11)}(-1)+\frac{(y+7)(y-2)(y-11)}{(0+7)(0-2)(0-11)}(1)+$$
$$+\frac{(y+7)(y-2)(y-0)}{(11+7)(11-2)(11-0)}\cdot(2)$$

Put $y=8$, then

$$x=\frac{6(8)(-3)}{(-9)(-7)(-18)}(-2)+\frac{(15)(8)(-3)}{(9)(2)(-9)}(-1)+$$
$$+\frac{(15)(6)(-3)}{7(-2)(-11)}(1)+\frac{(15)(6)8}{(18)(9)(11)}2$$
$$x=-\frac{8}{21}-\frac{20}{9}-\frac{135}{77}+\frac{80}{99}=-3.5483.$$

EXERCISE

Lagrange's interpolation

Use Lagrange's interpolation formula to solve the following problems:

1. Given (1, 2), (3, 5), (7, 12), (13, 20) find (4, ?).

   Hint: $-\frac{27}{144}(2)+\frac{81}{80}(5)+\frac{27}{144}(12)-\frac{9}{720}(20)$.

   Ans. 6.6875

2. Fit a polynomial of third degree and find $y(0.2)$ from the following data:

x_i :	0	0.1	0.3	0.5
y_i :	-0.5	0	0.2	1

   Ans. -0.15004, $\frac{125}{3}x^3-30x^2+\frac{73}{12}x-0.5$

3. Fit a polynomial of 3rd degree

x :	0	1	3	4
y :	-12	0	6	12

   Ans. $x^3-7x^2+18x-12$

4. Determine $p(v=21)$ given the following:

v :	10	15	22.5	33.75	50.625	75.937
p :	0.3	0.675	1.519	3.417	7.689	17.3

   Ans. 1.323

5. Compute $f(9)$ from the following data:

x :	5	7	11	13	17
$f(x)$:	150	392	1452	2366	5202

   Ans. $-\frac{50}{3}+\frac{3136}{15}+\frac{3872}{3}-\frac{2366}{3}+\frac{578}{5}=810$

6. Determine $f(323.5)$ from the data given below:

x :	321.0	322.8	324.2	325.0
$f(x)$:	2.50651	2.50893	2.51081	2.51188

   Ans. 2.50987

7. Given (300, 2.4771), (304, 2.4829), (305, 2.4843), (307, 2.4871), find (301, ?).

   Ans. 2.4786

8. Compute $f(27)$ from the data below:

x :	14	17	31	35
$f(x)$:	68.7	64.0	44.0	39.1

   Ans. 49.3.

Inverse interpolation

9. Determine $x(7)$ from the tabulated data:

x :	1	3	4
y :	4	12	19

   Ans. ?

10. Find $x(12)$ using Lagrange's technique

x :	1.2	2.1	2.8	4.1	4.9	6.2
y :	4.2	6.8	9.8	13.4	15.5	19.6

Ans. 3.55

11. Let $y(x) = \frac{2}{\sqrt{11}} \int_0^x e^{-t^2} dt$ be the probability integral. Given:

x :	0.46	0.47	0.48	0.49
$y(x)$:	0.4846555	0.4937452	0.5027498	0.5116683

Determine for what value of x, $y(x) = \frac{1}{2}$.

Ans. 0.476937

12. Find x corresponding to $y = 10$

x :	10	15	17	20
y :	3	7	11	17

Ans. $x = 16.641$

13. Compute the value of x when $y = 15$

x :	5	6	9	11
y :	12	13	14	16

Ans. $\frac{5}{4} - 6 + \frac{27}{2} + \frac{11}{4} = \frac{46}{4} = 11.5$

14. Determine $x(0)$ by inverse interpolation

x :	1	2	2.5	3
y :	−6	−1	5.625	16

Ans. 2.122.

32.9 DIVIDED DIFFERENCES

In constructing the finite (forward and backward) differences tables, it is assumed that the independent variable x is equally spaced, i.e., $x_{i+1} - x_i = h =$ constant for $i = 0, 1, 2, \ldots$. When x is unequally spaced, with variable interval, the concept of finite differences is generalized to "divided differences", which takes into consideration (account) of the changes in the values of the argument x.

Let the tabulated function $y = f(x)$ consists of $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n) \ldots$ with unequal intervals, $x_{i+1} - x_i \neq 0$ for $i = 0, 1, 2, \ldots$. Then the first order divided difference, denoted by $[x_i, x_{i+1}]$, is the ratio given by

$$[x_i, x_{i+1}] = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, \quad i = 0, 1, 2, \ldots$$

For example, $[x_0, x_1] = \dfrac{y_1 - y_0}{x_1 - x_0}$, $[x_1, x_2] = \dfrac{y_2 - y_1}{x_2 - x_1}$ etc.

Similarly, the second order divided differences

$$[x_i, x_{i+1}, x_{i+2}] = \frac{[x_{i+1}, x_{i+2}] - [x_i, x_{i+1}]}{x_{i+2} - x_i}$$

for $i = 0, 1, 2, \ldots$

For example, $$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}.$$

In general, the nth order divided differences:

$$[x_i, x_{i+1}, \ldots, x_{i+n}] = \frac{[x_{i+1}, \ldots, x_{i+n}] - [x_i, \ldots, x_{i+n-1}]}{x_{i+n} - x_i}$$

for $n = 1, 2, 3, \ldots,$; $\quad i = 0, 1, 2, \ldots$.

Special case

If $x_{i+1} - x_i = h =$ constant i.e., when x is equally spaced, then divided differences reduce to forward (or backward) differences. Consider

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{1}{h}(y_1 - y_0) = \frac{1}{h}\Delta y_0,$$

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = \frac{1}{2h}\left[\frac{\Delta y_1}{h} - \frac{\Delta y_0}{h}\right]$$

$$= \frac{1}{h^2 \cdot 2!}\Delta^2 y_0.$$

In general.

$$[x_0, x_1, x_2, \ldots, x_n] = \frac{1}{h^n n!}\Delta^n y_0 \quad \text{or}$$

$$[x_k, x_{k+i}, \ldots, x_{k+n}] = \frac{1}{h^n \cdot n!} \cdot \Delta^n y_k.$$

Properties

1. The divided differences operator denoted by $\Delta$is linear because $\Delta cf(x) = c\Delta f(x)$ and $\Delta(f(x) \pm g(x)) = \Delta f(x) \pm \Delta g(x)$, c being a constant.
2. The divided differences are symmetrical i.e.,

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{y_0}{x_0 - x_1} + \frac{y_1}{x_1 - x_0}$$

$$= \frac{y_0 - y_1}{x_0 - x_1} = [x_1, x_0]$$

Similarly,

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}$$

$$= \frac{1}{x_2 - x_0}\left\{\left[\frac{y_1}{x_1 - x_2} + \frac{y_2}{x_2 - x_1}\right] - \left[\frac{y_0}{x_0 - x_1} + \frac{y_1}{x_1 - x_0}\right]\right\}$$

$$= \frac{y_0}{(x_0 - x_1)(x_0 - x_2)} + \frac{y_1}{(x_1 - x_0)(x_1 - x_2)} + \frac{y_2}{(x_2 - x_0)(x_2 - x_1)}$$

$$= [x_1, x_2, x_0] = [x_2, x_0, x_1].$$

3. **Lemma:** If $y = P(x)$ is an nth degree polynomial, then its divided difference of nth order is constant. Consequently, the $(n + 1)$th order divided differences is zero i.e.,

$$[x, x_0, \ldots, x_{n-1}] = c = \text{constant and}$$

$$[x, x_0, \ldots, x_{n-1}, x_n] = \frac{c - c}{x - x_n} = 0$$

which follows from the fact that

$$[x, x_0] = \frac{P(x) - P(x_0)}{x - x_0}$$

is a polynomial of degree $(n - 1)$ in x, and

$$[x, x_0, x_1] = \frac{p(x, x_0) - p(x_0, x_1)}{(x - x_1)}$$

is a polynomial of degree $(n - 2)$ in x, and so on. Thus $[x, x_0, x_1, \ldots, x_n, x_{n-1}]$ is a polynomial of degree $(n - n)$ i.e., zero degree in x.

Note: Order of any divided difference is one less than the number of values of the argument in it i.e., 3rd order differences contain four values of x.

32.10 NEWTON'S DIVIDED DIFFERENCES FORMULA

When another interpolation point is added to the tabulated data, the Lagrangian coefficients are to be recalculated, resulting a different Lagrange's polynomial of higher degree. This difficulty is removed in the Newton's divided differences formula (or *Newton's general interpolation formula or Newton's interpolation formula for unequally spaced values of the argument*) in which polynomial of higher degree is obtained simply by addition of new terms (to the already existing formula).

Consider

$$[x, x_0] = \frac{y - y_0}{x - x_0} \quad \text{or}$$

$$y = y_0 + (x - x_0)[x, x_0] \tag{1}$$

Now $[x, x_0, x_1] = \dfrac{[x, x_0] - [x_0, x_1]}{x - x_1}$ or

$$[x, x_0] = [x_0, x_1] + (x - x_1)[x, x_0, x_1] \tag{2}$$

Substituting (2) in (1), we have

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1] \tag{3}$$

Again

$$[x, x_0, x_1, x_2] = \frac{[x, x_0, x_1] - [x_0, x_1, x_2]}{x - x_2} \quad \text{or}$$

$$[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2] \tag{4}$$

Substituting (4) in (3), we get

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2)[x, x_0, x_1, x_2].$$

Continuing this way, we arrive at the Newton's divided differences formula as

$$y = f(x) = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2)[x, x_0, x_1, x_2] + \cdots + (x - x_0)(x - x_1)\cdots(x - x_{n-1})[x_0, x_1, \ldots, x_n] \tag{5}$$

Note that the approximate relation (5) is obtained by suppressing the error term $E(x)$ where

$$E(x) = (x - x_0)\cdots(x - x_n)[x, x_0, x_1, \ldots, x_n].$$

Note: Various interpolation formulae obtained so far are 'different' forms of the same (unique) polynomial.

WORKED OUT EXAMPLES

Difference formula

Example 1: Find Newton's divided differences polynomial for the data in the table below. Also find $f(2.5)$

x :	-3	-1	0	3	5
$f(x)$:	-30	-22	-12	330	3458

Solution: The divided differences table is:

Table 32.11

x	$y = f(x)$	Divided differences of order			
		1	2	3	4
-3	-30	$\frac{-22-(-30)}{-1-(-3)} = \frac{8}{2} = 4$			
-1	-22	$\frac{-12-(-22)}{0-(-1)} = \frac{10}{1} = 10$	$\frac{10-4}{0-(-3)} = \frac{6}{3} = 2$		
0	-12	$\frac{330-(-12)}{3-0} = 114$	$\frac{114-10}{3-(-1)} = 26$	$\frac{26-2}{3-(-3)} = 4$	
3	330	$\frac{3458-330}{5-3} = 1564$	$\frac{1564-114}{5-0} = 290$	$\frac{290-26}{3-(-1)} = 44$	$\frac{44-4}{5-(-3)} = 5$
5	3458				

Here $x_0 = -3$, $x_1 = -1$, $x_2 = 0$, $x_3 = 3$, $x_4 = 5$, and $y_0 = -30$, $y_1 = -22$, $y_2 = -12$, $y_3 = 330$, $y_4 = 3458$. The divided differences are

$$[x_0, x_1] = 4, [x_1, x_2] = 10,$$

$$[x_2, x_3] = 114, [x_3, x_4] = 1564,$$

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = 2, [x_1, x_2, x_3] = 26,$$

$$[x_2, x_3, x_4] = 290. \quad \text{Now}$$

$$[x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0} = 4$$

$$[x_1, x_2, x_3, x_4] = \frac{[x_2, x_3, x_4] - [x_1, x_2, x_3]}{x_4 - x_1} = 44$$

and finally

$$[x_0, x_1, x_2, x_3, x_4] = \frac{[x_1, x_2, x_3, x_4] - [x_0, x_1, x_2, x_3]}{x_4 - x_0} = 5$$

The Newton's divided difference polynomial is

$$y = f(x) = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] +$$
$$+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] +$$
$$+ (x - x_0)(x - x_1)(x - x_2)(x - x_3)[x_0, x_1, x_2, x_3, x_4]$$

Substituting the above data

$$y = f(x) = -30 + (x + 3)(4) + (x + 3)(x + 1)(2) +$$
$$+ (x+3)(x+1)(x-0)(4) + (x+3)(x+1)x(x-3)(5)$$
$$y = f(x) = 5x^4 + 9x^3 - 27x^2 - 21x - 12$$

is the required 4th degree polynomial. Now when $x = 2.5$,

$$y = f(2.5) = 5(2.5)^4 + 9(2.5)^3 - 27(2.5)^2 - 21(2.5) - 12$$
$$= 102.6785.$$

EXERCISE

Newton's divided differences formula

Using Newton's divided differences formula (Newton's general interpolation formula) solve the following problems:

1. Find $f(8)$.

x:	4	5	7	10	11	13
$f(x)$:	48	100	294	900	1210	2028

Hint: Divided differences are 52, 15, 1, 0.

Ans. 448

2. Fit a polynomial and find $f(1)$ and $f(8)$.

x:	-1	0	3	6	7
$f(x)$:	3	-6	39	822	1611

Hint: Divided differences are -9, 6, 5, 1.

Ans. $x^4-3x^3+5x^2-6$, $f(1)=-3$, $f(8)=2874$.

3. Find value of y for $x = 5.6075$.

x:	5.6	5.602	.5605	.5607	.5608
y:	.77556588	.77682686	.77871250	.779965	.78059114

Hint: Divided differences are 0.6305, -0.38668, -0.9485714, -20.595237.

Ans. 0.77729893

4. Find $\log 323.5$.

x:	321.0	322.8	324.2	325.0
$\log x$:	2.50651	2.50893	2.51081	2.51188

Hint: Divided differences are 0.00134444, -0.00000158, -0.00000022.

Ans. 2.50987

5. Fit a cubic polynomial.

x:	0	1	2	5
$f(x)$:	2	3	12	147

Hint: Divided differences are 1, 4, 1.

Ans. $x^3 + x^2 - x + 2$

6. Obtain the Newton's divided difference interpolating polynomial and hence find $f(3)$.

x:	0	1	2	4	5	6
$f(x)$:	1	14	15	5	6	19

Hint: Divided differences are 13, -6, 1, 0, 0.

Ans. $x^3 - 9x^2 + 21x + 1$, $f(3) = 10$

7. Fit a cubic polynomial and find $f(6)$.

x:	3	7	9	10
$f(x)$:	168	120	72	63

Hint: Divided differences are -12, -2, 1.

Ans. $x^3 - 21x^2 + 119x - 27$, $f(6) = 147$

8. Form the table of divided differences:

x:	4	5	7	10	11	13
$f(x)$:	48	100	294	900	1210	2028

Extend the table to include values $x = 2$ and $x = 15$.

Hint: Divided differences: 52, 15, 1, 0, 0.

Ans. $x^3 - x^2$, $y(2) = 4$, $y(15) = 3150$.

9. Fit a polynomial to the data $(-4, 1245)$, $(-1, 33)$, $(0, 5)$, $(2, 9)$, $(5, 1335)$. Hence find $f(1)$ and $f(7)$.

Ans. $3x^4 - 5x^3 + 6x^2 - 14x + 5$, $f(1) = -5$, $f(7) = 5689$

10. Calculate $f(1.5)$.

x:	1	2	3	4	5
$f(x)$:	0	7	26	63	124

Hint: Divided differences are 7, 12, 6, 0.

Ans. $f(1.5) = 2.25$

11. Calculate $f(9)$.

x:	5	7	11	13	17
$f(x)$:	150	392	1452	2366	5202

Hint: Divided differences are 121, 24, 1.

Ans. 810

12. Determine the Newton's general interpolation polynomial and hence find $f(-3)$ and $f(9)$.

x:	-4	-1	0	2	5
$f(x)$:	1245	33	5	9	1335

Hint: Divided differences are -404, 94, -14, 3.

Ans. $3x^4 - 5x^3 + 6x^2 - 14x + 5$, $f(-3) = 479$, $f(9) = 16403$.

32.11 ERRORS IN POLYNOMIAL INTERPOLATION

Computing of values for a tabulated function at points *not* in the table is known as interpolation. Thus, interpolation means to estimate a missing function value by taking a weighted average of known (given) function values at neighbouring points.

Suppose the function $y = f(x)$ is known (given) at $(n + 1)$ points (x_0, y_0), $(x_1, y_1) \ldots (x_n, y_n)$. A poly-

nomial of degree n, $P_n(x)$ is known as the interpolating polynomial when $P_n(x)$ passes through (satisfies) these $(n+1)$ points. $P_n(x)$ can be constructed using only the numerical values x_k and y_k and does not need any order derivatives of $f(x)$. The approximation $P_n(x)$ is known as interpolated value when $x_0 < x < x_n$ and extrapolated value when $x < x_0$ or $x_n < x$.

The error is zero at tabulated points $x_0, x_1, \ldots, x_n$; while it may be nonzero at points *not* in the table. Error may increase when number of interpolation is increased (classical example: $f(x) = (1+25x^2)^{-1}$ in $[-1, 1]$).

The error term associated with the nth degree polynomial $P_n(x)$ is obtained by using the generalized Rolle's theorem and is given by

$$E(x) = f(x) - P_n(x)$$

$$= (x-x_0)(x-x_1)\ldots(x-x_n)\cdot\frac{f^{(n-1)}(\xi)}{(n+1)!} \quad (1)$$

Here ξ is any value in the smallest interval that contains $\{x, x_0, x_1, \ldots, x_n\}$. The disadvantage with this error expression is that the derivative of f is not known (since the function $f(x)$ itself is not known except at the tabulated points). Note that the error term (1) is same for Lagrange polynomial and interpolating polynomial obtained from divided-difference table.

Error estimation when $f(x)$ is unknown: When $f(x)$ is unknown, $\frac{f^{(n)}(x)}{n!}$ can be approximated by the nth order divided difference

i.e., $\frac{f^{(n)}(x)}{n!} = f[x_0, x_1, \ldots, x_n] = f_0^{[n]} = n$th order divided difference.

Thus the error of the interpolation is approximately given by the next term that would be added.

The *next-term* rule is a most valuable rule for estimating the error of interpolation as follows:

$E_n(x) \approx$ value of the next term
that would be added to $P_n(x)$.
$= f_0^{[n]}(x-x_0)(x-x_1)(x-x_2)\ldots(x-x_{n-1})$.

The interpolating polynomial that fits a divided difference table at $x = x_0, x_1, x_2, \ldots, x_n$ is given by

$$P_n(x) = f_0^{[0]} + (x-x_0)f_0^{[1]} + (x-x_0)(x-x_1)f_0^{[2]}$$
$$+(x-x_0)(x-x_1)(x-x_3)f_0^{[3]} + \ldots$$
$$+(x-x_0)(x-x_1)\ldots(x-x_{n-1})f_0^{[n]}$$

Thus for $P_1(x)$ the next term to be added is $f_0^{[2]}\cdot(x-x_0)(x-x_1)$.

WORKED OUT EXAMPLES

Example 1: Find the error of the interpolates for $f(1.25)$ using polynomial of degree 1, 2, 3, 4, (b). Estimate the error using the next-term rule. Here $f(x) = e^{x^2-1}$ and $x_0 = 1$, $x_1 = 1.1$, $x_2 = 1.2$, $x_3 = 1.3$, $x_4 = 1.4$

Solution: The divided difference table is

x_i	$f(x_i)$	$f_i^{[1]}$	$f_i^{[2]}$	$f_i^{[3]}$	$f_i^{[4]}$
1	1	2.3368	4.2675	6.105	7.65
1.1	1.23368	3.1903	6.099	9.165	
1.2	1.55271	4.4101	8.8485		
1.3	1.99372	6.1798			
1.4	2.61170				

Here

$$f_0^{[1]} = f[x_0, x_1] = \frac{f_1 - f_0}{x_1 - x_0},$$

$$f_0^{[2]} = f[x_0, x_1, x_2] = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}\text{etc.}$$

The interpolating polynomials of degree 1, 2, 3 are

$$P_1(x) = f_0 + f_0^{[1]}(x-x_0) = 1 + 2.3368(x-1)$$

$$P_2(x) = f_0 + f_0^{[1]}(x-x_0) + f_0^{[2]}(x-x_0)(x-x_1)$$

$$P_2(x) = 1 + 2.3368(x-1) + 4.2675(x-1)(x-1.1)$$

$$P_3(x) = f_0 + f_0^{[1]}(x-x_0) + f_0^{[2]}(x-x_0)(x-x_1)$$
$$+f_0^{[3]}(x-x_0)(x-x_1)(x-x_2)$$
$$= 1 + 2.3368(x-1) + 4.2675(x-1)(x-1.1)$$
$$+ 6.105(x-1)(x-1.1)(x-1.2)$$

$$P_4(x) = 1 + 2.3368(x-1) + 4.2675(x-1)(x-1.1)$$
$$+6.105(x-1)(x-1.1)(x-1.2)$$
$$+7.65(x-1)(x-1.1)(x-1.2)(x-1.3)$$

Interpolated values at $x = 1.25$ are

$P_1(1.25) = 1.5842$, $P_2(1.25) = 1.7442$,

$P_2(1.25) = 1.7556468$, $P_4(1.25) = 1.75496$;

The exact value $f(1.25) = e^{(1.25)^2-1}$
$= 1.7550546571$

Here Ist, 2nd, 3rd, 4th, 5th derivatives are

$f'(x) = 2xe^{x^2-1}$, $f'' = 2(1+2x)e^{x^2-1}$,

$f''' = 4(x+2x^2+1)e^{x^2-1}$,

$f^{\mathrm{IV}} = 4(4x^3+2x^2+6x+1)e^{x^2-1}$;

$f^{\mathrm{V}} = 8e^{x^2-1}(4x^4+2x^3+12x^2+3x+3)$.

Error using

$$E_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)$$

with $\xi = 1.3$.

$E_1(x) = 0.89717$

$E_2(x) = 0.0035388$, $E_3(x) = -0.000130682$

$E_4(x) = -0.0000133947$

Errors of interpolation for $f(1.25)$

Degree	Interpolated value	Actual error	Error	
			using $E_n(x)$	next-term rule
1	1.5842	0.17085467	0.89717	$(4.2675)(1.25-1)$ $\times (1.25-1.1) = 0.1600$
2	1.7442	0.01085467	− 0.0035388	0.0114
3	1.7556468	− 0.00059213	− 0.00013068	− 0.000717
4	1.75496	0.00009467	− 0.000013394	

Exact value: $f(1.25) = 1.755054657$

EXERCISE

1. Find the error of the interpolates for $f(1.75)$ using polynomials of degree 1, 2, 3. (b) Estimate the error using the next-term rule. Here $f(x) = x^2e^{-x/2}$.

Ans. Divided difference table

x_i	$f(x_i)$	$f_i^{[1]}$	$f_i^{[2]}$	$f_i^{[3]}$	$f_i^{[4]}$
1.10	0.6981	0.8593	−0.1755	0.0032	0.0027
2.00	1.4715	0.4381	−0.1631	0.0191	
3.50	2.1287	−0.0511	−0.0657		
5.00	2.0521	−0.2877			
7.10	1.4480				

Errors of interpolation for $f(1.75)$

Degree	Interpolated value	Actual error	Estimate from next-term rule	Max. f^{i+1}	Min. f^{i+1}	Upper bound	Lower bound
1.	1.25668	0.01996	0.02852	−.3679	.0594	.0299	−.00483
2.	1.28520	−0.00856	0.00091	−0.8661	0.1249	.0059	−0.0408
3.	1.28611	−0.00947	−0.00249	1.1398	−0.0359	.0014	−0.0439

2. Find an approximation to $\sin 0.34$ and find a bound for the error using a polynomial or degree 3 and given that $\sin 0.30 = 0.29552$, $\sin 0.32 = 0.31457$, $\sin 0.33 = 0.32404$, $\sin 0.35 = 0.34290$.

Ans. $\sin 0.34 = 0.33348$, Error bound: 1.2×10^{-9}

3. Find a bound for the error in approximation $f(1.09)$ using 3rd degree polynomial and the data $f(1.00) = 0.1924$, $f(1.05) = 0.2414$ $f(1.10) = 0.2933$, $f(1.15) = 0.3492$ for the function $f(x) = \log_{10} \tan x$.

Ans. $f(1.09) = 0.2826$, Error bound: 7.4×10^{-6}

4. Approximate $f(1.03)$ using interpolating polynomial of degree $1, 2, 3$ using $x_0 = 1$, $x_1 = 1.05$, $x_2 = 1.07$, $x_3 = 1.1$, $x_4 = 1.15$ compare the actual error to the error bound and to error obtained by next-term rule of $f(x) = 3xe^x - e^{2x}$.

5. Write the error term $E_3(x)$ for cubic Lagrange interpolation to $f(x) = x^5 - 5x^4$ where interpolation is to be exact at the four nodes $x_0 = -1$, $x_1 = 0$ $x_2 = 3$, $x_3 = 4$.

Ans. $f^{(4)}(c) = 120(c - 1)$ for all c, so that $E_3(x) = 5(x + 1)(x - 3)(x - 4)(c - 1)$

6. For $f(x) = \sin x$ on $[0, 1]$, determine the step size h so that linear Lagrange interpolation has an accuracy of 10^{-6}.

Ans. $|E_1(x)| = 5 \times 10^{-7}$, $|f^{(2)}(c)| \leq |-\sin(1)| = 0.84147098 = M_2$, $h^2 \frac{M_2}{8} = \frac{h^2}{8}(0.84147098) < 5 \times 10^{-7}$ or $h < 0.00218027$

7. Let $f(x) = \log(1 + x^2)$. Consider the following divided difference table. Estimate the error for interepolate $f(2)$ using a quadric interpolate with the points 1.91, 1.97, 2.02.

x_i	$f(x_i)$	$f_i^{[1]}$	$f_i^{[2]}$	$f_i^{[3]}$
1.88	1.511693			
1.91	1.536459	0.825333		
1.97	1.585330	0.8145167	−0.1224067	
2.02	1.625390	0.8012000	−0.1210609	0.96128×10^{-2}
2.11	1.696001	0.7845555	−0.1188893	1.0858×10^{-2}
2.18	1.749617	0.7659571	−0.1162400	1.26157×10^{-2}

Ans. $P_2(x) = 1.536459 + 0.8145167(x - 1.91)$
$-0.1210609(x - 1.91)(x - 1.97)$
Error $\simeq 10^{-2}(0.09)(0.03)(0.02) = 0.54 \times 10^{-6}$
Interpolated value: $P^2(2) = 1.609439$
Exact value: $f(2) = \log 5 = 1.6094379$

Note: Third order divided differences are all approximately equal to 10^{-2}.

8. Develop a divided differences table for the following data.

x:	0.3	0.42	0.5	0.58	0.66	0.72
y:	1.04403	1.08462	1.11803	1.15603	1.19817	1.23223

Estimate error at $x = 0.55$ using linear interpolation, at $x = 0.60$ using quadratic interpolation, at $x = 0.46$ using cubic interpolation. Exact values are $f(0.46) = 1.10073$, $f(0.55) = 1.14127$, $f(10.6) = 1.16619$.

Ans. $P_1(0.55) = 1.14178$ based on 0.5, 0.58
$P_2(0.6) = 1.16618$ based on 0.5, 0.58, 0.66
$P_3(0.46) = 1.04403$ based on 0.3, 0.42, 0.5, 0.58

32.12 SYMBOLIC RELATIONS AND SEPARATION OF SYMBOLS

Let $x_0, x_1, \ldots, x_n$ be a set of tabular points which are equally spaced i.e.,

$$x_i = x_0 + ih, i = 0, 1, 2, \ldots n.$$

Then

$$\nabla f(x_i) = f(x_i) - f(x_i - h)$$

is the backward-difference operator,

$$\Delta f(x_i) = f(x_i + h) - f(x_i)$$

is the forward-difference operator

$$\delta f(x_i) = f\left(x_i + \frac{h}{2}\right) - f\left(x_i - \frac{h}{2}\right)$$

is the central-difference operator. The shift operators are defined as

$$E(f(x_i)) = f(x_i + h)$$
$$E^{-1}(f(x_i)) = f(x_i - h)$$

entries in the difference table can be interpreted as either forward or backward differences.

Symbolic relations are established using these symbolic operators. Standing alone, all these operators are without any significance and are meaningless unless operated on a function, like $\int$ or $\frac{d}{dx}$ or $\sqrt{}$. These operators are linear operators i.e., say

$$\Delta(c_1 f_1 + c_2 f_2) = c_1 \Delta f_1 + c_2 \Delta f_2$$

and obey laws of algebra.

Relationship between the Operators

	Δ	∇	E	δ
Δ	Δ	$(1-\nabla)^{-1} - 1$	E^{-1}	$\frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$
∇	$1 - (1+\Delta)^{-1}$	∇	$1 - E^{-1}$	$-\frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$
E	$\Delta + 1$	$(1-\nabla)^{-1}$	E	$\frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2} + 1$
δ	$\Delta(1+\Delta)^{-\frac{1}{2}}$	$\nabla(1-\nabla)^{-\frac{1}{2}}$	$E^{\frac{1}{2}} - E^{-\frac{1}{2}}$	δ
μ	$(1+\frac{\Delta}{2})(1+\Delta)^{\frac{1}{2}}$	$(1-\frac{\nabla}{2})(1-\nabla)^{-\frac{1}{2}}$	$\frac{E^{\frac{1}{2}} + E^{-\frac{1}{2}}}{2}$	$\sqrt{1 + \frac{1}{4}\delta^2}$

The averaging operator is defined as

$$\mu(f(x_i)) = \frac{1}{2}\left[f\left(x_i + \frac{h}{2}\right) + f\left(x_i - \frac{h}{2}\right)\right]$$

Higher order differences are obtained by repeated applications of the difference operators. Let $f_i = f(x_i)$. Then

$$\nabla^n f(x_i) = \nabla^{n-1} f_i - \nabla^{n-1} f_{i-1}$$
$$\Delta^n f(x_i) = \Delta^{n-1} f_{i+1} - \Delta^{n-1} f_i$$
$$E^n f(x_i) = f(x_i + nh)$$
$$E^{-n} f(x_i) = f(x_i - nh)$$
$$\delta^n f(x_i) = \delta^{n-1} f_{i+\frac{1}{2}} - \delta^{n-1} f_{i-\frac{1}{2}}$$
$$= \sum_{j=0}^{n} (-1)^j \frac{n!}{j!(n-j)!} f_{i+\frac{n}{2}-j}$$

Note that the differences $\Delta^k f_0$ lie on a straight line sloping downward to the right. The differences $\nabla^k f_3$ lie on a straight line sloping upward to the right. The differences $\delta^{2k} f_2$ lie on a horizontal line. Thus the

Some of the above relations can be established as follows:

(a) $\Delta(f(x_0)) = f(x_0 + h) - f(x_0) \quad = E(f(x_0) - f(x_0)) = (E-1)(f(x_0))$
Thus $\Delta = (E-1)$

(b) $\nabla(f(x)) = f(x) - f(x+h) \; = f(x) \; - E^{-1}(f(x)) = (1 - E^{-1})(f(x))$
Thus $\nabla = (1 - E^{-1})$

(c) $\delta(f(x)) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right) =$
$= E^{\frac{1}{2}} f(x) - E^{-\frac{1}{2}} f(x) = \left(E^{\frac{1}{2}} - E^{-\frac{1}{2}}\right)(f(x))$
Thus $\delta = \left(E^{\frac{1}{2}} - E^{-\frac{1}{2}}\right)$

(d) $\mu(f(x)) = \frac{1}{2}\left[f\left(x + \frac{h}{2}\right) + f\left(x - \frac{h}{2}\right)\right]$

$$= \frac{1}{2}\left[E^{\frac{1}{2}} f(x) + E^{-\frac{1}{2}} f(x)\right]$$

$$= \frac{1}{2}\left[E^{\frac{1}{2}} + E^{-\frac{1}{2}}\right](f(x))$$

Thus $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

Similarly other results in the above table can be proved.

Now using laws of algebra, we can prove the following.

(i) $E\nabla = E(1 - E^{-1}) = E - 1 = \Delta$
$\nabla E = (1 - E^{-1})E = E - 1 = \Delta$
Thus $E^n \nabla^n = \nabla^n E^n = \Delta^n$

(ii) Since $\Delta = E - 1$

$$\Delta^n = (E-1)^n = E^n - nE^{n-1} + \binom{n}{2} E^{n-2} - \binom{n}{3} E^{n-3} + \dots$$

$$\Delta^n f(x) = E^n f - nE^{n-1} f + \binom{n}{2} E^{n-2} f - \binom{n}{3} E^{n-3} f + \dots$$

$$= f(x+nh) - nf[x+(n-1)h] + \binom{n}{2} f[x+(n-2)h] + \dots$$

Thus

$$\Delta^n f = f_n - nf_{n-1} + \binom{n}{2} f_{n-2} - \binom{n}{3} f_{n-3} + \dots$$

(iii) *Newton-Gregory forward formula*: We know that $E = 1 + \Delta$, so $E^n = (1 + \Delta)^n$
Then

$$f_n = E^n f_0 = (1+\Delta)^n f_0$$

$$= [1 + n\Delta + \binom{n}{2}\Delta^2 + \binom{n}{3}\Delta^3 + \dots] f_0$$

$$f_n = f_0 + n\Delta f_0 + \binom{n}{2}\Delta^2 f_0 + \binom{n}{3}\Delta^3 f_0 + \dots$$

(iv) *Newton-Gregory backward formula*
Since $E = (1 - \nabla)^{-1}$, $E^n = (1 - \nabla)^{-n}$. Then

$$f_n = E^n f_0 = (1-\nabla)^{-n} f_0$$

$$= \left[1 + n\nabla + \binom{n+1}{2}\nabla^2 + \binom{n+2}{3}\nabla^3 + \dots\right] f_0$$

$$f_n = f_0 + n\nabla f_0 + \binom{n+1}{2}\nabla^2 f_0 + \binom{n+2}{3}\nabla^3 f_0 + \dots$$

$$= f_0 + n\nabla f_{-1} + \binom{n+1}{2}\Delta^2 f_{-2} + \binom{n+2}{3}\Delta^3 f_{-3} + \dots$$

WORKED OUT EXAMPLES

Example 1: Prove that $\nabla^6 y_8 = \Delta^6 y_2$.

Solution: $\nabla^6 = (1 - E^{-1})^6$
$= (1 - 6E^{-1} + 15E^{-2} - 20E^{-3} + 15E^{-4} - 6E^{-5} + E^{-6})$

So

$$\nabla^6 y_8 = (1 - 6E^{-1} + 15E^{-2} - 20E^{-3} + 15E^{-4} - 6E^{-5} + E^{-6}) y_8$$

$$\nabla^2 y_8 = y_8 - 6y_7 + 15y_6 - 20y_5 + 15y_4 - 6y_3 + y_2$$

Also

$$\nabla^6 y_2 = (E-1)^6 y_2 = (E^6 - 6E^5 + 15E^4 - 20E^3 + 15E^2 - 6E + 1) y_2$$

$$= y_8 - 6y_7 + 15y_6 - 20y_5 + 15y_4 - 6y_3 + y_2$$

Thus $\nabla^6 y_8 = \Delta^6 y_2$

Example 2: Show that $\nabla^r f_k = \Delta^r f_{k-r}$

Solution:
RHS $= \Delta^r f_{k-r} = (E-1)^r f_{k-r}$ $(\because \Delta = E - 1)$
$= E^r (1 - E^{-1})^r f_{k-r}$ $(\because \nabla = 1 - E^{-1})$
$= E^r \nabla^r f_{k-r} = \nabla^r E^r f_{k-r} = \nabla^2 f_k$
$\because$ $E^r f_{k-r} = f_{k-r+r}$

Example 3: Show that $\delta = 2 \sin h \left(\frac{hD}{2}\right)$.

Solution: $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}} = e^{\frac{hD}{2}} - e^{-\frac{hD}{2}}$
$(\because E = e^{hD})$ $= 2 \sin h \left(\frac{hD}{2}\right)$.

Example 4: Prove that $(\Delta + 1)(1 - \nabla) = 1$

Solution: Since $\Delta = E - 1$, $\Delta + 1 = E$ and $\nabla = 1 - E^{-1}$ so $1 - \nabla = E^{-1}$
Then $(\Delta + 1)(1 - \nabla) = E \cdot E^{-1} = 1$

Example 5: Show that $\Delta\nabla = \delta^2$

Solution: $\Delta\nabla = (E-1)(1-E^{-1})$
$= E - 1 - 1 + E^{-1}$
$= (E - E^{-1})^2$
Now $\delta = (E^{\frac{1}{2}} - E^{-\frac{1}{2}})$

$$\delta^2 = (E^{\frac{1}{2}} - E^{-\frac{1}{2}})^2 = (E - 2 + E^{-1})$$
$$= (E - E^{-1})^2$$

Hence $\Delta\nabla = \delta^2$

Example 6: Show that $\frac{\Delta^2}{E}x^3 = 6xh^2$

Solution:
LHS $= \frac{\Delta^2}{E}x^3 = \frac{(E-1)^2}{E}x^3$

$$= \left(\frac{E^2 + 1 - 2E}{E}\right)x^3 = (E - 2 + E^{-1})x^3$$
$$= (x+h)^3 - 2x^3 + (x-h)^3$$
$$= (x^3 + 3x^2h + 3xh^2 + h^3) - 2x^3$$
$$+(x^3 - 3x^2h + 3xh^2 - h^3)$$
$$= 6xh^2$$

Example 7: Show that $(\Delta + \nabla)^2(x^2 + x) = 8$ with $h = 1$

Solution: Since $\Delta = E - 1$ and $\nabla = 1 - E^{-1}$. We have

$$(\Delta + \nabla)^2(x^2 + x) = (E - 1 + 1 - E^{-1})^2 = (E - E^{-1})^2$$
$$= E^2 + E^{-2} - 2$$

So

$$(\Delta + \nabla)^2 = (E^2 + E^{-2} - 2)(x^2 + x)$$
$$= E^2(x^2 + x) + E^{-2}(x^2 + x) - 2(x^2 + x)$$
$$[(x+2)^2 + (x+2)] + [(x-2)^2 + (x-2)]$$
$$-2(x^2 + x) = 8$$

Example 8: Show that $u_0 + Eu_0 \cdot x + E^2u_0 \cdot x^2 +$
$+ E^3u_0 \cdot x^3 + \ldots$
$= \left[\frac{1}{1-x} + \frac{x}{(1-x)^2}\Delta + \frac{x^2}{(1-x)^3}\Delta^2 + \ldots\right]u_0$

Solution: Rewriting LHS as

$$\text{LHS} = (1 + xE + x^2E^2 + x^3E^3 + \ldots)u_0$$
$$= \left(\frac{1}{1-xE}\right)u_0 = \left[\frac{1}{1-x(1+\Delta)}\right]u_0$$

since $E = 1 + \Delta$

$$= \left[\frac{1}{1-x-x\Delta}\right]u_0 = \frac{1}{1-x}\left[\frac{1}{1-\frac{x\Delta}{1-x}}\right]u_0$$
$$= \frac{1}{1-x}[1 + \frac{x\Delta}{1-x} + \frac{x^2\Delta^2}{(1-x)^2} + \ldots]u_0$$
$$= \left[\frac{1}{1-x} + \frac{x}{(1-x)^2}\Delta + \frac{x^2}{(1-x)^3}\Delta^2 + \ldots\right]u_0$$

Example 9: Show that $\delta E^{\frac{1}{2}} = \Delta$

Solution: $E^{\frac{1}{2}}f(x) = f(x + \frac{h}{2})$
$\delta E^{\frac{1}{2}}(f(x)) = \delta[f(x + \frac{h}{2})]$
$= f(x + \frac{h}{2} + \frac{h}{2}) - f(x + \frac{h}{2} - \frac{h}{2})$
$= f(x+h) - f(x)$
$= \Delta f(x)$

Example 10: Prove that $\nabla - \Delta = -\Delta\nabla$

Solution:
LHS : $\nabla - \Delta = (1 - E^{-1}) - (E - 1)$
$= 2 - E - E^{-1}$
RHS : $-\Delta\nabla = -(E-1)(1-E^{-1})$
$= -[E - 1 - 1 + E^{-1}]$
$= 2 - E - E^{-1}$

Example 11: Show that $\nabla + \Delta = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$

Solution: LHS $= \nabla + \Delta = (1 - E^{-1}) + (E - 1)$
$= E - E^{-1}$
Now

$$\frac{\Delta}{\nabla} = \frac{E-1}{1-E^{-1}} = \frac{E(E-1)}{(E-1)} = E$$
$$\frac{\nabla}{\Delta} = \frac{1-E^{-1}}{E-1} = \frac{1}{E}\left(\frac{1-E^{-1}}{1-E^{-1}}\right) = E^{-1}$$

RHS : $\frac{\Delta}{\nabla} - \frac{\nabla}{\Delta} = E - E^{-1}$

EXERCISE

1. Show that $e^x = \left(\frac{\Delta^2}{E}\right)e^x \cdot \frac{Ee^x}{\Delta^2 e^x}$ taking h as the interval of difference.

Hint: $\left(\frac{\Delta^2}{E}\right)e^x = \Delta^2 E^{-1}e^x = \Delta^2 e^{x-h} = e^{-h}\,\Delta^2 e^x$

RHS: $e^{-h}\Delta^2 e^x \cdot \frac{Ee^x}{\Delta^2 e^x} = e^{-h}e^{x+h} = e^x$

2. Prove that
   (a) $\Delta^3 y_2 = \nabla^3 y_5$
   (b) $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1+\frac{\delta^2}{4}}$
   (c) $2 + \Delta = (E^{\frac{1}{2}} + E^{-\frac{1}{2}})(1+\Delta)^{\frac{1}{2}}$
   (d) $hD = \log(1+\Delta) = -\log(1-\Delta) = \sinh^{-1}(\mu\delta)$
3. Prove that
   (a) $\delta = \Delta(1+\Delta)^{-\frac{1}{2}} = \nabla(1-\nabla)^{-\frac{1}{2}}$
   (b) $\mu^2 = 1 + \frac{\delta^2}{4}$
   (c) $\delta(E^{\frac{1}{2}} + E^{-\frac{1}{2}}) = \Delta E^{-1} + \Delta$
4. Show that $\nabla y_{n+1} = h[1 + \frac{1}{2}\nabla + \frac{5}{12}\nabla^2 + \ldots]y'_n$
5. Show that
   (a) $u_1 x + u_2 x^2 + u_3 x^3 + \ldots = \left(\frac{x}{1-x}\right)^2 \Delta u_1 + \left(\frac{x}{1-x}\right)^3 \Delta^2 u_1 + \ldots$
   (b) $u_0 + \frac{u_1 x}{1!} + \frac{u_2 x^2}{2!} + \frac{u_3 x^3}{3!} + \ldots = e^x(u_0 + x\Delta u_0 + \frac{x^2}{2!}\Delta^2 u_0 + \frac{x^3}{3!}\Delta^3 u_0 + \ldots)$
6. Prove that $E^{\frac{1}{2}} = \left(\frac{1}{2} + \frac{\delta^2}{4}\right)^{\frac{1}{2}} + \frac{\delta}{2}$
7. Show that $\nabla^2 = h^2D^2 - h^3D^3 + \frac{7}{12}h^4D^4 + \ldots$
8. Show that $E = e^{hD}$

   Hint: $E(f(x)) = f(x+h)$, expand by Taylor's series, $E(f(x)) = f(x) + hf' + \frac{h^2}{2!}f'' + \frac{h^3}{3!}f'' + \ldots$
   $= f(x) + hDf + \frac{h^2}{2!}D^2 f + \frac{h^3}{3!}D^3 f + \ldots$
   $= \left[1 + hD + \frac{h^2D^2}{2!} + \frac{h^3D^3}{3!} + \ldots\right] f$
   $= e^{hD}(f(x))$

   Prove the following.
9. $\Delta(f_i g_i) = f_i \Delta g_i + g_{i+1}\Delta f_i$
10. $\Delta f_i^2 = (f_i + f_{i+1})\Delta f_i$
11. $\Delta\left(\frac{1}{f_i}\right) = -\frac{\Delta f_i}{f_i f_{i+1}}$
12. $\Delta\left(\frac{f_i}{g_i}\right) = \frac{(g_i\Delta f_i - f_i\Delta g_i)}{g_i g_{i+1}}$

32.13 NUMERICAL DIFFERENTIATION

Numerical differentiation or approximate differentiation is used when the function $y = f(x)$ is (i) given in tabular form (ii) function is highly complex. The basic idea in numerical differentiation is to replace the given function $y = f(x)$ on the interval $[a, b]$ by an interpolating polynomial $P(x)$ and set $f'(x) = P'(x)$; $f''(x) = P''(x)$ etc. Numerical differentiation is less exact than interpolation. Although $f(x_1) = P(x_1)$, $f'(x_1)$ need not be equal to $p'(x_1)$. (see figure).

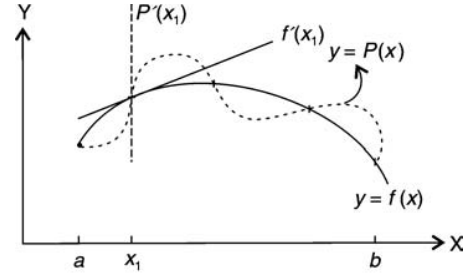


Fig. 32.13

Numerical Differentiation Using Newton's Forward Formula

Suppose $y = f(x)$ is specified in an interval $[a, b]$ at equally spaced points $x_i = x_0 + ih (i = 1, 2, \ldots n)$ by means of the values $y_i = f(x_i)$. Replace the tabulated function $y = f(x)$ by Newton's forward interpolation polynomial passing through the $(n+1)$ points $x_0, x_1, \ldots, x_n$. Thus

$$y(x) = y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_0 + \frac{q(q-1)(q-2)}{3!}\cdot\Delta^3 y_0 + \frac{q(q-1)(q-2)(q-3)}{4!}\Delta^4 y_0 + \cdots \quad (1)$$

where $q = \frac{x-x_0}{h}$ and $h = x_{i+1} - x_i$, for $i = 0, 1, \ldots$ Here q is a function of x and $\frac{dq}{dx} = \frac{1}{h}$. Rewriting (1)

$$y(x) = y_0 + q\Delta y_0 + \frac{q^2-q}{2}\Delta^2 y_0 + \frac{q^3 - 3q^2 + 2q}{6}\cdot\Delta^3 y_0 + \frac{q^4 - 6q^3 + 11q^2 - 6q}{24}\Delta^4 y_0 + \cdots (1^*)$$

Differentiating (1*) w.r.t. x, we have

$$\frac{dy}{dx} = \frac{dy}{dq}\cdot\frac{dq}{dx} = \frac{1}{h}\frac{dy}{dq} =$$
$$= \frac{1}{h}\left[\Delta y_0 + \frac{2q-1}{2}\Delta^2 y_0 + \frac{3q^2-6q+2}{6}\Delta^3 y_0 + \frac{4q^3-18q^2+22q-6}{24}\Delta^4 y_0 + \cdots\right] \quad (2)$$

Similarly, differentiating (2) once again w.r.t. x, we get

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{1}{h}\frac{d}{dq}\left(\frac{dy}{dx}\right)$$
$$= \frac{1}{h^2}\left[\Delta^2 y_0 + (q-1)\Delta^3 y_0 + \frac{6q^2-18q+11}{12}\Delta^4 y_0 + \cdots\right] \quad (3)$$

Higher order derivatives can be computed similarly.

Special case

When derivative is required at a basic tabulated point x_i, then choose $x = x_0$, so $q = 0$ (since each tabular value may be taken as the initial value x_0). Thus

$$\left.\frac{dy}{dx}\right|_{x=x_0} = y'(x_0) = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} + \frac{\Delta^5 y_0}{5} - \cdots\right] \quad (4)$$

and

$$\left.\frac{d^2y}{dx^2}\right|_{x=x_0} = y''(x_0) = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \cdots\right] \quad (5)$$

Similarly,

$$y'''(x_0) = \frac{1}{h^3}\left[\Delta^3 y_0 - \frac{3}{2}\Delta^4 y_0 + \frac{7}{4}\Delta^5 y_0 + \cdots\right] \quad (6)$$

$$y''''(x_0) = \frac{1}{h^4}\left[\Delta^4 y_0 - 2\Delta^5 y_0 + \cdots\right] \quad (7)$$

Thus formulas (2), (3) (and (4), (5) in special case) given the derivatives of $y(x)$.

Numerical Differentiation Using Newton's Backward Formula

In this case, we replace $y(x)$ by Newton backward interpolation formula:

$$y(x) = y_n + q\nabla y_n + \frac{q(q+1)}{2!}\nabla^2 y_n + \frac{q(q+1)(q+2)}{3!}\nabla^3 y_n + \cdots$$

Differentiating w.r.t. x,

$$y'(x) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}(2q+1)\nabla^2 y_n + \frac{3q^2+6q+2}{6}\nabla^3 y_n + \cdots\right] \quad (8)$$

$$y''(x) = \frac{1}{h^2}\left[\nabla^2 y_n + (q+1)\nabla^3 y_n + \frac{6q^2+18q+11}{12}\nabla^4 y_n + \cdots\right] \quad (9)$$

Special case: When $x = x_n$, then $q = 0$.

$$y'(x_n) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \cdots\right] \quad (10)$$

$$y''(x_n) = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \cdots\right] \quad (11)$$

Worked Out Examples

Example: Compute $f'(x)$ and $f''(x)$ at (a) $x = 16$; (b) $x = 15$; (c) $x = 24$; (d) $x = 25$ from the following table:

x:	15	17	19	21	23	25
$f(x) = \sqrt{x}$:	3.873	4.123	4.359	4.583	4.796	5.8

Compare with the exact values.

Solution:

a. To find f', f'' at $x = 16, 15$, use the forward differences results.

$$f'(x) = \frac{1}{h}\left[\Delta y_0 + \frac{2q-1}{2}\Delta^2 y_0 + \frac{3q^2-6q+2}{6}\cdot\Delta^3 y_0 +\right.$$

Table 32.12: The finite difference table

x	$y=\sqrt{x}$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
15	3.873					
		0.250				
17	4.123		−0.014			
		0.236		0.002		
19	4.359		−0.012		−0.001	
		0.224		0.001		0.002
21	4.583		−0.011		0.001	
		0.213		0.002		
23	4.796		−0.009			
		0.204				
25	5.000					

$$+\frac{4q^3-18q^2+22q-6}{24}\Delta^4 y_0\Bigg]$$

Here $q=\frac{x-x_0}{h}$, $x_0=15$, $h=2$ and $x=16$, then $q=\frac{16-15}{2}=\frac{1}{2}=0.5$,
Also $\Delta y_0=0.250$, $\Delta^2 y_0=-0.014$, $\Delta^3 y_0=0.002$, $\Delta^4 y_0=-0.001$.

$$\therefore\ f'(16)=\frac{1}{2}\left[0.250+0+\frac{3(.5)^2-6(.5)+2}{6}(.002)\right.$$
$$\left.+\frac{4(.5)^3-18(.5)^2+22(.5)-6}{24}(-0.001)\right]$$
$$f'(16)=\ 0.1249375\quad\text{(Exact value: 0.125)}$$

Similarly,

$$f''(16)=\frac{1}{2^2}\left[-0.014+(0.5-1)(.002)\right.$$
$$\left.+\frac{6(.5)^2-18(.5)+11}{12}(-.001)\right.$$
$$f''(16)=-0.0038229\quad\text{(Exact value}=-0.00390625)$$

b. When $x=15$ (which is a tabulated point), then $q=0$

$$f'(15)=\frac{1}{2}\left[0.250-\frac{1}{2}(-0.014)+\frac{1}{3}(0.002)-\frac{1}{4}(-0.001)\right.$$
$$f'(15)=0.128958\quad\text{(Exact value: 0.12909)}$$
$$f''(15)=\frac{1}{4}\left[-0.014-0.002+\frac{11}{12}(-0.001)\right]$$
$$f''(15)=-0.004229\quad\text{(Exact value: }-0.0043033).$$

For (c) and (d) use Newton's backward formulae since $x=24$ and $x=25$ are at the end of the table.

c. Here $q=\frac{x-x_n}{h}$, $x=24$, $x_n=25$, $h=2$, $q=\frac{24-25}{2}=-0.5$, $q=-0.5$. Here $\nabla y_n=.204$, $\nabla^2 y_n=-0.009$, $\nabla^3 y_n=0.002$, $\nabla^4 y_n=0.001$.

$$f'(24)=\frac{1}{2}\left[.204+\frac{1}{2}(2(.5)+1)(-0.009)\right.$$
$$\left.+\frac{3(-.5)^3+6(-.5)+2}{6}(0.002)\right]$$
$$=0.09727\quad\text{(Exact value: 0.10206)}$$
$$f''(24)=\frac{1}{2^2}\left[-0.009+(-.5+1)(0.002)\right.$$
$$\left.+\frac{6(-.5)^2+18(-.5)+11}{12}(0.001)\right]$$
$$=-0.00242708\ \text{(Exact value : }-0.002126293)$$

d. When $x=25$ which is a tabulated value, $q=0$. Then

$$f'(25)=\frac{1}{2}\left[.204+\frac{1}{2}(-0.009)+\frac{1}{3}(.002)+\frac{1}{4}(.001)\right]$$
$$=0.100208\quad\text{(Exact value: 0.100)}$$
$$f''(25)=\frac{1}{2^2}\left[-0.009+0.002+\frac{11}{12}(.001)\right]$$
$$=-0.00225\ \text{(Exact value: }-0.002)$$

Note: $y=\sqrt{x}$, $y'(x)=\frac{1}{2\sqrt{x}}$, $y''=\frac{-1}{4x^{3/2}}$.

EXERCISE

1. Using forward differences find the first and second derivatives of y at $x=2$ for the data given below:

x:	2	4	6	8	10
y:	0	0	1	0	0

Hint: Differences are 0, 1, −3, 6.

Ans. −1.5, 2.375

2. Compute $y'(1)$, $y''(1)$ and $y'(3)$ from the following data:

x:	1	2	3	4	5	6	7	8
y:	2.105	2.808	3.614	4.604	5.857	7.451	9.467	11.985

Hint: Differences: .703, .103, .081, −0.002.

Ans. $y'(1)$=0.6925, $y''(1)$=0.0201; $y'(3)$=.883

3. Determine $y'(2)$, $y''(2)$, $y'''(2)$, $y''''(2)$, $y'(1.5)$, $y''(1.5)$ from the data below, compare with exact values:

x:	0	1	2	3	4	5
y:	0	1	8	27	64	125

Hint: Differences: 1, 6, 6, 0, 0.

Ans. $y'(2) = 12$, $y''(2) = 12$, $y'''(2) = 6$, $y''''(2) = 0$, $y'(1.5) = 6.75$, $y''(1.5) = 9$. Exact value: $y = x^3$, $y' = 3x^2$, $y'' = 6x$, $y''' = 6$, $y'''' = 0$.

4. Compute $y'(1.05)$, $y''(1.05)$, $y'(1.25)$, $y''(1.25)$ from the following data:

x:	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y:	1.00	1.0247	1.04881	1.07238	1.09544	1.11803	1.14017

Ans. $y'(1.05) = 0.48763$, $y''(1.05) = -0.2144$, $y'(1.25) = 0.44733$, $y''(1.25) = -0.158332$

5. Find $y'(50)$ of the tabulated function $y = \log_{10} x$, given below:

x:	50	55	60	65
y:	1.6990	1.7404	1.7782	1.8129

Hint: Forward differences: 414, −36, 5, $h = 5$. Exact value: 0.0087.

Ans. 0.0087 $\left(= \frac{1}{5}(0.0414 + 0.008 + 0.0002)\right)$

6. The following table contains the path $y = f(t)$ traversed in time t by a point moving in a straight line. Using finite forward differences upto order five inclusive, find the velocity $v = \frac{dy}{dt}$ and acceleration $A = \frac{d^2y}{dt^2}$ at the point of times: $t = 0, 0.01, 0.02, 0.03, 0.04$.

time t in sec:	0	0.01	0.02	0.03	0.04	0.05
Path $y(t_i)$ in cm:	0	1.519	6.031	13.397	23.396	35.721

t:	0.06	0.07	0.08	0.09
$y(t_i)$	50	65.798	82.635	100

Hint: Forward differences are 2.993, −0.139, −0.082, −0.004. Exact: $y = 100(1 - \cos\frac{50\pi t}{9})$, $V = \frac{dy}{dt} = \frac{5000\pi}{9}\sin\left(\frac{50\pi t}{9}\right)$, $W = \frac{d^2y}{dt^2} = \frac{250000\pi^2}{81}\cos\left(\frac{50\pi t}{9}\right)$.

Ans.

Table 32.13

t	V	W	V_{exact}	W_{exact}
0.00	0.4	30600	0.0	30462
0.01	303.6	29780	303.08	30001
0.02	596.3	28780	596.98	28625
0.03	873.2	26250	872.66	26381
0.04	1121.7	23360	1121.9	23340

7. Compute $y'(1.1)$, $y''(1.1)$, $y'(1.6)$, $y''(1.6)$ given:

x:	1.0	1.1	1.2	1.3	1.4	1.5	1.6
y:	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Hint: Forward differences: .414, −0.36, 0.006, −0.002, 0.002, −0.003, backward differences .281, −0.018, 0.005, −0.001, −.001, −0.003.

Ans. $y'(1.1) = 3.946$, $y''(1.1) = -3.545$, $y'(1.6) = 2.727$, $y''(1.6) = -1.703$

8. Determine $y'(0)$, $y''(0)$ from following data:

x:	0	1	2	3	4	5
y:	4	8	15	7	6	2

Ans. $y'(0) = -27.9$, $y''(0) = 117.67$.

32.14 NUMERICAL INTEGRATION

Let $f(x)$ be continuous on the interval $[a, b]$ and its antiderivative $F(x)$ is known. Then the definite integral of $f(x)$ from a to b may be evaluated using Newton-Leibnitz formula

$$\int_a^b f(x)dx = F(b) - F(a) \qquad (1)$$

where $f'(x) = f(x)$.

However, computation of the definite integral by (1) becomes difficult or practically impossible when (i) the antiderivative $F(x)$ can not be found by elementary means or is too involved (ii) when the integrand $f(x)$ is specified in tabular form.

The numerical integration of a single integral is known as **mechanical quadrature** and uses the geometrical interpretation of the definite integral $\int_a^b f(x)dx$ as the area under the curve $y = f(x)$ between the ordinates $x = a$ and $x = b$. The basic integration rule is to replace $f(x)$ by a simple polynomial $\phi(x)$, say Lagrange's interpolation polynomial in $[a, b]$. Thus

$$\int_a^b f(x)dx \sim \int_a^b \phi(x)dx \tag{2}$$

Choosing interval size $h = \frac{b-a}{n}$, divide the interval $[a, b]$ into n intervals by means of $(n + 1)$ equally spaced points $x_0 = a,\ x_i = x_0 + ih,\ i = 1, 2, 3, \ldots, n-1,\ x_n = b$. Let $y_i = f(x_i)$ for $i = 0, 1, 2, \ldots, n$.

Integrating the integral in the R.H.S. of (2), we get Newton-Cotes formula of the form

$$\int_a^b f(x)dx \sim c \cdot h \cdot \sum_{i=0}^{n} w_i \cdot f(x_i) \tag{3}$$

[Since the end points x_0, x_n are used in (3), these formula are known as 'closed' type.]

Here m is the degree of the polynomial $\phi(x)$, c is the coefficient and w_i are weights, given in the following table:

Table 32.14

Case	m	c	w_0	w_1	w_2	w_3	Name of the rule
I.	0	1	1				Rectangular and mid point rules
II.	1	$\frac{1}{2}$	1	1			Trapezoidal rule
III.	2	$\frac{1}{3}$	1	4	1		Simpson's $\frac{1}{3}$ rule
IV.	3	$\frac{3}{8}$	1	3	3	1	Simpson's $\frac{3}{8}$ rule

The degree m of the polynomial $\phi(x)$ used in the derivation of the above formula are 0, 1, 2, 3 respectively.

Derivation of Trapezoidal, Simpson's $\frac{1}{3}$ and Simpson's $\frac{3}{8}$ Rule Formula:

The basic idea in numerical integration is to replace the unknown tabulated function $y = f(x)$ by an nth degree polynomial $P_n(x)$ say Newton-Gregory forward interpolation formula and carry on the integration. Thus

$$\begin{aligned} I = \int_{x_0}^{x_n} f(x)dx &\simeq \int_{x_0}^{x_n} P_n(x)dx \\ &\simeq \int_{q=0}^{n} \left[y_0 + q\Delta y_0 + \frac{q(q-1)}{2!}\Delta^2 y_0 \right. \\ &\quad + \frac{q(q-1)(q-2)}{3!}\Delta^3 y_0 + \cdots \\ &\quad \left. + \frac{q(q-1)(q-2)\cdots(q-n+1)}{n!}\Delta^n y_0 \right] dq \end{aligned}$$

Here the new variable is $q = \frac{x-x_0}{h}$, so $dq = \frac{dx}{h}$.

Trapezoidal Rule

Take $n = 1$ (two points, one internal)

$$\begin{aligned} \int_0^1 y\,dx &= h\int_0^1 (y_0 + q\Delta y_0)dq = h\left[y_0 q + \Delta y_0 \cdot \frac{q^2}{2} \right]\Big|_{q=0}^{1} \\ &= h\left[y_0 + \frac{\Delta y_0}{2} \right] = h\left[y_0 + \frac{y_1 - y_0}{2} \right] \\ &= \frac{h(y_1 + y_0)}{2} = \frac{h}{2}(y_1 + y_0). \end{aligned}$$

Applying this to successive intervals, we get

$$\begin{aligned} \int_{x_0}^{x_n} y\,dx &= \int_{x_0}^{x_1} y\,dx + \int_{x_1}^{x_2} y\,dx + \cdots + \int_{n_{n-1}}^{x_n} y\,dx \\ &= \frac{h}{2}(y_0 + y_1) + \frac{h}{2}(y_1 + y_2) + \cdots + \frac{h}{2}(y_{n-1} + y_n) \end{aligned}$$

$$\boxed{I = \frac{h}{2}\left[y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n \right]}$$

Simpson's $\frac{1}{3}$ Rule

Take $n = 2$ (three points, two intervals, curve (polynomial) parabola).

$$\begin{aligned} \int_{x_0}^{x_2} y\,dx &= h\int_0^2 \left[y_0 + q\Delta y_0 + \frac{q(q-1)}{2}\Delta^2 y_0 \right] dq \\ &= h\left[y_0 \cdot q + \frac{q^2}{2}\Delta y_0 + \frac{\Delta^2 y_0}{2}\left(\frac{q^3}{3} - \frac{q^2}{2} \right) \right]\Big|_{q=0}^{2} \\ &= h\left[2y_0 + 2\Delta y_0 + \frac{1}{3}\Delta^2 y_0 \right] \end{aligned}$$

$$= h\left[2y_0 + 2(y_1 - y_0) + \frac{1}{3}(y_2 - 2y_1 + y_0)\right]$$

$$= \frac{h}{3}[y_0 + 4y_1 + y_2].$$

Applying for successive intervals (*the number of intervals must be even or the number of points is odd*).

$$\int_{x_0}^{x_n} y\,dx = \int_{x_0}^{x_2} y\,dx + \int_{x_2}^{x_4} y\,dx + \cdots + \int_{x_{n-2}}^{x_n} y\,dx$$

$$= \frac{h}{3}[y_0 + 4y_1 + y_2] + \frac{h}{3}[y_2 + 4y_3 + y_4] + \cdots$$

$$+ \frac{h}{3}[y_{n-2} + 4y_{n-1} + y_n]$$

$$\boxed{I = \frac{h}{3}\left[y_0 + 2(y_2 + y_4 + \cdots + y_{n-2}) + 4(y_1 + y_3 + \cdots + y_{n-1}) + y_n\right]}$$

Simpson's $\frac{3}{8}$ Rule

Take $m = 3$, number of intervals 3 and number of points 4: (number of intervals should be multiples of 3 i.e., $3N$).

$$\int_{x_0}^{x_3} y\,dx = h\int_0^3 \left[y_0 + q\Delta y_0 + \frac{q(q-1)\Delta^2 y_0}{2} + \frac{q(q-1)(q-2)}{3!}\Delta^3 y_0\right] dq$$

$$= h\left[3y_0 + \frac{9}{2}\Delta y_0 + \frac{9}{4}\Delta^2 y_0 + \frac{3}{8}\Delta^3 y_0\right.$$

$$= h\left[3y_0 + \frac{9}{2}(y_1 - y_0) + \frac{9}{4}(y_2 - 2y_1 + y_0) + \right.$$

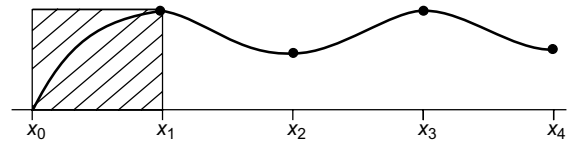
$$\left. + \frac{3}{8}(y_3 - 3y_2 + 3y_1 - y_0)\right]$$

$$= \frac{3}{8}h[y_0 + 3y_1 + 3y_2 + y_3].$$

Applying repeatedly

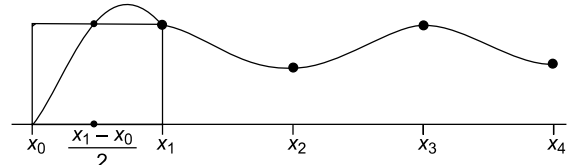
$$\int_{x_0}^{x_n} y\,dx = \frac{3}{8}h\left[y_0 + 2(y_3 + y_6 + \cdots + y_{n-3}) + 3(y_1 + y_2 + y_4 + y_5 + \cdots + y_{n-2} + y_{n-1}) + y_n\right].$$

Thus



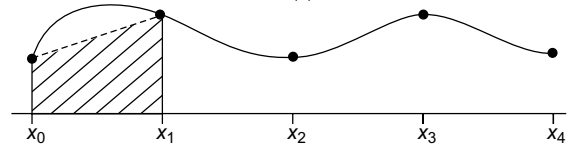
Rectangular rule on $[x_0, x_1]$

(i)



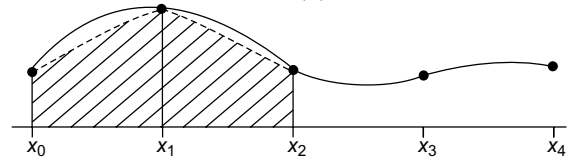
Mid point rule on $[x_0, x_1]$

(ii)



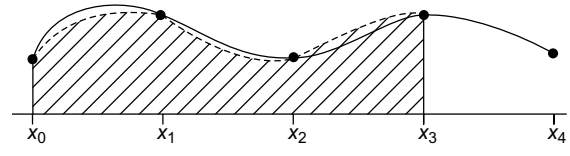
Trapezoidal rule on $[x_0, x_1]$

(iii)



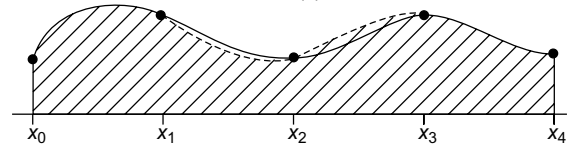
Simpson's $\frac{1}{3}$ rule on $[x_0, x_2]$

(iv)



Simpson's $\frac{3}{8}$ th rule on $[x_0, x_3]$

(v)



Boole's rule on $[x_0, x_4]$

(vi)

Fig. 32.14

Trapezoidal Rule

When $n = 1$, $h = b - a$, then

$$\int_a^b f(x)dx = \int_a^b y\,dx \simeq \frac{h}{2}[y_0 + y_1]. \qquad (4)$$

Generalized or Composite or Multiple Segment Trapezoidal Rule

When $h = \frac{b-a}{n}$ (i.e., interval is divided into n intervals).

$$\int_a^b y\,dx = \frac{h}{2}[y_0+y_1]+\frac{h}{2}[y_1+y_2]+\cdots+\frac{h}{2}[y_{n-1}+y_n]$$
$$= \frac{h}{2}[y_0 + 2(y_1 + y_2 + \cdots + y_{n-1} + y_n)]. \quad (5)$$

Simpson's $\frac{1}{3}$ Rule

When $n = 2$,

$$\int_a^b y\,dx = \frac{h}{3}[y_0 + 4y_1 + y_2].$$

Generalized Simpson's $\frac{1}{3}$ Rule

When $n = 2m$ (*even*) number of intervals

$$\int_a^b y\,dx = \frac{h}{3}[(y_0+4y_1+y_2)] + \frac{h}{3}[y_2+4y_3+y_4] + \cdots$$
$$+\frac{h}{3}[y_{n-2} + 4y_{n-1} + y_n]$$
$$= \frac{h}{3}\left[(y_0 + y_n) + 2(y_2 + y_4 + \cdots + y_{n-2})\right.$$
$$\left. + 4(y_1 + y_3 + \cdots + y_{n-1})\right].$$

Simpson's $\frac{3}{8}$ Rule

When $n = 3$

$$\int_a^b y\,dx = \frac{3}{8}h[y_0 + 3y_1 + 3y_2 + y_3].$$

Generalized Simpson's $\frac{3}{8}$ Rule

$$\int_a^b y\,dx = \frac{3}{8}h[y_0 + 3y_1 + 3y_2 + y_3]$$
$$+\frac{3}{8}h[y_3 + 3y_4 + 3y_5 + y_6] + \cdots$$
$$+\frac{3}{8}h[y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n]$$
$$= \frac{3}{8}h\left[(y_0 + y_n) + 2(y_3 + y_6 + \cdots + y_{n-3})\right.$$
$$\left. + 3(y_1 + y_2 + y_4 + y_5 + \cdots + y_{n-2} + y_{n-1})\right].$$

Weddle Rule

When $(n = 6)$,

$$\int_a^b y\,dx = \frac{3h}{10}[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6].$$

Generalized Weddle Rule

$$\int_{a=x_0}^{b=x_0+nh} y\,dx = \frac{3h}{10}[y_0+5y_1+y_2+6y_3+y_4+5y_5+y_6]$$
$$+\frac{3h}{10}[y_6+5y_7+y_8+6y_9+y_{10}+5y_{11}$$
$$+y_{12}]+\cdots+\frac{3h}{10}[y_{n-6}+5y_{n-5}+y_{n-4}$$
$$+6y_{n-3}+y_{n-2}+5y_{n-1}+y_n].$$

Note: While there is no restriction for the number of intervals in trapezoidal rule, number of sub intervals n in the case of Simpson's $\frac{1}{3}$ rule must be even, for Simpson's $\frac{3}{8}$ rule must be multiple of 3, for Weddle's rule must be multiple of 6.

WORKED OUT EXAMPLES

Example 1: Given that $y = \log x$, and

x:	4.0	4.2	4.4	4.6	4.8	5.0	5.2
y:	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

evaluate $I = \int_4^{5.2} \log x\,dx$ by (a) Trapezoidal rule; (b) Simpson's $\frac{1}{3}$ rule; (c) Simpson's $\frac{3}{8}$ rule; (d) Weddle's rule; (e) Compare it with exact value.

Solution:

a. By trapezoidal rule, $(h = 0.2)$

$$I = \frac{0.2}{2}[1.3863 + 1.6487 + 2(1.4351 + 1.4816$$
$$+1.5261 + 1.5686 + 1.6094)]$$

$$I = 1.8276551$$

b. By Simpson's $\frac{1}{3}$ rule

$$I = \frac{0.2}{3}[1.3863 + 1.6487 + 2(1.4816 + 1.5686) + 4(1.4351 + 1.5261 + 1.6094)]$$

$$I = 1.8278472$$

c. By Simpson's $\frac{3}{8}$ rule

$$I = \frac{3}{8}(0.2)[1.3863 + 1.6487 + 2(1.5261) + 3(1.4351 + 1.4816) + (1.5686 + 1.6094)]$$

$$I = 1.8278470$$

d. By Weddle's rule

$$I = \frac{3}{10}(0.2)[1.3863 + 5(1.4351 + 1.6094) + 6(1.5261) + 1.4816 + 1.5686 + 1.6487]$$

$$I = 1.8278474$$

e. By integrating by parts

$$I = \int_5^{4.2} \ln x\, dx = x \cdot \ln x - \int x \cdot \frac{1}{x} dx = x \cdot \ln x - x \Big|_4^{5.2}$$

$$= (5.2 \ln 5.2 - 5.2) - (4 \ln 4 - 4) = 3.373 - 1.54517$$

$$I = 1.827822556.$$

Example 2: The half ordinates in fect of the mid ship section of a vessel are 12.5, 12.8, 12.9, 13, 13, 12.8, 12.4, 11.8, 10.4, 6.8, 0.5 and the ordinates are 2 feet apart. Find the centre of gravity of the section.

Solution: Centre of gravity $(\overline{x}, 0)$.

x:	0	2	4	6	8	10
half ordinates:	12.5	12.8	12.9	13	13	12.8
y:	25	25.6	25.8	26	26	25.6
xy:	0	51.2	103.2	156	208	256
x:	12	14	16	18	20	
half ordinates:	12.4	11.8	10.4	6.8	0.5	
y:	24.8	23.6	20.8	13.6	1	
xy:	297.6	330.4	332.8	244.8	20	

Here $\overline{x} = \dfrac{\int_0^{20} xy\, dx}{\int_0^{20} y\, dx} = \dfrac{I_1}{I_2}$

where by Simpson's $\frac{1}{3}$ rule

$$I_1 = \frac{1}{3}(2)[20 + 332.8 + 4(51.2 + 156 + 256 + 330.4 + 244.8) + 2(103.2 + 208 + 297.6 + 332.8)]$$

$$I_1 = 4037.8666$$

$$I_2 = \frac{1}{3}(2)[25 + 1 + 4(25.6 + 26 + 25.6 + 23.6 + 13.6) + 2(25.8 + 26 + 24.8 + 20.8)]$$

$$I_2 = 452.2666$$

$$\therefore \overline{x} = \frac{4037.8666}{452.2666} = 8.928066154 \approx 8.93.$$

Example 3: The speeds of an electric train at various times after leaving one station until it stops at the next station are given in the following table (Table 11.15). Find the distance between the two stations.

Table 32.15

Speed in mph	0	13	33	$39\frac{1}{2}$	40	40	36	15	0
Time in minutes	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3	$3\frac{1}{4}$	$3\frac{1}{2}$

Solution: Let $v = \frac{ds}{dt}$ be the velocity of the train at any time t. Then s, the distance between the two stations (with a lapse of $3\frac{1}{2}$ minutes) is

$$s = \int_{t=0}^{3\frac{1}{2}} \cdot \frac{ds}{dt} dt = \int_{t=0}^{3\frac{1}{2}} v\, dt = \int_0^3 v\, dt + \int_3^{3\frac{1}{2}} v\, dt$$

since the time interval are not the same. By applying Simpson's $\frac{1}{3}$ rule with $h = \frac{1}{2}$ and $h = \frac{1}{4}$, we have

Table 32.16

Speed in mph	0	13	33	$39\frac{1}{2}$	40	40	36	15	0
Time in hours	0	$\frac{1}{120}$	$\frac{1}{60}$	$\frac{1}{40}$	$\frac{1}{30}$	$\frac{1}{24}$	$\frac{1}{20}$	$\frac{13}{240}$	$\frac{7}{480}$

$$s = s_1 + s_2 = \frac{1}{3} \cdot \frac{1}{120}\left[0 + 36 + 4(13 + 39\frac{1}{2} + 40) + 2(33 + 40)\right] + \frac{1}{3} \cdot \frac{1}{240}[36 + 0 + 4(15)]$$

$$= \frac{23}{15} + \frac{2}{15} = \frac{25}{15} = \frac{5}{3} = 1\frac{2}{3} = 1.666 \text{ miles}$$

Thus the distance between the two stations is 1.666 miles.

EXERCISE

Evaluate approximately the following integrals:

Trapezoidal rule

1. Use trapezoidal rule to evaluate (a) $\int_1^{10} \frac{dx}{x}, n = 10$; (b) $\int_0^1 \frac{dx}{1+x^2}, n = 4$; (c) $\int_0^1 \frac{\sin x}{x} dx, n = 4$.

Ans. **a.** $\frac{1}{2}\left[1 + 0.1 + 2\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9}\right)\right] = 2.3788$

b. $\frac{1}{4} \cdot \left[\frac{1}{2}\left(1 + \frac{1}{2}\right) + \frac{16}{17} + \frac{4}{5} + \frac{16}{25}\right] = 0.7828$

c. $\frac{1}{4} \cdot \frac{1}{2}\,[1 + 0.84 + 2(.9898 + 0.95885 + .90885)] = 0.9445$

2. Using the following data evaluate by T.R.

a.

x:	1	2	3	4	5	6	7
y:	2.105	2.808	3.614	4.604	5.857	7.451	9.467

b.

x:	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8
$f(x)$:	1.543	1.668	1.811	1.971	2.151	2.352	2.577	2.828	3.107

with (i) $h = 0.1$; (ii) $h = 0.2$; (iii) $h = 0.4$.

Ans. **a.** $\int_1^7 y\,dx = \frac{1}{2}[2.105 + 2(2.808 + 3.614 + 4.604 + 5.851 + .7451) + 9.467] = 30.120$

b. $\int_1^{1.8} f(x)dx = 1.7683 (h = 0.1)$, $1.7728 (h = 0.2)$, $1.7904 (h = 0.4)$.

3. Integrate $\int_0^{\pi} \sin x\,dx, n = 10$.

Hint:

x:	0	$\frac{\pi}{10}$	$\frac{2\pi}{10}$	$\frac{3\pi}{10}$	$\frac{4\pi}{10}$	$\frac{5\pi}{10}$	$\frac{6\pi}{10}$	$\frac{7\pi}{10}$	$\frac{8\pi}{10}$	$\frac{9\pi}{10}$	π
$\sin x$:	0	.309	.5878	.809	.951	1	.951	.809	.5878	.309	0

Ans. 6.3138

4. Evaluate $\int_0^1 \frac{dx}{1+x^2}, n = 10$.

Hint:

x:	0	.1	.2	.3	.4	.5	.6	.7	.8	.9	1
$f(x)$:	1	.99	.96	.917	.862	.8	.735	.67	.61	.55	.5

Ans. 0.6547

Simpson's $\frac{1}{3}$ rule

5. Integrate approximately $\int_1^{1.04} f(x)dx$ from the following data:

x:	1	1.01	1.02	1.03	1.04
$f(x)$:	3.953	4.066	4.182	4.300	4.421

Hint: $h = 0.1$.

Ans. 0.16734, $I = \frac{0.1}{3}[(3.953) + (4.421) + 4(4.066 + 4.182 + 4.3)]$

6. Evaluate $\log e^7$ by Simpson's $\frac{1}{3}$ rule.

Hint: $I = \int_0^6 \frac{dx}{1+x}$.

Ans. 1.9588

7. Compute $\int_0^{\frac{\pi}{2}} \sqrt{\sin x}\,dx$.

Hint: $n = 6, h = \frac{\pi}{12}$.

Ans. 1.18728, $I = \frac{\pi}{3} \cdot \frac{1}{12}[0 + 1 + 4(.50874 + 0.8409 + 0.98282) + 2(0.70711 + 0.9306)]$

8. Compute $I = \int_0^1 \frac{dx}{1+x}, n = 10, h = 0.1$.

Ans. 0.69315

9. Integrate approximately $\int_0^{\pi} \sin x\,dx, n = 10$.

Ans. 3.0778

10. Compute $\int_0^1 \frac{dx}{1+x^2}, n = 10$.

Ans. 0.785425

11. A river is 80 feet wide. The depth d in feet at a distance x feet from one bank is given by:

x:	0	10	20	30	40	50	60	70	80
d:	0	4	7	9	12	15	14	8	3

Find approximately the area of cross section of the river.

Hint: $I = \int_0^{80} d\,dx = \frac{1}{3}(10)[0 + 3 + 4(4 + 9 + 15 + 8) + 2(7 + 12 + 14)]$

Ans. 710 sq. feet

12. A body of weight w tons is acted upon by a variable force F tons weight. It acquires a speed of v mph after travelling y yards where $v^2 = \frac{89.8}{w} \int F(y)dy$. If $w = 600$ and F is given by the table below, estimate the velocity at the end of 400 yards from the starting point.

y:	0	50	100	150	200	300	400
$f(y)$:	90	62	45	34	26	15	8

Hint: $\int_0^{400} F\,dy = \int_0^{200} + \int_{200}^{400}$ with $h_1 = 50$, $h_2 = 100$.

Ans. $v = 44.16$ mph

13. Find the approximate mileage travelled by a train between 11.50 AM to 12.30 PM from the following data:

time t:	11.50 AM	12	12.10	12.20	12.30
speed (mph):	24.2	35	41.3	42.8	39.2

Hint: $I = s\int_{t=11.50}^{12.30} v\,dt = \frac{1}{6}\left(\frac{1}{3}\right)[24.2 + 39.2 + 4(35.0 + 42.8) + 2(41.3)]$.

Ans. s = distance = 25.4 miles

Simpson's $\frac{3}{8}$ rule

14. Evaluate $\log_e 7$ by Simpson's $\frac{3}{8}$th rule.

Hint: $I = \int_0^6 \frac{dx}{1+x} = \log_e(1+x)\big|_0^6 = \log_e 7$

x:	0	1	2	3	4	5	6
$\frac{1}{1+x}$:	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$

Ans. 1.966

15. Integrate $\int_0^1 \frac{dx}{1+x^2}$, $n = 12$.

Hint:

x:	0	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$	$\frac{1}{2}$
$f(x) = \frac{1}{1+x^2}$:	1	.993	.973	.94	.9	.85	.8
x:	$\frac{7}{12}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{5}{6}$	$\frac{11}{12}$	1	
$f(x) = \frac{1}{1+x^2}$:	.8	.746	.692	.64	.59	.543	0.5

Ans. 0.78539

16. Evaluate $\int_0^1 e^x dx$ approximately.

Hint: $n = 9$, $h = \frac{1}{9} = 0.1111$

x:	0	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{4}{9}$	$\frac{5}{9}$
e^x:	1	1.1175	1.2488	1.3956	1.5596	1.743
x:	$\frac{6}{9}$	$\frac{7}{9}$	$\frac{8}{9}$	1		
e^x:	1.9477	2.176	2.432	2.718		

Ans. 1.7182888

17. Integrate $\int_0^{40} a(t)dt$ from the data given below:

t:	0	5	10	15	20	25	30	35	40	45
$a(t)$:	40	45.25	48.5	51.25	54.35	59.48	61.5	64.3	68.7	70

Hint: $5\left(\frac{3}{8}\right)$ [40 + 70 + 2(51.25 + 61.5) + 3(45.25 + 48.5 + 54.35 + 59.48 + 64.3 + 68.7)].

Ans. $5\left(\frac{3}{8}\right)(1357.24) = 2544.825$

Weddle's rule

18. Integrate approximately $\int_0^{1.1} e^x dx$, $n = 12$.

Hint: $I = \frac{3}{10}\left(\frac{1.1}{12}\right)[1 + 5(1.1051) + 1.2214 + 6(1.3498) + 1.4918 + 5(1.6487) + 2(1.822) + 5(2.0137) + 2.2225 + 6(2.4596) + 5(2.71828) + 3.004166]$

Exact value: $e^x|_0^{1.1} = 2.004166$

Ans. 2.0040219

19. Evaluate approximately $\int_1^7 y\,dx$, $n = 6$.

x:	1	2	3	4	5	6	7
y:	2.157	3.519	4.198	4.539	4.708	4.792	4.835

Ans. 25.4061

20. Integrate $\int_{0.2}^{1.4}(\sin x - \ln x + e^x)dx$, $n = 12$.

Hint:

x:	0.2	0.3	0.4	0.5	0.6	0.7
y:	3.0295	2.8493	2.797	2.821	2.8976	3.01465
x:	0.8	0.9	1.0	1.1	1.2	1.3	1.4
y:	3.166	3.348	3.559	3.8	4.0698	4.3705	4.3704

$I = \frac{3}{10}(0.1)$ [21.0584 + 5(13.5828) + 6(6.62137) + 2(3.16605)]

Exact value: $-\cos x - x(\ln x - 1) + e^x\big|_{0.2}^{1.4} = 4.05095$

Ans. 4.05098

21. Evaluate $\int_0^1 \frac{dx}{1+x^2}$, $n = 12$.

Hint:

x:	0	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$	$\frac{1}{2}$
$f(x)$:	1	0.993	0.9729	0.94	0.9	0.852	0.8
x:	$\frac{7}{12}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{5}{6}$	$\frac{11}{12}$	1	
$f(x)$:	.746	.6923	.64	.5906	.5434	0.5	

Ans. 0.7603725

Boole's Rule of Integration

The goal in numerical integration is to approximate $\int_a^b f(x)\,dx$ using $(N+1)$ sample points (x_0, f_0), $(x_1, f_1)\ldots, (x_N, f_N)$ where f_k denotes $f(x_k)$.

Boole's rule is obtained by using a 4th degree Lagrange interpolating polynomial based on the

nodes x_0, x_1, x_2, x_3 and x_4. Consider the approximation formula

$$\int_0^4 g(t)dt = w_0 g(0) + w_1 g(1) + w_2 g(2) + w_3 g(3) + w_4 g(4) \quad (1)$$

we determine the weights w_0, w_1, w_2, w_3, w_4 assuming that (1) is exact for the functions $g(t) = 1, t, t^2, t^3, t^4$. Thus when $g(t) = 1$ then $g(0) = g(1) = g(2) = g(3) = g(4) = 1$. From (1)

$$\int_1^4 1 \cdot dt = 4 = w_0 + w_1 + w_2 + w_3 + w_4 \quad (2)$$

For $g(t) = t$, $g(0) = 0$, $g(1) = 1$, $g(2) = 2$, $g(3) = 3$, $g(4) = 4$ so from (1)

$$\int_0^4 t\,dt = \left.\frac{t^2}{2}\right|_0^4 = 8 = 0 + w_1 + 2w_2 + 3w_3 + 4w_4 \quad (3)$$

Similarly for $g(t) = t^2, t^3, t^4$ from (1) we get

$$\frac{64}{3} = w_1 + 4w_2 + 9w_3 + 16w_4 \quad (4)$$

$$64 = w_1 + 8w_2 + 27w_3 + 64w_4 \quad (5)$$

$$\frac{1024}{5} = w_1 + 16w_2 + 81w_3 + 256w_4 \quad (6)$$

Solving equations (1) to (6) we get $w_0 = \frac{14}{45}$, $w_1 = \frac{64}{45}$, $w_2 = \frac{24}{45}$, $w_3 = \frac{64}{45}$, $w_4 = \frac{14}{45}$.

Thus the Boole's rule for the interval $[x_0, x_4]$

$$\int_{x_0}^{x_4} f(x)dx \approx \frac{2h}{45}[7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4] \quad (7)$$

Composite Boole's Rule

When Boole's rule is repeated over 4 M subintervals of $[a, b]$, we get the composite Boole's rule as

$$\int_a^b f(x)dx \approx \frac{2h}{45}\sum_{k=1}^{M}[7f_{4k-4} + 32f_{4k-3} + 12f_{4k-2} + 32f_{4k-1} + 7f_{4k}] \quad (8)$$

Note: To apply Boole's rule, the number of subintervals should be multiples of 4 (i.e., 4, 8, 12 . . .) and the numbers of nodes must be odd.

WORKED OUT EXAMPLES

Example 1: Evaluate $\int_1^5 \frac{dx}{x}$ by Boole's rule.

Solution: Take $h = \frac{5-1}{4} = 1$ so that there 4 subintervals.

x	1	2	3	4	5
$f(x) = \frac{1}{x}$	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$

Using Boole's rule

$$\int_1^5 \frac{dx}{x} \approx \frac{2.1}{45}\left[7.1 + 32\left(\frac{1}{2}\right) + 12\left(\frac{1}{3}\right) + 32\left(\frac{1}{4}\right) + 7 \cdot \frac{1}{5}\right]$$

$$= \frac{364}{225} = 1.61777$$

while the exact value is $\ln 5 - \ln 1 = 1.60943$

Example 2: Find the distance travelled by a submarine under polar ice cap from the following table showing the velocity at variance times

t time (hr)	0.00	0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00
Velocity $v(t)$ km/hr	6.0	7.5	8.0	9.0	8.5	10.5	9.5	7.0	6.0

Solution: We know that

$$d = \text{distance travelled} = \int_{t=0}^{2} v(t)dt$$

We evaluate this definite integral by Boole's rule with $4 \cdot 2 = 8$ subintervals. Here $h = \frac{2-0}{8} = \frac{1}{4} = 0.25$ (and 9 node points). Applying the composite Boole's rule we get

$$d = \frac{2h}{45}[7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4] + \frac{2h}{45}[7f_4 + 32f_5 + 12f_6 + 32f_7 + 7f_8]$$

$$= \frac{2(0.25)}{45}[7(6) + 32(7.5) + 12(8) + 32(9) + 7(8.5)] + \frac{2(0.25)}{45}[7(8.5) + 32(10.5) + 12(9.5) + 32(7.0) + 7(6.0)]$$

$$= 16.6777$$

EXERCISE

1. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by Boole's rule with 12 subintervals.

Ans. Exact value: $\frac{\pi}{4} = 0.7854$, by Boole's rule: 0.8236

Hint: $\frac{2}{45} \cdot \frac{1}{12}[7\{1 + 2(0.9 + 0.746 + 0.692) + 0.5\} + 32\{0.993 + 0.94 + 0.85 + 0.746 + 0.64 + 0.543\} + 12\{0.973 + 0.8 + 0.59\}] = 0.8236$

2. Evaluate $\int_{0.2}^{1.4} (\sin x - \ln x + e^x)dx$

Ans. Exact value: 4.05025, $n = 12$, $h = 0.1$, by Boole's: 4.04235

Hint: $\frac{2(0.1)}{45}[7\{3.0295 + 2(2.8976 + 3.559) + 4.3704\} + 32\{2.8493 + 2.821 + 3.01465 + 3.348 + 3.8 + 4.3705\} + 12(2.797 + 3.166 + 4.0698)] = 4.04235$

3. A river is 80 feet wide. The depth d in feet at a distance x from one bank (side) of the river is given below.

x :	0	10	20	30	40	50	60	70	80
d :	0	4	7	9	12	15	14	8	3

Find approximately the area of cross section of the river.

Ans. 708

Hint: $n = 8$, $h = 10$
Area $= \frac{2(10)}{45}[7\{0 + 2(12) + 3\} + 32\{4 + 9 + 15 + 8\} + 12(7 + 14)] = 708$

4. Find the distance between two stations from the following data consisting of the speeds of an electric train at various times after leaving one station until it stops at the next station.

Speed of train $v(t)$ in mph	0	13	33	$39\frac{1}{2}$	40	40	36	15	0
t in mt	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3	$3\frac{1}{2}$	4

Ans. 1.788148 miles

Hint: Distance $= \int_0^4 v(t)dt$, $n = 8$, $h = \frac{1}{2}$mt $= \frac{1}{120}$hr, $= \frac{2}{45} \cdot \frac{1}{120}[7(0 + 40 + 40 + 0) + 32(13 + 39\frac{1}{2} + 40 + 15) + 12(33 + 36)]$

32.15 SPLINE INTERPOLATION

In analyzing experimental data, in ascertaining the relations among variables and in design work, the common problem encountered is fitting a curve through specified points in a plane. *Interpolating curve* is a curve that passes through a set of points in a plane and the curve is said to *interpolate* at these points.

It is observed that approximation of arbitrary functions on a closed interval becomes oscillatory for higher-degree polynomials. By dividing the given interval into a collection of subintervals, *piecewise polynomial approximation* consists of constructing a (generally) different approximating polynomial on *each* subinterval. *Cubic spline interplation* is the most common piecewise approximation using cubic polynomial, known as *spline function*, connecting each pair of data points (or nodes or knots). Cubic splines are third-order curves employed to connect each pair of data points.

Consider a set of $(n + 1)$ data points, which need not be evenly spaced, $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2)) \ldots (x_n, f(x_n))$

or (x_i, f_i) for $i = 0, 1, 2, \ldots n$. (1)

Between each pair of adjacent points from x_i to x_{i+1}, we fit an nth degree polynomial $s_i(x)$. Thus the spline (function) curve can be of any degree. By far, cubic splines are the most popular and are useful version in engineering practice since the cubic spline interpolation gives an interpolation formula that is smooth in the first derivative and continuous, both within an interval and its boundaries (end points x_0 and x_n). The name 'spline' is originated from the draftman's spline which is a drafting aid consisting of a thin flexible rod or strip of wood that is bent to draw a smooth curve through the points to be interpolated.

Linear Splines

Linear splines are first-order splines defined by linear functions. Geometrically they are straight lines connecting two adjacent data points. For a given set of $(n + 1)$ data points (1), the set of linear functions defining linear splines in the 'n' intervals are:

$S_1(x) = f(x_0) + m_0(x - x_0)$; $x_0 \le x \le x_1$.
$S_2(x) = f(x_1) + m_1(x - x_1)$; $x_1 \le x \le x_2$.

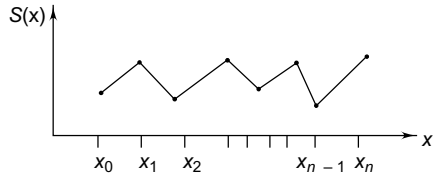


Fig. 32.15 Piecewise linear interpolation

.....

$S_n(x) = f(x_n - 1) + m_{n-1}(x - x_{n-1})$;
$x_{n-1} \le x \le x_n$

Here m_i is the slope of the straight line connecting the data points (or nodes) x_i and x_{i+1} and is given by

$$m_i = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \qquad (2)$$

Fitting of linear splines is piecewise linear interpolation and the interpolating curve is broken curve joining 'n' straight line segments. Disadvantage of linear splines is that at the point (known as *knot*) where two splines meet, the slope changes abruptly i.e., first derivative is discontinuous.

Quadratic Splines

Interpolation of second-order polynomials leads to quadratic splines having continuous first derivatives at the knots.

For each interval we fit a parabola given by

$$s_i(x) = a_i x^2 + b_i x + c_i$$

For $n + 1$ data points we have n intervals. So we should determine $3 \times n$ unknown constants (a's, b's and c's). The $3n$ condition to find the $3n$ unknowns are given by the following equations.

1. At the interior knots, the function values must be equal.

$$a_{i-1}x_{i-1}^2 + b_{i-1}x_{i-1} + c_{i-1} = f(x_{i-1})$$

$$a_i x_{i-1}^2 + b_i x_{i-1} + c_i = f(x_{i-1})$$

for $i = 2, n$ giving raise to $2(n - 1) = 2n - 2$ conditions.

2. The first and last functions must pass through the two end points x_0 and x_n.

$$a_1 x_0^2 + b_1 x_0 + c_1 = f(x_0)$$

$$a_n x_n^2 + b_n x_n + c_n = f(x_n)$$

giving 2 conditions.

3. At the interior knots, the first derivatives $f'(x) = 2ax + b$ must be equal.

$$2a_{i-1}x_i + b_{i-1} = 2a_i x_i + b_i$$

for $i = 2$ to n giving $n - 1$ conditions

4. Assume that the second derivative is zero at the first point x_0. Then the first two points are connected by a straight line, so $a_1 = 0$.

Thus solving $(2n - 2) + 2 + (n - 1) + 1 = 3n$ conditions above we get the $3n$ unknown constants a, b, c's. The drawback is the straight line connecting the first two points.

Cubic Splines

A spline of at least $(m + 1)$ order should be used in order to make sure that the mth derivatives are continuous at knots. Cubic splines employing third order polynomials are very popular in practice because they ensure that the first and second derivatives are continuous.

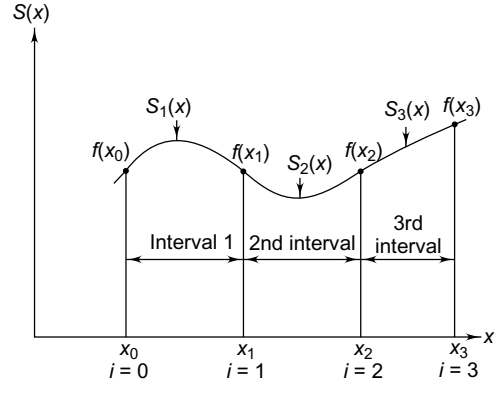


Fig. 32.16 Piecewise cubic spline approximation

Let us fit a (different) third order polynomial

$$S_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i \qquad (3)$$

for **each** of the n intervals between the $(n + 1)$ data points (knots) $(i = 0, 1, 2, \ldots n)$. The problem is to find the $4 \times n = 4n$ unknown constants

($a_i's$, $b_i's$, $c_i's$ and $d_i's$). We have the following conditions

a. $2n-2$ conditions since the function values must be equal at the $(n-1)$ interior knots.
b. 2 conditions since the first and last functions must pass through the two end points x_0 and x_n.
c. $(n-1)$ conditions since the first derivatives must be equal at the $(n-1)$ interior points.
d. $(n-1)$ conditions since the second derivatives must be equal at the $(n-1)$ interior knots.

The above conditions are summed to $(2n-2)+(2)+(n-1)+(n-1)=4n-2$, thus short of 2 more conditions to solve the $4n$ unknown constants. Two end point constraint are assumed as follows:

Natural spline for which the end point constraints are assumed as "second derivatives at the end knots are zero" i.e., $S''(x_0)=S''(x_n)=0$. Natural splines are more frequently used in which the end cubic (in the first and last intervals) approach linearity (straight line) at their extremities. This matches precisely with the drafting spline.

The cubic spline for *each* interval (x_{i-1}, x_i) is given by

$$S_i(x)=\frac{f''(x_{i-1})}{6(x_i-x_{i-1})}(x_i-x)^3 + \frac{f''(x_i)}{6(x_i-x_{i-1})}(x-x_{i-1})^3 + \left[\frac{f(x_{i-1})}{(x_i-x_{i-1})}-\frac{f''(x_{i-1})(x_i-x_{i-1})}{6}\right](x_i-x)+ + \left[\frac{f(x_i)}{(x_i-x_{i-1})}-\frac{f''(x_i)\cdot(x_i-x_{i-1})}{6}\right](x-x_{i-1}). \tag{4}$$

Equation (4) contains only two unknowns $f''(x_{i-1})$ and $f''(x_i)$, the second derivatives at the end points of the interval (x_{i-1}, x_i), which are determined from the following equation.

$$(x_i-x_{i-1})f''(x_{i-1})+2(x_{i+1}-x_{i-1})f''(x_i)+(x_{i+1}-x_i)f''(x_{i+1}) = \frac{6}{(x_{i+1}-x_i)}[f(x_{i+1})-f(x_i)] + \frac{6}{(x_i-x_{i-1})}[f(x_{i-1})-f(x_i)] \tag{5}$$

Equation (5) written for the $(n-1)$ interior points results in $(n-1)$ simultaneous equations for the second derivatives. Note that these $(n-1)$ equations form a tridiagonal system which can be solved in an extremely efficient manner using computer algorithms.

WORKED OUT EXAMPLES

Example 1: Fit a linear spline to the following data:

x:	1	3	6	8
$y=f(x)$:	4	5.5	7	9.5

Estimate the value at $x=2, 4, 7$.

Solution: There are $n=4$ points and 3 intervals. Assume that

$x_0=1, \quad x_1=3, \quad x_2=6, \quad x_3=8,$
$f(x_0)=4, \quad f(x_1)=5.5, \quad f(x_2)=7, \quad f(x_3)=9.5$

The three first order linear splines in the 3 intervals are:

$f(x)=f(x_0)+m_0(x-x_0),\ x_0\le x\le x_1$
$f(x)=f(x_1)+m_1(x-x_1),\ x_1\le x\le x_2$
$f(x)=f(x_2)+m_2(x-x_2),\ x_2\le x\le x_3$

Here $m_i=\frac{f(x_{i+1})-f(x_i)}{x_{i+1}-x_i}$ is the slope of straight line connecting the two points. For the given data

$$m_0=\frac{f(x_1)-f(x_0)}{x_1-x_0}=\frac{5.5-4}{3-1}=\frac{1.5}{2}=0.75$$

$$m_1=\frac{f(x_2)-f(x_1)}{x_2-x_1}=\frac{7-5.5}{6-3}=\frac{1.5}{3}=0.5$$

$$m_2=\frac{f(x_3)-f(x_2)}{x_3-x_2}=\frac{9.5-7}{8-6}=\frac{2.5}{2}=1.25$$

Thus the three linear splines are

$f(x)=4+0.75(x-1)$, when $1\le x\le 3$
$f(x)=5.5+0.5(x-3)$, when $3\le x\le 6$
$f(x)=7+1.25(x-6)$, when $6\le x\le 8$

Since $x=2$ lies in the first interval $1\le x\le 3$, we

get $f(2)$ from the first equation.

$$f(2) = 4 + 0.75(2 - 1) = 4.75$$

Similarly $f(7) = 7 + 1.25(7 - 6) = 8.25$.
Since $x = 4$ lies in the second interval $3 \le x \le 6$, we get $f(4)$ from the second equation:

$$f(4) = 5.5 + 0.5(4 - 3) = 6$$

Example 2: Fit quadratic splines to the following data. Use the results to estimate the value at $x = 2, 4, 7$.

x:	1	3	6	8
$f(x)$:	4	5.5	7	9.5

Solution: For the 4 data points in 3 intervals we fix three quadratic splines of the form

$$S_i(x) = a_i x^2 + b_i x + c_i$$

we have to determine $3 \times 3 = 9$ unknown $a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3$. We need $3 \cdot 3 = 9$ equations or conditions to determine these 9 unknowns. There are two interior points x_1 and x_2.

1. Function values must be equal at the two interior points, giving rise to 4 equations.

$$a_{i-1}x_{i-1}^2 + b_{i-1}x_i + c_{i-1} = f(x_{i-1})$$
$$a_i x_{i-1}^2 + b_i x_{i-1} + c_i = f(x_{i-1})$$

for i = 2 to 3 .

For $i = 2$, $a_1 x_1^2 + b_1 x_1 + c_1 = f(x_1)$

$$a_2 x_1^2 + b_2 x_1 + c_2 = f(x_1)$$

i.e., $9a_1 + 3b_1 + c_1 = 5.5$ (1)

$$9a_2 + 3b_2 + c_2 = 5.5 \quad (2)$$

For $i = 3$, $a_2 x_2^2 + b_2 x_2 + c_2 = f(x_2)$

$$a_3 x_2^2 + b_3 x_2 + c_3 = f(x_2)$$

i.e., $36a_2 + 6b_2 + c_2 = 7$ (3)

$$36a_3 + 6b_3 + c_3 = 7 \quad (4)$$

2. The first and last functions must pass through the end points x_0 and x_0 giving rise to 2 equations.

$$a_1 x_0^2 + b_1 x_0 + c_1 = f(x_0)$$
$$a_n x_n^2 + b_n x_n + c_n = f(x_n)$$

i.e.,

$$a_1 + b_1 + c_1 = 4 \quad (5)$$

$$64a_3 + 8b_3 + c_3 = 9.5 \quad (6)$$

3. The first derivatives at the interior knots must be equal.

$$f'(x) = 2ax + b$$
$$2a_{i-1}x_i + b_{i-1} = 2a_i x_i + b_i$$

for i = 2, 3

$$2a_1 x_2 + b_1 = 2a_2 x_2 + b_2$$
$$2a_2 x_3 + b_2 = 2a_3 x_3 + b_3$$

i.e., $6a_1 + b_1 = 6a_2 + b_2$ (7)

$$12a_2 + b_2 = 12a_3 + b_3 \quad (8)$$

4. Assume that the second derivative is zero at the first point i.e.,

$$a_1 = 0 \quad (9)$$

Using (9) $a_1 = 0$ in (1) and (5) we get

$$3b_1 + c_1 = 5.5$$
$$b_1 + c_1 = 4$$

solving $b_1 = 0.75$, $c_1 = 3.25$

From (2) and (3) we get

$$\begin{array}{r} 9a_2 + 3b_2 + c_2 = 5.5 \\ 36a_2 + 6b_2 + c_2 = 7 \\ \hline 27a_2 + 3b_2 \quad = 1.5 \end{array}$$

Form (7) $6a_2 + b_2 = 0 + b_1 = 0.75$
Solving $a_2 = \frac{-0.75}{9}$, $b_2 = 1$, $c_2 = 4$
From (4) and (6)

$$\begin{array}{r} 36a_3 + 6b_3 + c_3 = 7 \\ 64a_3 + 8b_3 + c_3 = 9.5 \\ \hline \end{array}$$

Subtracting $28a_3 + 2b_3 = 2.5$

From (8): $12a_3 + b_3 = 12a_2 + b_2 = 0$
Solving $a_3 = \frac{2.5}{4}$, $b_3 = -7.5$, $c_3 = 29.5$
Thus the values of the 9 coefficients are
$a_1 = 0$, $b_1 = 0.75$, $c_1 = 3.25$,
$a_2 = \frac{-0.75}{9}$, $b_2 = 1$, $c_2 = 4$,
$a_3 = \frac{2.5}{4}$, $b_3 = -7.5$, $c_3 = 29.5$,
Then the three required quadratic splines are
$S_1(x) = 0 + 0.75x + 3.25$, in $1 \le x \le 3$
$S_2(x) = \frac{-0.75}{9}x^2 + x + 4$, in $3 \le x \le 6$

$S_3(x) = \frac{2.5}{4}x^2 - 7.5x + 29.5$, in $6 \le x \le 8$
At $x = 2$, $S_1(2) = 4.75$
At $x = 4$, $S_2(4) = 6.6666$
At $x = 7$, $S_3(7) = 7.625$

Example 3: Fit cubic splines to the following data and estimate the value at $x = 2, 4, 7$ and $y'(4)$.

x:	1	3	6	8
$y = f(x)$:	4	5.5	7	9.5

Solution: Assume a cubic spline a third order polynomial

$$S_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i$$

For the 4 data points and 3 intervals, we have to determine $4 \times 3 = 12$ unknown constants $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2; a_3, b_3, c_3, d_3$.

1. The second derivatives at the end knots are zero; i.e., $f''(1) = f''(8) = 0$.
2. The second derivatives at the end of each interval (at the two interior points) are given by

$$\begin{aligned}&(x_i - x_{i-1})f''(x_{i-1}) + 2(x_{i+1} - x_{i-1})f''(x_i)\\ &+(x_{i+1} - x_i)f''(x_{i+1})\\ &= \frac{6}{(x_{i+1} - x_i)}[f(x_{i+1}) - f(x_i)] +\\ &+\frac{6}{(x_i - x_{i-1})}[f(x_{i-1}) - f(x_i)] \qquad (1)\end{aligned}$$

For $i = 1$, from (i) we have

$$\begin{aligned}&(x_i - x_0)f''(x_0) + 2(x_2 - x_0)f''(x_1) + (x_2 - x_1)f''(x_2)\\ &= \frac{6}{(x_2 - x_0)}[f(x_2) - f(x_1)] + \frac{6}{(x_1 - x_0)}[f(x_0) - f(x_1)]\end{aligned}$$

with $x_0 = 1, x_1 = 3, x_2 = 6$, $f_0 = 4$, $f_1 = 5.5$, $f_2 = 7$, we have

$$\begin{aligned}&(3-1)f''(1) + 2(6-1)f''(3) + (6-3)f''(6)\\ &= \frac{6}{(6-1)}[7 - 5.5] + \frac{6}{(3-1)}[4 - 5.5].\end{aligned}$$

For natural spline, $f''(1) = 0$. So above equation reduces to

$$10f''(3) + 3f''(6) = -2.7 \qquad (2)$$

Similarly for $i = 2$, from (1) we have

$$\begin{aligned}&(x_2 - x_1)f''(x_1) + 2(x_3 - x_1)f''(x_2) + (x_3 - x_2)\, f''(x_3)\\ &= \frac{6}{(x_3 - x_2)}[f(x_3) - f(x_2)] +\\ &+\frac{6}{(x_2 - x_1)}[f(x_1) - f(x_2)]\end{aligned}$$

with $x_1 = 3, x_2 = 6, x_3 = 8$, $f_1 = 5.5$, $f_2 = 7$, $f_3 = 9.5$, we get $(6-3)f''(3) + 2(8-3)f''(6) + (8-6)f''(8)$

$$= \frac{6}{(8-6)}[9.5 - 7] + \frac{6}{(6-3)}[5.5 - 7].$$

For natural spline $f''(8) = 0$. So above equation reduces to

$$3f''(3) + 10f''(6) = 4.5 \qquad (3)$$

solving (2) and (3) we get
$f''(3) = -0.4451$, $f''(6) = 0.58352$

The equation of the cubic spline is

$$\begin{aligned}S_i = &\frac{f''(x_{i-1})}{6(x_i - x_{i-1})}(x_i - x)^3 + \frac{f''(x_i)}{6(x_i - x_{i-1})}(x - x_{i-1})^3\\ &+\left[\frac{f(x_{i_{-1}})}{(x_i - x_{i-1})} - \frac{f''(x_{i-1})(x_i - x_{i-1})}{6}\right](x_i - x)\\ &+\left[\frac{f(x_i)}{(x_i - x_{i-1})} - \frac{f''(x_i)(x_i - x_{i-1})}{6}\right](x - x_{i-1})\end{aligned} \qquad (4)$$

For $i = 1$,

$$\begin{aligned}S_1(x) = &\frac{f''(x_0)}{6(x_1 - x_0)}(x_1 - x)^3 + \frac{f''(x_1)}{6(x_1 - x_0)}(x - x_0)^3\\ &+\left[\frac{f(x_0)}{x_1 - x_0} - \frac{f''(x_0)(x_1 - x_0)}{6}\right](x_1 - x)\\ &+\left[\frac{f(x_1)}{x_1 - x_0} - \frac{f''(x_1)(x_1 - x_0)}{6}\right](x - x_0)\end{aligned}$$

with $x_0 = 1, x_1 = 3, x_2 = 6$, $f_0 = 4$, $f_1 = 5.5$, $f_2 = 7$, $f''(1) = 0$, $f''(3) = -0.4451$, $f''(6) = 0.58352$ the above equation reduces to

$$S_1(x) = 0 + \frac{(-0.4451)}{6(3-1)}(x - 1)^3 +$$

$$+\left[\frac{4}{(3-1)}-0\right](3-x)$$
$$+\left[\frac{5.5}{(3-1)}-(-0.4451)\frac{(3-1)}{6}\right](x-1) \qquad (5)$$

Thus the cubic spline in the first interval [1, 3] is

$$S_1(x) = -0.0371(x-1)^3 + 2(3-x) + 2.898(x-1)$$

For $i = 2$, form (4) we get

$$S_2(x) = \frac{f''(x_1)}{6(x_2-x_1)}(x_2-x)^3 + \frac{f''(x_2)}{6(x_2-x_1)}(x-x_1)^3$$
$$+\left[\frac{f(x_1)}{(x_2-x_1)} - \frac{f''(x_1)(x_2-x_1)}{6}\right](x_2-x)$$
$$+\left[\frac{f(x_2)}{x_2-x_1} - \frac{f''(x_2)(x_2-x_1)}{6}\right](x-x_1)$$

with the substitution of data, we get

$$S_2(x) = \frac{-0.4451}{6(6-3)}(6-x)^3 + \frac{0.58352}{6(6-3)}(x-3)^3$$
$$+\left[\frac{5.5}{(6-3)} - \frac{(-0.4451)(6-3)}{6}\right](6-x)$$
$$+\left[\frac{7}{6-3} - \frac{(0.58352)(6-3)}{6}\right](x-3) \qquad (6)$$

Thus the second cubic spline in the second interval [3, 6] is

$$S_2(x) = -0.02473(6-x)^3 + 0.03242(x-3)^3$$
$$+(2.05588)(6-x) + (2.04154)(x-3)$$

Similarly for $i = 3$, from (4) we get

$$S_3(x) = \frac{f''(x_2)}{6(x_3-x_2)}(x_3-x)^3 + \frac{f''(x_3)}{6(x_3-x_2)}$$
$$+\left[\frac{f(x_2)}{x_3-x_2} - \frac{f''(x_2)}{6}(x_3-x_2)\right](x_3-x)$$
$$+\left[\frac{f(x_3)}{x_3-x_2} - \frac{f''(x_3)(x_3-x_2)}{6}\right](x-x_2)$$

with the given data

$$f_3(x) = \frac{0.58352}{6(8-6)}(8-x)^3 + 0$$
$$+\left[\frac{7}{8-6} - \frac{0.58352}{6}(8-6)\right](8-x)$$
$$+\left(\frac{8}{8-6} - 0\right)(x-6)$$

Thus the third cubic spline in the 3rd interval [6, 8] is

$$S_3(x) = 0.04863(8-x)^3 + 3.3055(8-x)$$
$$+4(x-6) \qquad (7)$$

Now using (5), (6), (7), we get

$$f_1(2) = 4.8609$$
$$f_2(4) = 5.98788$$
$$f_3(7) = 7.35413$$

Since 4 is in the interval [3, 6] differentiate $f_2(x)$ to find $f'(4)$. So

$$f_1'(x) = +0.07419(6-x)^2 + 0.09726(x-3)^2$$
$$-2.05588 + 2.04154$$

Thus $f'(4) = 0.37968$

EXERCISE

1. Obtain the natural cubic spline approximation for the following data

x:	0	1	2	3
$f(x)$:	1	2	33	244

Ans. $f_1(x) = -4x^3 + 5x + 1$ in [0, 1]
$f_2(x) = 50x^3 - 162x^2 + 167x - 53$ in [1, 2]
$f_3(x) = -46x^3 + 414x^2 - 985x + 715$ in [2, 3]

2. Fit a natural cubic spline to the following data

x:	1	2	3	4
y:	1	5	11	8

Hence compute $y(1.5)$, $y'(2)$

Ans. $y(1.5) = 103/40$, $y'(2.0) = 94/15$
$f_1(x) = \frac{1}{15}[17x^3 - 51x^2 + 94x - 45]$ in $1 \le x \le 2$
$f_2(x) = \frac{1}{15}[-55x^3 + 381x^2 - 770x + 531]$ in $2 \le x \le 3$
$f_3(x) = \frac{1}{15}[38x^3 - 456x^2 + 1741x - 1980]$ in $3 \le x \le 4$

3. Fit a natural cubic spline and evaluate the spline at 0.66, 1.75.

x:	0.0	1.0	1.5	2.25
$f(x)$:	2.000	4.4366	6.7134	13.9130

Ans. $f_1(x) = 0.3820(x-0)^3 + 0(x-0)^2$
$+2.0546(x-0) + 2.000$ in $[0, 1.0]$
$f_2(x) = 3.1199(x-1)^3 + 1.146(x-1)^2$
$+3.2005(x-1) + 4.4366$ in $[1, 1.5]$
$f_3(x) = -2.5893(x-1.5)^3 + 5.8259(x-1.5)^2$
$+6.6866(x-1.5) + 6.7134$ in $[1.5, 2.25]$
$f_1(0.66) = 3.4659$ Exact: 3.4340
$f_3(1.75) = 8.7087$ Exact: 8.4467
Exact $f(x) = 2e^x - x^2$

4. Fit a natural cubic spline to the following data

x:	0	1	2	3
y:	0	0.5	2	1.5

Ans. $S_1(x) = 0.4x^3 + 0.1x$ in $0 \le x \le 1$
$S_2(x) = -(x-1)^3 + 1.2(x-1)^2+$
$+1.3(x-1) + 0.5$ in $1 \le x \le 2$
$S_3(x) = 0.6(x-2)^3 - 1.8(x-2)^2+$
$+0.7(x-2) + 2.0$ in $2 \le x \le 3$.

5. Fit a cubic natural spline

x:	0	1	2
y:	4	1	2

Ans. $s_1(x) = 4 - 4x + x^3$ in $[0, 1]$
$s_2(x) = 1 - (x-1) + 3(x-1)^2 - (x-1)^3$ in $[1, 2]$

6. Fit (a) linear spline (b) quadratic spline (c) cubic natural spline to the following data.

x:	3.0	4.5	7.0	9.0
$y = f(x)$:	2.5	1.0	2.5	0.5

Evaluate the function at $x = 5$.

Ans. (a) Linear splines
$f_1(x) = 2.5 - (x-3)$ in $[3, 4.5]$
$f_2(x) = 1.0 + 0.6(x-4.5)$ in $[4.5, 7]$
$f_3(x) = 2.5 - 1(x-7)$ in $[7, 9]$
$f_2(5) = 1.3$
(b) Quadratic splines
$f_1(x) = -x + 5.5$ in $[3, 4.5]$
$f_2(x) = 0.64x^2 - 6.76x + 18.46$ in $[4.5, 7.0]$
$f_3(x) = -1.6x^2 + 24.6x - 91.3$ in $[7.0, 9.0]$
$f_2(5) = 0.66$
(c) Cubic natural splines
In $[3, 4.5]$
$f_1(x) = 0.193939(x-3)^3 + 1.66667(4.5-x)$
$+0.23030(x-3)$
$f_2(x) = 0.1163647(7-x)^3 - 0.116364(x-$
$4.5)^3 \quad -0.327273(7-x) + 1.727273(x-4.5)$
in $[4.5, 7]$
$f_3(x) = -0.145455(9-x)^3 + 1.831818(9-x)$
$+0.25(x-7)$ in $[7, 9]$
$f_2(5) = 1.1255$

7. Fit natural cubic splines for the following data.

x:	−10	0	10	20	30
y:	.99815	.99987	.99973	.99823	.99567

Ans. In $-10 \le x \le 0$.
$f_1(x) = -0.00000042(x+10)^3 +$
$0.000214(x+10) + .99815$
In $0 \le x \le 10$
$f_2(x) = 0.00000024x^3 - 0.0000126x^2 +$
$.000088x + .99987$
In $10 \le x \le 20$
$f_3(x) = -0.00000004(x-10)^3 -$
$0.000054(x-10)^2 - 0.000092(x-10) + .9973$
In $20 \le x \le 30$
$f_4(x) = .00000022(x-20)^3 - 0.0000066(x-$
$20)^2 - 0.000212(x-20) + .99823$

32.16 NUMERICAL METHODS IN LINEAR ALGEBRA: GAUSS-SEIDEL METHOD

Solutions to system of non-homogeneous linear equations in n unknowns can be obtained by direct (or reduction or elimination) methods such as Cramer's rule, matrix inversion method, Gaussian elimination, Gauss-Jordan method, Cholosky's (Crout's) method. The solutions can also be obtained by indirect (or iterative methods) which include the Gauss-Seidel method, (South Well's) relaxation method, Sweep method.

Diagonal:

A system of simultaneous linear equations is called a diagonal system if in each equation the coefficient of a different unknown is greater in absolute value than the sum of the absolute values of the other coefficients.

Usually, the large coefficient appears in the main diagonal position a_{ii}. Gauss-Seidel method is an iterative method which gives approximate solution by successive approximations. This method will always converge rapidly if the system is diagonal.

Gauss-Seidel method

Step I: Solve each of the n equations of the system for the unknown with the largest coefficient. Suppose a_{11} is the largest coefficient of x_1 in the first equation, then divide Equation (1) throughout by a_{11}, resulting in a new equation with x_1 expressed in terms of the remaining variables $x_2, x_3, \ldots, x_n$. Similarly, dividing the second equation by a_{22}, 3rd equation by $a_{33}\ldots$ and nth equation by a_{nn}, Then

$$\left.\begin{aligned} x_1 &= a_{12}^* x_2 + a_{13}^* x_3 + \cdots + a_{1n}^* x_n + k_1 \\ x_2 &= a_{21}^* x_1 + a_{23}^* x_3 + \cdots + a_{2n}^* x_n + k_2 \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ x_n &= a_{n1}^* x_1 + a_{n2}^* x_2 + \cdots + a_{n,n-1}^* x_{n-1} + k_n \end{aligned}\right\} \quad (1)$$

Step II Assume an initial solutions (guess) $(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, \ldots, x_n^{(0)})$. Here 0 indicates the starting solution. It may be taken as $(0, 0, 0, \ldots, 0)$.

Step III Substitute $x_i^{(0)}$ in the right hand side members of (1) which results in the new values $x_i^{(1)}$, which is first approximation. Now substitute $x_i^{(1)}$ in the R.H.S. of (1) which yields the second approximation $x_i^{(2)}$. Repeating this procedure, we arrive at $(x_1^{(m)}, x_2^{(m)}, \ldots, x_n^{(m)})$ at the mth iteration. Step III is repeated until required accuracy is obtained.

Note 1: Always use the last calculated (i.e., latest) values for the other unknown, say

$$x_2^{(k+1)} = a_{21}^* x_1^{(k+1)} + a_{23}^* x_3^{(k)} + \cdots + a_{2n}^* x_n^{(k)} + k_1$$

Note 2: Choice of initial value will not affect the solution (we get same solution) but will affect the number of iterations for convergence.

Worked Out Examples

Example: Using Gauss-Seidel method, solve the following system of equations starting with initial solution as (a) (0, 0, 0) (b) $\left(\frac{9}{5}, \frac{4}{5}, \frac{6}{5}\right)$.

$$\begin{bmatrix} 5 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ -6 \end{bmatrix}$$

Solution: **(a)** System is

$$\begin{aligned} 5x_1 - x_2 &= 9 \\ -x_1 + 5x_2 - x_3 &= 4 \\ -x_2 + 5x_3 &= -6 \end{aligned}$$

Step I: Rewriting the system

$$\left.\begin{aligned} x_1 &= \frac{9}{5} + \frac{1}{5}x_2 \\ x_2 &= \frac{4}{5} + \frac{1}{5}x_1 + \frac{1}{3}x_3 \\ x_3 &= -\frac{6}{5} + \frac{1}{5}x_2 \end{aligned}\right\} (*)$$

Step II: Assume initial solution as

$$(x_1 = 0, x_2 = 0, x_3 = 0)$$

Step III: Using the initial solution in R.H.S. of $(*)$

$$x_1^{(1)} = \frac{9}{5} + \frac{1}{5}\cdot 0 = \frac{9}{5} = 2.032$$

$$x_2^{(1)} = \frac{4}{5} + \frac{1}{5}\cdot\frac{9}{5} = \frac{29}{25} = 1.0128$$

$$x_3^{(1)} = \frac{-6}{5} + \frac{1}{5}\cdot\frac{29}{25} = -\frac{121}{125} = -0.99744$$

Note that in the calculation of $x_2^{(1)}$, the latest (available) value $x_1^{(1)} = \frac{9}{5}$ is used (but not $x_1^{(0)} = 0$). Similarly, in the calculation of $x_3^{(1)}$ the latest value $x_2^{(1)} = \frac{29}{25}$ is used (but not $x_2^{(0)} = 0$). Thus the first approximation is $\left(\frac{9}{5}, \frac{29}{25}, \frac{-121}{125}\right)$.
Second iteration:

$$x_1^{(2)} = \frac{9}{5} + \frac{1}{5}\left(\frac{29}{25}\right) = \frac{254}{125} = 2.032$$

$$x_2^{(2)} = \frac{4}{5} + \frac{1}{5}\left(\frac{254}{125}\right) - \frac{1}{5}\left(\frac{121}{125}\right) = \frac{633}{625} = 1.0128$$

$$x_3^{(2)} = \frac{-6}{5} + \frac{1}{5}\cdot\frac{633}{625} = \frac{-3117}{3125} = -0.99744$$

Third iteration:

$$\begin{cases} x_1^{(3)} = \dfrac{9}{5} + \dfrac{1}{5}(1.0128) = 2.00256 \\ x_2^{(3)} = \dfrac{4}{5} + \dfrac{1}{5}(2.00256) - \dfrac{1}{3}(-0.99744) = 0.868032 \\ x_3^{(3)} = \dfrac{-6}{5} + \dfrac{1}{5}(.868032) = -1.0263936. \end{cases}$$

Similarly, we get at the 4th iteration $x_1^{(4)} = 2.002048$, $x_2^{(4)} = 1.0000829$, $x_3^{(4)} = -0.99998$

At 5th iteration: $x_1^{(5)} = 2.0000165$, $x_2^{(5)} = 1.0000073$, $x_3^{(5)} = -0.99999$

(b) Initial approximation: $x_1^{(0)} = \frac{9}{5}$, $x_2^{(0)} = \frac{4}{5}$, $x_3^{(0)} = \frac{-6}{5}$

1st iteration : 1.96, 0.952, −1.0096

2st iteration : 1.9904, 0.99616, −1.000768

3st iteration : 1.999232, 0.99969, −1.000064

4st iteration : 1.999938, 0.99997, 1.0000049

5st iteration : 1.999994, 0.99999, −1.0000004

Note 1: Exact solution is 2, 1, −1

EXERCISE

Solve the following system of equations by Gauss-Seidel method:

1. $10x_1 + 8x_2 + 6x_3 = 16.4$, $10x_2 + 8x_3 + 4x_4 = -3.8$, $2x_1 + 10x_3 + 2x_4 = 36.9$, $x_1 + 6x_3 + 10x_4 = 30.9$

Ans. $x_1 = 2.4$, $x_2 = -3.2$, $x_3 = 3$, $x_4 = 1.05$. At 6th iteration: $x_1 = 2.3721$, $x_2 = -3.19096$, $x_3 = 3.00525$, $x_4 = 1.049632$

2. $8x_1 + x_2 - x_3 = 8$, $2x_1 + x_2 + 9x_3 = 12$, $x_1 - 7x_2 + 2x_3 = -4$.

Ans. $x_1 = 1, x_2 = 1, x_3 = 1$

3. Start with (2, 2, −1) and solve $5x_1 - x_2 + x_3 = 10$, $2x_1 + 4x_2 = 12$, $x_1 + x_2 + 5x_3 = -1$.

Ans. $x_1 = 2.5555$, $x_2 = 1.7222$, $x_3 = -1.0555$

4. $10x_1 + x_2 + x_3 = 12$, $2x_1 + 10x_2 + x_3 = 13$, $2x_1 + 2x_2 + 10x_3 = 14$.

Ans. $x_1 = 1, x_2 = 1, x_3 = 1$

5. $10x_1 - 2x_2 - x_3 - x_4 = 3$, $-2x_1 + 10x_2 - x_3 - x_4 = 15$, $-x_1 - x_2 + 10x_3 - 2x_4 = 27$, $-x_1 - x_2 - 2x_3 + 10x_4 = -9$.

Ans. $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 0$

6. $4x_1 + 2x_2 + x_3 = 11$, $-x_1 + 2x_2 = 3$, $2x_1 + x_2 + 4x_3 = 16$.

Ans. $x_1 = 1, x_2 = 2, x_3 = 3$

7. $20x_1 + x_2 - 2x_3 = 17$, $3x_1 + 20x_2 - x_3 = -18$, $2x_1 - 3x_2 + 20x_3 = 25$.

Ans. $x_1 = 1, x_2 = -1, x_3 = 1$.

8. $3x_1 - 0.1x_2 - 0.2x_3 = 7.85$, $0.1x_1 + 7x_2 - 0.3x_3 = -19.3$, $0.3x_1 - 0.2x_2 + 10x_3 = 71.4$.

Ans. $x_1 = 3, x_2 = -2.5, x_3 = 7$

9. $27x_1 + 6x_2 - x_3 = 85$, $6x_1 + 15x_2 + 2x_3 = 72$, $x_1 + x_2 + 54x_3 = 110$.

Ans. $x_1 = 2.4255$, $x_2 = 3.5730$, $x_3 = 1.9260$

10. $x_1 - 8x_2 + 3x_3 = -4$, $2x_1 + x_2 + 9x_3 = 12$, $8x_1 + 2x_2 - 2x_3 = 8$.

Ans. $x_1 = 1, x_2 = 1, x_3 = 1$.

32.17 LARGEST EIGEN VALUE AND THE CORRESPONDING EIGEN VECTOR: BY POWER METHOD

Bounds for eigen values λ of an arbitrary n rowed square matrix $A = (a_{jk})$ are given by circular disks from **Gershgorin theorem**

$$|a_{kk} - \lambda| \le |a_{k1}| + |a_{k2}| + \cdots + |a_{k,k-1}| + |a_{k,k+1}| + \cdots + |a_{kn}|$$

for some k where $1 \le k \le n$. Thus all the eigen values lie within these n circular disks (some may be identical) with centres at the diagonal elements a_{kk} and radii equal to the sum of the absolute values of the elements of the kth row baring a_{kk} (except a_{kk}).

Ex: $A = \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$, eigen values 4, 6. Gershgorin circular disks : 2 : centres 5, 5, radii 1, 1.

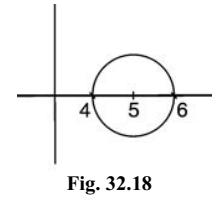


Fig. 32.18

However, to find the approximate values of all the eigen values and eigen vactors, iteration mathod or power mathod is used. Power method is particularly used when only the largest and/or the smallest eigen

values of a matrix are desired. The advantage of this method is that the eigen vector associated with the largest eigen value is also obtained simultaneously. Also the intermediate (remaining) eigen values can also be found, after "sweeping" the already found largest and/or smallest eigen values.

Power Method

Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigen values and $V_1, V_2, \ldots V_n$ be the corresponding eigen vactors of a n-rowed square matrix A. Assume that λ_1 is the absolutely largest eigen value of A i.e.,

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \ldots, \geq |\lambda_n|.$$

Consider $V = \alpha_1 V_1 + \alpha_2 V_2 + \cdots + \alpha_n V_n$ where $\alpha_1, \alpha_2, \cdots \alpha_n$ are scalars. Then

$$\begin{aligned} AV &= \alpha_1 AV_1 + \alpha_2 AV_2 + \cdots + \alpha_n AV_n \\ &= \alpha_1\lambda_1 V_1 + \alpha_2\lambda_2 V_2 + \cdots + \alpha_n\lambda_n V_n \end{aligned}$$

Pre multiplying by A

$$A^2V = \alpha_1\lambda_1^2 V + \alpha_2\lambda_2^2 V_2 + \cdots + \alpha_n\lambda_n^2 V_n.$$

Thus for any positive integer p

$$A^pV = \alpha_1\lambda_1^p V_1 + \alpha_2\lambda_2^p V_2 + \cdots + \alpha_n\lambda_n^p V_n$$

$$A^pV = \lambda_1^p \left[\alpha_1 V_1 + \alpha_2 \left(\frac{\lambda_2}{\lambda_1}\right)^p V_2 + \cdots + \alpha_n \left(\frac{\lambda_n}{\lambda_1}\right)^p V_n\right]$$

provided $\alpha_1 \neq 0$. All terms in R.H.S. except the first term have limit zero since λ_1 is the dominant value and p is large.

Now

$$\frac{A^{P+1}V}{A^pV} = \frac{\lambda_1^{p+1} \cdot \alpha_1 V_1}{\lambda_1^p \alpha_1 V_1} = \lambda_1$$

Also the required eigen vector corresponding to this largest eigen value is

$$\lambda_1^{-p} A^p V = \alpha_1 V_1.$$

Power method procedure

Step I: Choose an arbitrary real vector $x_0 \neq 0$. Generally, X_0 is chosen as

$$X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

or any other value like $X_0 = [10000]^T$ etc.

Step II: Compute $X_1 = AX_0$, $X_2 = AX_1$, $X_3 = AX_2, \ldots, X_s = AX_{s-1}$. Put $X = X_{s-1}$, $Y = X_s$

Step III: Compute $m_0 = x^T x, m_1 = x^T y, m_2 = Y^T Y$

Step IV: Largest eigen value $= \lambda_1 = \frac{m_1}{m_0}$,

ϵ, the error in $\lambda_1 : |\epsilon| \leq \sqrt{\frac{m_2}{m_0} - \lambda_1^2}$,

Eigen vector corresponding to λ_1 is $Y = X_s$.

[This eigen vector can be normalized either by dividing the eigen vector by the magnitude of its first component or by its largest component or by normalizing to a unit length.]

Determination of Smallest Eigen Value

$$\begin{aligned} AX &= \lambda X \\ A^{-1}AX &= A^{-1}\lambda X \\ \frac{1}{\lambda}X &= A^{-1}X \end{aligned}$$

Thus if λ is eigen value of A, then the reciprocal $\frac{1}{\lambda}$ is the eigen value of A^{-1}. Then the reciprocal of the largest eigen value of A^{-1} will be the smallest eigen value of A.

Procedure:

Step I: Determine A^{-1}

Step II: Calculate λ^* dominant eigen value of A^{-1}

Step III: $\frac{1}{\lambda^*}$ is the smallest eigen value of A.

Determination of Intermediate (remaining) Eigen Values

$$Ax = \lambda x$$

$$(A - \lambda_1 I)x = (\lambda - \lambda_1)x$$

Then $(\lambda - \lambda_1)$ is an eigen value of $(A - \lambda_1 I)$. Thus 'sweeping' the dominant eigen value, $(A - \lambda_1 I)$ will have remaining eigen values dominant, say λ_2 which can be found by power method.

WORKED OUT EXAMPLES

Example 1: Find the largest eigen value and the corresponding eigen vector of the matrix

$$A = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$$

Find the error in the value of the largest eigen value.

Solution: Choose the initial vector $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$,

Then $X_1 = AX_0 = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$.

Now $X_2 = AX_1 = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$.

Then $X_3 = Ax_2 = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 4 \end{bmatrix}$.

continuing these iterations, we get

$$X_4 = AX_3 = \begin{bmatrix} -4 \\ 6 \\ -4 \end{bmatrix}, \quad X_5 = AX_4 = \begin{bmatrix} 12 \\ -14 \\ 4 \end{bmatrix},$$

$$X_6 = AX_5 = \begin{bmatrix} -28 \\ 30 \\ -4 \end{bmatrix}, \quad X_7 = AX_6 = \begin{bmatrix} 60 \\ -62 \\ 4 \end{bmatrix},$$

$$X_8 = AX_7 = \begin{bmatrix} -124 \\ 126 \\ -4 \end{bmatrix} \text{ and } X_9 = AX_8 = \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix}.$$

Put $X = X_8$ and $Y = X_9$. Then

$$m_0 = X^T X = [-124 \;\; 126 \;\; -4] \begin{bmatrix} -124 \\ 126 \\ -4 \end{bmatrix} = 31268$$

$$m_1 = X^T y = [-124 \;\; 126 \;\; -4] \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix} = -63268$$

$$m_2 = Y^T Y = [252 \;\; -254 \;\; 4] \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix} = 128036.$$

The largest eigen value $= \lambda_1 = \frac{m_1}{m_0} = \frac{-63268}{31268}$

$$\lambda_1 = -2.0234105$$

(at the 9th iteration).
The corresponding eigen vector is

$$X_9 = \begin{bmatrix} 252 \\ -254 \\ 4 \end{bmatrix} \quad \text{or} \quad X_9 = \begin{bmatrix} 1 \\ -1.007936 \\ 0.0015 \end{bmatrix}$$

The error in the largest eigen value is ϵ

$$|\epsilon| \le \sqrt{\frac{m_2}{m_0} - \lambda_1^2} = \sqrt{\frac{128036}{31268} - (2.023415)^2}$$
$$= \sqrt{4.094793 - 4.094208} = \sqrt{0.00058473}$$
$$= 0.02418.$$

Note that the exact three eigen values of A are $-2, -1, 0$.

Example 2: Find the absolutely smallest eigen value of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Solution: We know that if λ is an eigen value of A then the reciprocal $\frac{1}{\lambda}$ is the eigen value of A^{-1}. Thus the largest eigen value of A^{-1} is the smallest eigen value of A.

Choose $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and apply power method, for A^{-1}. Now

$$B = A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X_1 = BX_0 = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 6 \\ 8 \\ 6 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}.$$

$$X_2 = B^2 X_0 = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 20 \\ 28 \\ 20 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 5 \\ 7 \\ 5 \end{bmatrix}$$

$$X_3 = BX_2 = B^3 X_0 = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 5 \\ 7 \\ 5 \end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix}34\\48\\34\end{bmatrix} = \frac{1}{4}\begin{bmatrix}17\\24\\17\end{bmatrix}$$

$$X_4 = B^4X_0 = \frac{1}{4}\begin{bmatrix}3&2&1\\2&4&2\\1&2&3\end{bmatrix}\frac{1}{4}\begin{bmatrix}17\\24\\17\end{bmatrix}$$

$$= \frac{1}{16}\begin{bmatrix}116\\260\\116\end{bmatrix} = \frac{1}{16}\begin{bmatrix}116\\164\\116\end{bmatrix}$$

$$X_4 = \frac{1}{4}\begin{bmatrix}29\\41\\29\end{bmatrix},$$

$$X_5 = bX_4 = \frac{1}{4}\begin{bmatrix}3&2&1\\2&4&2\\1&2&3\end{bmatrix}\frac{1}{4}\begin{bmatrix}29\\41\\29\end{bmatrix} = B^5X_0$$

$$X_5 = \frac{1}{16}\begin{bmatrix}198\\280\\198\end{bmatrix}.$$

$$X_6 = BX_5 = B^6X_0 = \frac{1}{64}\begin{bmatrix}1352\\1912\\1352\end{bmatrix} = \frac{1}{8}\begin{bmatrix}169\\239\\169\end{bmatrix}$$

Put $X = X_5, Y = X_6, m_0 = X^TX = \frac{1}{64}\cdot(39202)$
$m_1 = x^Ty = \frac{1}{64}(66922)$.
Then the largest eigen value of B is $\frac{m_1}{m_0} = 1.70710$
The reciprocal $\frac{1}{1.70710} = 0.58578$ is the smallest eigen value of A.

Example 3: Find the bounds for eigen values of A.
where $A = \begin{bmatrix}1&0&-1\\1&2&1\\2&2&3\end{bmatrix}$.

Solution: The bounds for eigen values are given by three Gershgorin circular disks c_1, c_2, c_3
c_1: centre 1 and radius $|0| + |-1| = 1$,
c_2: centre 2 and radius $1 + 1 = 2$
c_3: centre 3 and radius $2 + 2 = 4$
The actual eigen values 1, 2, 3 of A lies within these three circular disks c_1, c_2 and c_3.

EXERCISE

Find the largest eigen value of the matrix A and the error in the largest eigen value $|\epsilon| \le \sqrt{\frac{m_2}{m_0} - \lambda_1^2}$.

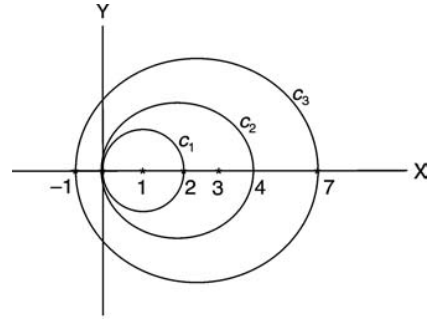


Fig. 32.19

(Ex. 1 to 5)

1. $A = \begin{bmatrix}2&2&1\\1&3&1\\1&2&2\end{bmatrix}$

Hint: $X_0 = \begin{bmatrix}1\\1\\1\end{bmatrix}$, $X_1 = AX_0 = \begin{bmatrix}5\\5\\5\end{bmatrix} = 5\begin{bmatrix}1\\1\\1\end{bmatrix}$,
$X_2 = 25\begin{bmatrix}1\\1\\1\end{bmatrix}$ $X_3 = 125\begin{bmatrix}1\\1\\1\end{bmatrix}$, $X_4 = 625\begin{bmatrix}1\\1\\1\end{bmatrix}$,
$X_5 = 3125X_0$, $X_6 = 15625X_0X_7 = 78125X_0$.
Put $m_0 = X_6$, $m_1 = X_7$, $\lambda_1 = \frac{m_1}{m_0} = \frac{(12625)(78125)(3)}{(12625^2)(3)} = 5$, $\frac{m_2}{m_0} = 25$, $|\epsilon| = 0$

Ans. 5 (Actual eigen values are 1, 1, 5)

2. $\begin{bmatrix}-3&2&1\\1&-2&1\\1&2&-3\end{bmatrix}$

Hint: $X_0 = \begin{bmatrix}1\\1\\1\end{bmatrix}$, $X_1 = \begin{bmatrix}-3\\1\\1\end{bmatrix}$, $X_2 = \begin{bmatrix}12\\-4\\-4\end{bmatrix}$,
$X_3 = \begin{bmatrix}-48\\16\\16\end{bmatrix}$, $X_4 = \begin{bmatrix}192\\-64\\-64\end{bmatrix}$, $m_2 = 45056$,
$m_1 = -11264$, $m_0 = 2816$, $\frac{m_2}{m_0} = 16$, $\frac{m_1}{m_0} = \lambda_1 = -4$, $|\epsilon| = 0$

Ans. -4

3. $A = \begin{bmatrix}1&0&-1\\1&2&1\\2&2&3\end{bmatrix}$

Hint: $X_1 = \begin{bmatrix} 0 \\ 4 \\ 7 \end{bmatrix}$, $X_2 = \begin{bmatrix} -7 \\ 15 \\ 29 \end{bmatrix}$, $X_3 = \begin{bmatrix} -36 \\ 52 \\ 103 \end{bmatrix}$, $X_4 = \begin{bmatrix} -139 \\ 171 \\ 341 \end{bmatrix}$, $X_5 = \begin{bmatrix} -480 \\ 544 \\ 1087 \end{bmatrix}$, $X_6 = \begin{bmatrix} -1567 \\ 1695 \\ 3389 \end{bmatrix}$.

$m_2 = 16813835$, $m_1 = 5358083$, $m_0 = 1707905$, $q = \frac{m_1}{m_0}$. Exact eigen values are 1, 1, 3.

Ans. 3.131722.

4. $A = \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 1 & 3 & 4 & -2 \\ 0 & 1 & -2 & 0 \end{bmatrix}$

Hint: $X_1 = [3\ 4\ 6\ -1]^T$, $X_2 = [12\ 17\ 41\ -8]^T$, $X_3 = [65\quad 115\quad 243\quad -65]$, $m_2 = 80724$, $m_1 = 13218$, $m_0 = 2178$, $|\epsilon| = 0.48184$.

Ans. 6.0688705

5. $A = \begin{bmatrix} 9 & 10 & 8 \\ 10 & 5 & -1 \\ 8 & -1 & 3 \end{bmatrix}$

Hint:

$X_1 = [27\ 14\ 10]^T$, $X_2 = [463\ 330\ 232]^T$,

$X_3 = [9323\ 6048\ 4070]^T$,

$X_4 = [176947\ 119400\ 80746]^T$

$X_5 = [3432491\ 2285724\ 1538414]^T$

$X_6 = [66056971\ 44215116\ 29789446]$

$X_7 = [1.2749 \times 10^9,\ 8.5185 \times 10^8,\ 5.7360 \times 10^8]$

$m_0 = 7.2059 \times 10^{15},\ m_1 = 1.3896 \times 10^{17}$.

Ans. 19.284197

6. Determine the largest eigen value and the corresponding eigen vector of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Hint: $x_0 = [1\ 0\ 0]^T$, $X_1 = [1, -0.5\ 0]^T$, $X_2 = 2.8[1\ -1\ .43]^T$ $X_3 = 3.43[.87\ -1\ 0.54]^T$, $X_4 = 3.41[.8\ -1\ .61]^T$, $X_5 = 3.41[.76\ -1\ .65]^T$, $X_6 = 3.41[.74\ -1\ .67]$

Ans. 3.41, $[0.74, -1, 0.67]^T$

7. Find the largest eigen value and the corresponding eigen vector of the matrix A where

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Hint: [1 1 1], [5 4 2], [24 15 6], [111 60 21], [504 252 81], [2268 1089 333], [10161 4779 1492], [45433, 21141, 6201], [202833, 93906, 27342], [905238, 417987, 121248], [4038939, 1862460, 539235].

Ans. 4.46, $[0.9\ 0.42\ .12]^T$

8. $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Ans. 3.0, [1 1 0]

9. $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

Hint: Exact value: 4, $\left[\frac{2}{3}\ 1\right]^T$

Ans. 3.987, $[.668\ 1]^T$

10. $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Hint: $X_0 = [1\ 0\ 0]$, $X_1 = [1\ 1\ 0]$, $X_2 = 3[2.3\ 1\ 0]$, $X_3 = 4[2.1\ 1.1\ 0]$, $X_4 = 4[2.2\ 1.1\ 0]$, $X_5 = 4.4[2\ 1\ 0]$, $X_6 = 4[2\ 1\ 0]$, $X_7 = 4[2\ 1\ 0]$.

Ans. 4, $[2\ 1\ 0]^T$

11. Determine the bounds for the eigen values of the matrix A where

(a) $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Ans. $\lambda = 0$; Circles c_1, c_2; centres 0, 0 and radii 1, 0 respectively.

(b) $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$

Ans. $\lambda = \pm 5$; centres: 3, −3; radii 4, 4 respectively

(c) $A = \begin{bmatrix} 6 & 0 & -3 \\ 0 & 6 & 3 \\ -3 & 3 & 2 \end{bmatrix}$

Ans. $\lambda = 6, 4 \pm \sqrt{22} = 6, 8.6904, -0.6904$ circular disks with centres at 6, 6, 2 and radii $|-3|, 3, |-3|+3$ respectively.

(d) $A = \begin{bmatrix} 26 & -2 & 2 \\ 2 & 21 & 4 \\ 4 & 2 & 28 \end{bmatrix}$

Ans. $\lambda = 30, 25, 20$; circles: centers 26, 21, 28 radii: $|-2|+2, 2+4, 4+2$.

Chapter 33

Numerical Solutions of ODE and PDE

INTRODUCTION

Analytical solutions can be obtained only for selected class of ODE & PDE, using series solutions, Laplace transform, Fourier transform, separation of variables technique etc. For certain problems, analytical solutions can not be obtained. However numerical solutions can be obtained to the desired degree of accuracy using computers, in all the above cases. The advantage is that numerical solutions can be obtained for problems involving irregularly shaped boundaries and is easy to program on computer. The only serious disadvantage with these numerical solutions is that they lack the generality of the analytical solutions. When the initial conditions are changed, the numerical solution must be calculated again. Taylor's series, Picard's, Modified Euler's, Runge-Kutta 4th order, Milne's predictor-corrector and Adame-Bashforth-Milne's methods for solution of first order ODE are considered in this chapter. Numerical solutions of PDE: one dimensional heat equation, wave equation and two-dimensional Laplace's equation are studied.

33.1 NUMERICAL SOLUTIONS OF FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS

Differential equation is the most important mathematical model for physical phenomena such as the motion of objects, fluids, heat flow, bending and cracking of materials, vibrations, chemical reactions. Since an nth order D.E. can in general, be reduced to a system of n first order D.E., in essence, it would be sufficient to examine a first order O.D.E.

$$\frac{dy}{dx} = f(x, y) \tag{1}$$

together with an initial condition (I.C.)

$$y(x_0) = y_0 \tag{2}$$

Numerical solutions of first order O.D.E. (1) plays a vital role, since analytical solutions of (1) are rare (except in standard forms).

Taylor's Series Method

Consists of expanding the function $y(x)$ in powers of $(x - x_0)$:

$$y(x) = y(x_0) + y'(x_0) \cdot (x - x_0) + \frac{y''(x_0)}{2!}(x - x_0)^2 + \frac{y'''(x_0)}{3!}(x-x_0)^3 + \cdots + \frac{y^{(n)}(x_0)}{n!}(x-x_0)^n + \cdots \tag{3}$$

In (3), $y(x_0)$ is known from I.C. (2). The remaining coefficiently $y'(x_0)$, $y''(x_0)$, ..., $y^{(n)}(x_0)$ etc. are obtained by successively differentiating (1) and evaluating at x_0. Substituting these values in (3), $y(x)$ at any point can be calculated from (3), provided $h = x - x_0$ is small.

When $x_0 = 0$, then Taylor's series (3) takes simpler form

$$y(x)=y(0)+y'(0) \cdot x+y''(0) \cdot \frac{x^2}{2!}+\cdots+y_{(0)}^{(n)} \cdot \frac{x^n}{n!}+\cdots \tag{4}$$

Modified Euler's Method

For small changes Δx in x, the change in y is $\Delta y \approx \frac{dy}{dx}\Delta x$. Thus

$$f(y+\Delta y) - f(y) = \Delta y = \frac{dy}{dx}\Delta x.$$

Using $\frac{dy}{dx} = f(x, y)$ at (x_i, y_i), we have

$$y_i' = f(x_i, y_i) \tag{5}$$

$$\boxed{\tilde{y}_{i+1} = y_i + h\, f(x_i, y_i)} \quad \text{Predictor} \tag{6}$$

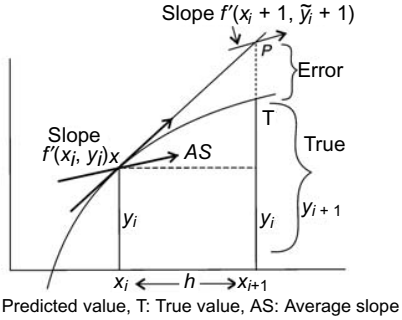
Equation (6) is known as *Euler's method*. Using (5) as a predictor equation find

$$\tilde{y}'_{i+1} = f(x_i + 1, \tilde{y}_{i+1}). \tag{7}$$

Using the average of the slopes (5) and (7), we get

$$y_{i+1} = y_i + \frac{y_i' + \tilde{y}'_{i+1}}{2}.$$

$$\boxed{y_{i+1} = y_i + h\frac{f(x_i, y_i) + f(x_{i+1}, \tilde{y}_{i+1})}{2}} \quad \text{Corrector} \tag{8}$$



P: Predicted value, T: True value, AS: Average slope

Fig. 33.1

The Equation (8) is known as the corrector equation. Thus the method using the predictor (6) and corrector (8) is known as the *modified Euler's method* or Heun's method or Euler-Cauchy method or Euler-Trapezoidal method or improved Euler method or Euler's predictor-corrector method.

Runge-Kutta 4th Order Method

The one-step methods like Taylor series method, Euler's method or modified Euler's method evaluate y at a pivota point x_i, thus giving the starting values for the multi-step methods which use recurrence equations, at these preceding pivotal points. Runge-Kutta method is another one such one-step (or single step) method, which is also self-starting like the Taylor's Euler's, methods. Runge-Kutta methods achieves the accuracy of a Taylor series method without requiring the calculation of higher derivatives. A two parameter h, k family of formulas of fourth order accuracy are known as Runge-Kutta fourth order method and are given by the following formulae:

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{9}$$

where

$$\left.\begin{aligned} k_1 &= h\, f(x_i, y_i) \\ k_2 &= h\, f\left(x_i + \tfrac{h}{2}, y_i + \tfrac{k_1}{2}\right) \\ k_3 &= h\, f\left(x_i + \tfrac{h}{2}, y_i + \tfrac{k_2}{2}\right) \\ k_4 &= h\, f(x_i + h, y_i + k_3) \end{aligned}\right\} \tag{10}$$

Note: When f is independent of y, (9) reduce's to Simpson's $\frac{1}{3}$ rule,

$$\left(\frac{h}{2}\right)\frac{1}{3}\left[f(x_i) + 4f\left(x_i + \frac{h}{2}\right) + f(x_i + h)\right].$$

The only disadvantage with R-K method is the repeated evaluation of function $f(x, y)$ at several points.

Milne's Predictor-Corrector Method

Is a multi-step method which uses the previously calculated four values of the dependent variable y and its derivative y' in the subsequent steps. Integrating

$$\int_{x_{n-3}}^{x_{n+1}} \frac{dy}{dx}dx = \int_{x_{n-3}}^{x_{n+1}} f(x, y)dx$$

where R.H.S. is evaluated by Simpson's $\frac{1}{3}$ rule for the extended interval x_{n-2} to x_n to x_{n-3} to x_{n+1}.

$$y_{n+1} - y_{n-3} = \frac{4h}{3}\left(2y'_{n-2} - y'_{n-1} + 2y'_n\right) \quad \text{or}$$

$$y_{n+1} = y_{n-3} + \frac{4h}{3}(2y^{\mathrm{i}}_{n-2} - y^{\mathrm{i}}_{n-1} + 2y^{\mathrm{i}}_n) \tag{11}$$

(11) is known as the predictor formula. Using (11), calculate $y^{\mathrm{i}}_{n+1} = f(x_{n+1}, y_{n+1})$. Integrating between x_{n-1} to x_{n+1} using Simpson's $\frac{1}{3}$ rule.

$$y_{n+1} = y_{n-1} + \frac{h}{3}(y^{\mathrm{i}}_{n-1} + 4y^{\mathrm{i}}_n + y^{\mathrm{i}}_{n+1}) \tag{12}$$

(12) is known as the corrector formula. The next pair of values y_{n+2}, y^{i}_{n+2} are obtained in a similar way first using (11) to find y_{n+2}, then y^{i}_{n+2} by $f(x_{n+2}, y_{n+2})$.

With this value the corrector value of y_{n+2} is obtained from (12).

WORKED OUT EXAMPLES

Taylor's series method

Example 1: Using Taylor series expansion evaluate the integral of $y' - 2y = 3e^x$, $y(0) = 0$ at (a) $x = 0.1(0.1)0.3$; (b) $x = 1.0; 1.1$.

Solution: Rewriting $y' = 2y + 3e^x$, $y(0) = 0$. Differentiating and evaluating at $x = 0$,

$$y'(0) = 2y(0)+3e^0=2\cdot 0+3\cdot 1=3$$
$$y'' = 2y'+3e^x, \quad y''(0)=2y'(0)+3=2\cdot 3+3=9$$
$$y''' = 2y''+3e^x, \quad y'''(0)=2y''(0)+3=2\cdot 9+3=21$$
$$y'''' = 2y'''+3e^x, \quad y''''(0)=2y'''(0)+3=2\cdot 21+3=45$$
$$y''''' = 2y''''+3e^x, \quad y'''''(0)=2y''''+3=2\cdot 45+3=93$$

In general,

$$y^{(n+1)}(x) = 2y^{(n)}(x) + 3e^x \text{ or } y^{(n+1)}(0) = 2y^{(n)}(0) + 3.$$

The Taylor's series expansion of $y(x)$ about 0 is

$$y(x) = y(0) + xy'(0) + x^2\frac{y''(0)}{2!} + x^3\frac{y'''(0)}{3!} + x^4\frac{y''''(0)}{4!} + x^5\frac{y'''''(0)}{5!} + \cdots$$

Substituting the values of $y'(0), y''(0), \ldots$

$$y(x) = 0 + 3x + \frac{9}{2}x^2 + \frac{21}{6}x^3 + \frac{45}{24}x^4 + \frac{93}{120}x^5 + \cdots$$
$$y(x) = 3x + 4.5x^2 + 3.5x^3 + \frac{15}{8}x^4 + \frac{31}{40}x^5 + \cdots \quad (1)$$

Now put $x = 0.1$ in (1)

$$y(0.1) = 3(0.1)+4.5(0.1)^2+3.5(0.1)^3+\frac{15}{8}(0.1)^4+\frac{31}{40}(.1)^5$$
$$y(0.1) = 0.34869$$

Similarly, put $x = 0.2$ in (1)

$$y(0.2) = 0.6 + 0.18 + 0.028 + 0.003 + 0.000248 = 0.811244.$$
$$y(0.3) = 0.9 + 0.405 + .0945 + .015187 + .0018832 = 1.4165702$$

Put $x = 1$ in (1)

$$y(1) = 3 + \frac{9}{2} + \frac{7}{2} + \frac{15}{8} + \frac{31}{40} = 13.65$$

Put $x = 1.1$ in (1)

$$y(1.1) = 3.3 + 5.445 + 4.6585 + 2.745 + 1.248 = 17.39.$$

Modified Euler's method

Example 2: Intensity of radiation is directly proportional to the amount of remaining radioactive substance. The DE is $y' = -ky$, where $k = 0.01, t_0 = 0$, $y_0 = 100$ g. Determine how much substance will remain at the moment $t = 100$ sec. Find the solution by (a) Euler's method, (b) Modified Euler's method with $h = 25$.

Solution: Here $h = 25$, $f(t, y) = -ky$, $f(t, y) = -0.01\,y$, since $\frac{dy}{dt} = f(t, y) = -ky$. (a) Euler's method (see Table 33.1).

Table 33.1 Euler's method

t	y_i	$f_i = f(t_i, y_i)$	$\Delta y_i = hf_i$	$y_{i+1} = y_i + \Delta y_i$
0	100	-1	-25	$100 + (-25) = 75$
25	75	-0.75	-18.75	$75 - 18.75 = 56.25$
50	56.25	-0.5625	-14.0625	$56.25 - 14.0625 = 42.1875$
75	42.1875	-0.421875	-10.546875	$42.1875 - 10.5468 = 31.640625$
100	31.640625			

(b) Modified Euler's method: $h = 25$, $f(t, y) = -0.01y$ (see Table 33.2).

Runge-Kutta 4th order method

Example 3: Using Runge-Kutta 4th order method find the solution of $\frac{dy}{dx} = y - x$ with initial condition

Table 33.2 Modified Euler's method

t	y_i	$f_i = f(t_i, y_i)$	$\tilde{y}_{i+1} = y_i + hf_i$	$\tilde{f}_{i+1} = f(x_{i+1}, \tilde{y}_{i+1})$	$\tilde{\Delta y}_i = \frac{h}{2}(f_i + \tilde{f}_{i+1})$	$y_{i+1} = y_i + \tilde{\Delta y}_i$
0	100	-1	$100 + 25(-1)$ $= 75$	$(-0.01)(75)$ $= -0.75$	$\frac{25}{2}(-1 - 0.75)$ $= -21.875$	$100 - 21.875$ $= 78.125$
25	78.125	-0.78125	$78.125 + 25(-0.78)$ $= 58.59$	$(-0.01)(58.59)$ $= -0.5859$	$\frac{25}{2}(-0.78 - 0.59)$ $= -17.09$	$78.125 - 17.09$ $= 61.034$
50	61.034	-0.61	45.7755	-0.4578	-13.35	47.68
75	47.68	-0.4768	35.76	-0.3576	-10.43	37.25
100	37.25					

Note: Exact solution: $y = 100\,e^{-kt}$, At $t = 100$, $y = 36.7879$.

$y(0) = 1.5$ on $[0, 1]$.

Solution: Here choose $h = 0.2$ given $f(x, y) = y - x$, $x_0 = 0$, $y_0 = 1.5$.
See Table 33.3 on p. 33.5.

Milne's predictor-corrector method

Example 4: Using Milne's predictor-corrector method evaluate the integral of $y' - 4y = 0$ at $x = 0.4,\ 0.5$ given that $y(0) = y_0 = 1,\ y_1 = y(0.1) = 1.492,\ y_2 = y(0.2) = 2.226,\ y_3 = y(0.3) = 3.320$.

Solution: Here $\frac{dy}{dx} = 4y$, so $f(x, y) = y' = 4y$, $h = 0.1, x_0 = 0, x_1 = 0.1, x_2 = 0.2, x_3 = 0.3, x_4 = 0.4, x_5 = 0.5$.

Predictor: $y_{n+1} = y_{n-3} + \frac{4h}{3}(2y^{\mathrm{i}}_{n-2} - y^{\mathrm{i}}_{n-1} + 2y^{\mathrm{i}}_n)$

with $n = 3$,

$$y_4 = y_0 + \frac{4h}{3}(2y^{\mathrm{i}}_1 - y^{\mathrm{i}}_2 + 2y^{\mathrm{i}}_3).$$

Here $h = 0.1$, $y_0 = 1$, $y^{\mathrm{i}}_1 = 4y_1 = 4(1.492) = 5.968$.

$$y^{\mathrm{i}}_2 = 4y_2 = 4(2.226) = 8.904,\ y^{\mathrm{i}}_3 = 4y_3 = 4(3.32) = 13.28$$

$$\therefore\ y_4 = y(x_4) = y(0.4) = 1 + \frac{4(0.1)}{3}[2(5.968) - 8.904 + 2(13.28)]$$

$$y_4 = 1 + \frac{4(0.1)}{3}(29.592) = 1 + 3.9456 = 4.9456$$

So $y^{\mathrm{i}}_4 = 4y_4 = 4(4.9456) = 19.7824$

Corrector

$$y_{n+1} = y_{n-1} + \frac{h}{3}(y^{\mathrm{i}}_{n-1} + 4y^{\mathrm{i}}_n + y^{\mathrm{i}}_{n+1})$$

with $n = 3$,

$$y_4 = y_2 + \frac{h}{3}(y^{\mathrm{i}}_2 + 4y^{\mathrm{i}}_3 + y^{\mathrm{i}}_4)$$

$$\therefore\quad y_4 = y(x = 0.4) = 2.226 + \frac{0.1}{3}(2.226 + 4(13.28) + 19.7824)$$

$$y_4 = 2.226 + 2.72688 = 4.95288$$

with $n = 4$, predictor

$$y_5 = y_1 + \frac{4h}{3}[2y^{\mathrm{i}}_2 - y^{\mathrm{i}}_3 + 2y^{\mathrm{i}}_4]$$

Since $y' = 4y$, $y^{\mathrm{i}}_i = 4y_i$

$$y_5 = 1.492 + \frac{4(0.1)}{3}4[2y_2 - y_3 + 2y_4]$$

$$= 1.492 + \frac{16(0.1)}{3}[2(2.226) - 3.320 + 2(4.95288)]$$

$$y_5 = 7.3788$$

Correct with $n = 4$,

$$y_5 = y_3 + \frac{h}{3}[y^{\mathrm{i}}_3 + 4y^{\mathrm{i}}_4 + y^{\mathrm{i}}_5]$$

$$y_5 = y_3 + \frac{4h}{3}[y_3 + 4y_2 + y_5] \qquad (\text{since } y'_i = 4y_i)$$

$$= 3.320 + \frac{4(0.1)}{3}[3.320 + 4(4.95288) + 7.3788]$$

$$y_5 = y(x = 0.5) = 3.320 + 4.06804 = 7.3880426.$$

Note: The exact solution is $y = e^{4x} + c$, with $y(0) = 1, c = 0$ so $y = e^{4x}$. Then $y(0.4) = e^{4(0.4)} = 4.953032425$ and $y(0.5) = e^{4(0.5)} = 7.389056099$.

Table 33.3 Runge-Kutta 4th order method

	x	y	$k_v = h\,f(x, y)$	Correction
	x_0	y_0	$k_1 = h\,f(x_0, y_0)$	$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
	$x_0 + \frac{h}{2}$	$y_0 + \frac{k_1}{2}$	$k_2 = h\,f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$	
	$x_0 + \frac{h}{2}$	$y_0 + \frac{k_2}{2}$	$k_3 = h\,f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$	
	$x_0 + h$	$y_0 + k_3$	$k_4 = h\,f(x_0 + h, y_0 + k_3)$	
1st iteration	x	y	$k_v = 0.2(x + y)$	$\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
	0.0	0.0	$k_1 = 0$	$k_1 = 0$
	0.1	0.0	$k_2 = 0.02$	$2k_2 = 0.04$
	0.1	0.01	$k_3 = 0.022$	$2k_3 = 0.044$
	0.2	0.022	$k_4 = 0.0444$	$k_4 = 0.0444$
				$\Delta y = 0.0214$
IInd iteration		$\therefore y = 0 + 0.0214 = 0.0214$		
	0.2	0.0214	0.04428	
	0.3	0.04354	0.068708	
	0.3	0.055754	0.071151	
	0.4	0.092551	0.098510	
				$\Delta y = 0.070418$
		$\therefore y = 0.0214 + 0.070418 = 0.091818$		
IIIrd iteration	0.4	0.0918	0.0983	
	0.5	0.141	0.1282	
	0.5	0.1559	0.1311	
	0.6	0.223	0.1646	
				$\Delta y = 0.13028$
		$\therefore y = 0.0918 + 0.13028 = 0.222107$		
IV iteration	0.6	0.222	0.1644	
	0.7	0.3043	0.2008	
	0.7	0.3225	0.2045	
	0.8	0.4266	0.245	
				$\Delta y = 0.20341$
Vth iteration		$\therefore y = 0.222 + 0.20341 = 0.425522$		
	0.8	0.42552	0.2451	
	0.9	0.548	0.2896	
	0.9	0.5703	0.2940	
	1.0	0.71958	0.3439	
				$\Delta y = 0.29273$

$\therefore y(1) = 0.425522 + 0.29273 = 0.718253.$

Note: Exact solution: $y(x) = e^x - 1 - x$, $y(1) = e^1 - 1 - 1 = 0.718281828$.

EXERCISE

Taylor series

1. Evaluate the integral of $y'' + yy' - x^2 = 0$, $y(0) = 1$, $y'(0) = 1$ at $x = 0.1(0.1)0.3$ by Taylor series.
   Hint: $y(x) = 1 + x - \frac{x^2}{2!} + \frac{5x^4}{4!} - \frac{8x^5}{5!} + \cdots$.

Ans. $y(0.1) = 1.095$, $y(0.2) = 1.180$, $y(0.3) = 1.257$

2. Find the first five terms of the expansion in a power series of the solution $y = y(x)$, $z = z(x)$ of the system $y'(x) = y\cos x - z\sin x$, $z^i(x) = y\sin x + 2\cos x$, with initial conditions $y(0) = 1$, $z(0) = 0$.

Ans. $y(x) = 1 + x + \frac{x^2}{2} - 5\frac{x^4}{4!} + \cdots$, $z(x) = \frac{x^2}{2} + \frac{x^3}{2} + \frac{5}{4!}x^4 + \cdots$

3. Find $y(0.1)$ by Taylor's series expansion when $y' = x - y^2$, $y(0) = 1$.
   Hint: $y(x) = 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 - \frac{17}{12}x^4 - \frac{31}{20}x^5 + \cdots$.

Ans. $y(0.1) = 0.9138$

4. Given $y' = x^2 + y^2$, $y(0) = 0$, determine the first three non-zero terms in Taylor series and hence obtain $y(1)$.

Ans. $y(x) = \frac{1}{3}x^3 + \frac{1}{63}x^7 + \frac{2}{2079}x^{11}$, $y(1) = 0.3502$.

5. Determine the three terms in the Taylor's series solution to the Blasius equation $y''' + yy'' = 0$, $y(0) = 0$, $y'(0) = 0$, $y''(0) = 1$.

Ans. ?

6. Compute $y(0.1)$ and $y(0.2)$ by Taylor's series method if $y' = x^2y - 1$, $y(0) = 1$.
   Hint: $y(x) = 1 - x + \frac{x^3}{3} - \frac{x^4}{4}\cdots$.

Ans. $y(0.1) = 0.9033$, $y(0.2) = 0.80227$.

Euler's method

7. Evaluate the integral of $y' - y^2 = 0$ by Euler's predictor-corrector method at $x = 0.1,\ 0.2$ with initial condition $y(0) = 1$.

Ans. $y(0.1) = 1.1105$, $y(0.2) = 1.24827$.

8. Evaluate by Euler's method
   a. $y' - y^2 = 0$, $y(0) = 1$, at $x = 0.1, 0.2, 0.3$
   b. $y' = y - x$, $y(0) = 2$, at $x = 0.2, 0.4, 0.6$.

Ans. **a.** $y(0.1)$=1.1, $y(0.2)$=1.221, $y(0.3)$=1.37
b. $y(0.2) = 2.4$, $y(0.4) = 2.84$, $y(0.6) = 3.328$

9. Use Heun's method to integrate $y' = 4e^{0.8x} - 0.5y$ from $x = 0$ to 4 with a step size of 1 and with $y(0) = 2$.
   Hint: $y = \frac{4}{1.3}(e^{0.8x} - e^{-0.5x}) + 2e^{-0.5x}$ is exact solution.

Ans. $y(1) = 6.70108$, $y(2) = 16.319$, $y(3) = 37.199$, $y(4) = 83.377$

10. Solve the following initial value problems by modified Euler's method.
    a. $y' = x + y$, $y(0) = 1$, $h = 0.1$, $x = 0(0.1)0.3$
    b. $y' = \ln(x + y)$, $y(0) = 2$, $h = 0.2$, $x = 0(0.2)0.8$
    c. $y' = x + |\sqrt{y}|$, $y(0) = 1$, $h = 0.2$, $x = 0(0.2)0.6$

Ans. **a.** 1.1105, 1.2432, 1.4004
b. 2.0656, 2.1416, 2.2272, 2.3217
c. 1.2309, 1.5253, 1.8861

Runge-Kutta 4th order method

11. Integrate the D.E. $y' = y - \frac{2x}{y}$, $y(0) = 1$ from 0 to 2 by R-K method with $h = 0.2$.

Ans. $y(0.2) = 1.1832$, $y(0.4) = 1.3416$, $y(0.6) = 1.4832$, $y(0.8) = 1.61251$, $y(1.0) = 1.7321$, $y(1.2) = 1.844$, $y(1.4) = 1.9495$, $y(1.6) = 2.049$, $y(1.8) = 2.145$, $y(2.0) = 2.2366$

12. Integrate D.E. $y' = x + y$, $y(0) = 0$, $h = 0.2$ by R-K method, from 0 to 1.

Ans. $y(0.2) = 0.0214$, $y(0.4) = 0.0918$, $y(0.6) = 0.222$, $y(0.8) = 0.4255$, $y(1.0) = 0.718$

13. Using R-K method, find the solution of
    $$y' = 0.25\,y^2 + x^2$$
    with initial condition $y(0) = -1$ on [0, 0.5] with $h = 0.1$.

Ans. −0.97528, −.94978, −.92154, −.8887, −.84945

14. Integrate D.E. by R-K method given $y' = -2xy^2$, $y(0) = 1$, $h = 0.2$ on [0, 1].

Ans. $y(0.2) = .9615$, $y(0.4) = .862$, $y(.6) = .73527$, $y(.8) = .6097$, $y(1.0) = .500073$

15. Apply R-K method to solve

a. $y' = x + y$, $y(0) = 1$, $h = 0.2$, and find $y(0.2)$

b. $y' = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1, h = 0.2, x = 0(0.2)0.4$

Ans. **a.** $y(0.2) = 0.2428$

b. $y(0.2) = 1.196$, $y(0.4) = 1.3752$.

Milne's predictor-corrector method

16. Use Milne's predictor-corrector method to integrate $y' = 4e^{0.8x} - 0.5y$ from $x = 0$ to $x = 4$ with a step size of 1 and with $y(0) = 2$.

Ans. 6.20485, 14.86, 33.7242, 75.4329

17. Compute $y(0.8)$ and $y(1.0)$ by Milne's method if $y' = 1 + y^2$, $y(0) = 0$, $y(0.2) = 0.2027$, $y(0.41) = 0.4228$, $y(0.6) = 0.6841$.

Hint: $y'(0) = 1$, $y'(0.2) = 1.0411$, $y'(0.4) = 1.1787$, $y'(0.6) = 1.4681$, $y_{\text{predictor}}(0.8) = 1.0239$, $y'(0.8) = 2.048$

Ans. $y(0.8) = 1.0294$, $y(1) = 1.5549$

18. Find $y(0.4)$ and $y(0.5)$ by Milne's method if $y' + y = 2e^x$, $y_0 = 2$, $y_1 = 2.010$, $y_2 = 2.04$, $y_3 = 2.09$, $h = 0.1$.

Ans. $\tilde{y}_4 = 2.162$, $y_4 = 2.162$, $\tilde{y}_5 = 2.255$, $y_5 = 2.2546$

19. Calculate y at $x = 0.4$, and $x = 0.5$, if $y' = x + y$ and $y_0 = 1$, $y_1 = 1.1103$, $y_2 = 1.2428$, $y_3 = 1.3997$, $h = 0.1$.

Ans. $\tilde{y}_4 = 1.5836$, $y_4 = 1.5836$, $\tilde{y}_5 = 1.7974$, $y_5 = 1.7974$

20. Determine $y(0.8)$, $y(1.0)$ by Milne's P-C method when $y' = x - y^2$, $y(0) = 0$.

Hint: $y_1 = 0.02$, $y_1^i = 0.1996$, $y_2 = 0.0795$, $y_2^i = 0.3937$, $y_3 = 0.1762$, $y_3^i = 0.5689$ $\left(y = \frac{x^2}{2} - \frac{x^5}{20} + \frac{x^8}{160} - \frac{x^{11}}{4400}\right)$.

Ans. $y(0.8) = 0.3046$, $y(1.0) = 0.4555$.

33.2 PICARD'S METHOD OF SUCCESSIVE APPROXIMATION*

A problem involving ODEs is not completely specified by its equations. Boundary conditions (B.C.s), which are algebaric conditions on the values of the function $y(x)$, play crucial role in determining how to attack the problem numerically. Initial value problems involve condition(s) specified at some starting value x_0, while in (two-point) boundary value problems boundary conditions are specified at more than one point x.

Picard's method of successive approximation and Taylor's series are single step methods in which the solutions are obtained in analytical form as power serier in x.

Picard's Method of Successive Approximation

The particular solution of the initial value problem consisting of an ordinary first order differential equation

$$\frac{dy}{dx} = f(x, y) \tag{1}$$

with initial condition $y(x_0) = y_0$ is obtained by integrating (1). Then

$$\int_{y_0}^{y} dy = \int_{x_0}^{x} f(x, y)dx$$

or

$$y(x) = y_0 + \int_{x_0}^{x} f(x, y)dx \tag{2}$$

In the Picard's method, the first approximation y_1 is obtained by replacing y in $f(x, y)$ in RHS of (2) by y_0 and then evaluating the integral (which is now a function of x) wrt x. The second approximation y_2

*Emile Picard (1856–1941), French mathematician.

is then obtained by replacing y in $f(x, y)$ of RHS of (2) by y_1 and integrating wrt x. Successive approximations are obtained similarly. In practice, Picard's method is restricted to a limited class of problems in which the integral in RHS of (2) can be evaluated easily. Thus

$$y_1 = y_0 + \int_{x_0}^{x} f(x, y_0)dx$$
$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)dx$$
$$y_3 = y_0 + \int_{x_0}^{x} f(x, y_2)dx$$

and so on yielding a sequence of functions $y_1, y_2, y_3, \ldots$ giving rise to a better approximation.

WORKED OUT EXAMPLES

Example 1: Solve the differential equation

$$\frac{dy}{dx} = x^2 - y,\ y(0) = 1$$

by Picard's method of successive approximations to get the value of y at $x = 1$. Use terms through x^5, compare it with the exact analytical solution.

Solution: Integrating the D.E., we obtain

$$\int_{y_0}^{y} dy = \int_{x_0}^{x} (x^2 - y)dx$$

$y(x) = y(x_0) + \int_{x_0}^{x} (x^2 - y)dx = 1 + \int_0^x (x^2 - y)dx$
The first approximation y_1 is obtained by replacing y by $y_0 = 1$ in the integrand in RHS integral. Thus

$$y_1 = 1 + \int_0^x (x^2 - 1)dx = 1 + \frac{x^3}{3} - x$$

To obtain the second approximation y_2, replace y by $1 - x + \frac{x^3}{3}$ in the integrand of the RHS integral. Thus

$$y_2 = 1 + \int_0^x \left[x^2 - (1 - x + \frac{x^3}{3})\right]dx$$
$$y_2 = 1 + \frac{x^3}{3} - x + \frac{x^2}{x} - \frac{x^4}{12}$$

Now the third approximation y_3 is obtained by replacing y by $y_2 = 1 - x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{12}$.
Thus

$$y_3 = 1 + \int_0^x \left[x^2 - \left\{1 - x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{12}\right\}\right]dx$$

Integrating we get

$$y_3 = 1 + \frac{x^3}{3} - x - \frac{x^4}{12} + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^5}{60}$$
$$y_3 = 1 - x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{60}$$

Similarly replacing y by y_3 and integrating wrt x we get the fourth approximation y_4 as

$$y_4 = 1 + \int_0^x \left[x^2 - \left\{1 - x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{60}\right\}\right]dx$$
$$y_4 = 1 - x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24} + \frac{x^5}{60} - \frac{x^6}{360} \quad (*)$$

The exact solution of given DE is

$$y = x^2 - 2x + 2 - e^{-x}$$

Expanding e^{-x} in x, the exact solution is

$$y(x) = x^2 - 2x + 2 - \left[1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} - \frac{x^4}{4!} + \frac{x^5}{5!} - \frac{x^6}{6!} + \ldots\right]$$
$$y(x) = 1 - x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{24} + \frac{x^5}{120} - \frac{x^6}{720} + \ldots \quad (**)$$

Thus the fourth approximation $y_4(*)$ concides with the exact analytical solution $y(**)$ up to the 4th power of x. Now

$$y_4(x = 1) = 1 - 1 + \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{60} - \frac{1}{360} = 0.638888$$
$$y(x = 1) = 1 - 1 + \frac{1}{2} + \frac{1}{6} - \frac{1}{24} + \frac{1}{120} - \frac{1}{720} = 0.6319444$$

Example 2: Obtain the Picard's second approximation for the initial value problem

$$\frac{dy}{dx} = \frac{x^2}{y^2+1},\ y(0) = 0.$$

Find $y(1)$.

Solution: The first approximation is

$$y_1 = y_0 + \int_{x_0}^{x} \frac{x^2}{y^2+1}dx = 0 + \int_0^x \frac{x^2}{y^2+1}dx$$

Relpacing y by $y_0 = 0$, we get

$$y_1 = \int_0^x \frac{x^2}{0^2+1}dx = \frac{x^3}{3}$$

Replacing y by $y_1 = \frac{x^3}{3}$ we get the second approximation y_2 as

$$y_2 = \int_0^x \frac{x^2}{\left(\frac{x^3}{3}\right)^2+1}dx = \tan^{-1}\left(\frac{x^3}{3}\right) - 0$$

In the third approximation

$$y_3 = \int_0^x \frac{x^2}{\left(\tan^{-1}\frac{x^3}{3}\right)^2+1}$$

The integration is difficult. This is the drawback of the method.

Expanding $\tan^{-1}$, we get from second approximation

$$y_2 = \tan^{-1}\left(\frac{x^3}{3}\right) = \left(\frac{x^3}{3}\right) - \left(\frac{x^3}{3}\right)^3 \frac{1}{3} + \left(\frac{x^3}{3}\right)^5 \frac{1}{5}\cdots$$

$$y_2 = \frac{x^3}{3} - \frac{x^9}{81} + \frac{x^{15}}{1215}\cdots$$

At $x = 1$, $y_2(1) = \frac{1}{3} - \frac{1}{81} + \frac{1}{1215} = 0.321810699$.

EXERCISE

Using Picard's successive approximation, solve the following initial value problems.

1. $y' = x - y^2$, $y(0) = 1$. Find $y(0.1)$.

Ans. $y_2 = 1 - x + \frac{3}{2}x^2 - \frac{2}{3}x^3 + \frac{x^4}{4} - \frac{x^5}{20}$, $y(0.1) = 0.914357$

2. $y' = x^2 + y^2$, $y(0) = 0$, find $y(0.4)$

Ans. $y_3 = \frac{x^3}{3} + \frac{1}{63}x^7 + \frac{2}{2079}x'' + \frac{1}{59535}x^{15}$, $y(0.4) = 0.02135938$

3. $y' = x + y$, $y(0) = 1$. Find $y(1)$. Compare with exact solution.

Ans. $y_5 = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{720}$

Exact solution: $y = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{360} + \ldots$

$y_5(1) = 3.434$, $y(1) = 3.44$

4. $y' = \frac{y-x}{y+x}$, $y(0) = 1$. Find $y(0.1)$.

Ans. $y_1 = 1 - x + 2\log(1+x)$, y_2 is difficult for evaluation. $y_1(0.1) = 0.9828$

5. $y' = x + y + xy$, $y(0) = 1$, find $y(0.1)$

Ans. $y_3 = 1 + x + x^2 + \frac{x^3}{3} + \frac{5}{12}x^4 + \frac{11}{60}x^5 + \frac{x^6}{24}$

$y_3(0.1) = 1.1103768$

33.3 ADAMS-BASHFORTH-MOULTON METHOD (ABM METHOD)

Predictor-corrector is often loosely used to denote the multistep or multivalue integration technique for ODEs. The most popular predictor-corrector methods are probabily the Adams-Bashforth-Moulton schemes, which have good stability properties unlike the Milne's method which is unstable due to the corrector. The predictor-corrector method consists of three separate processes. The predictor step P, evaluation of derivative y'_{n+1} from the latest value of y, call as E and the corrector step, call as C. Iterating m times with the corrector in this notation, may be written as $P(EC)^m$. Since E is superior, the strategy is PECE i.e., predict, evaluate, correct and evaluate.

Adams-Bashforth-Moulton (ABM) Method is a non self-starting four step method which uses four initial points (x_0, y_0), (x_1, y_1), (x_2, y_2) and (x_3, y_3), obtained already by any one of the single-step methods to generate the points (x_n, y_n) for $n \geq 4$. It requires only two function evaluations of $f(x, y)$ per step.

A Lagrange cubic polynomial based on the four points from x_{n-3} to x_n, is integrated over one step

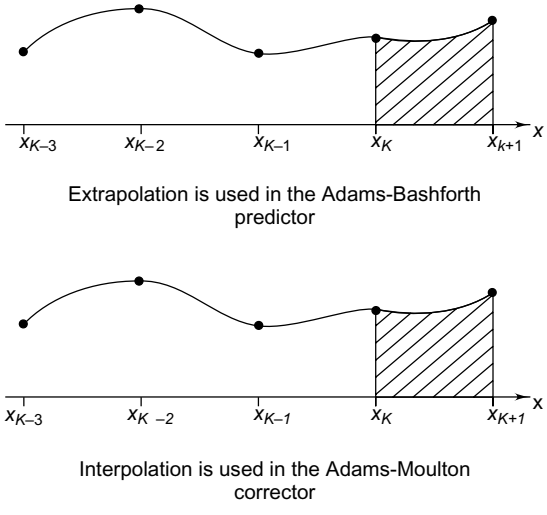


Fig. 33.2 Integration on the interval x_k, x_{k+1}

from x_n to x_{n+1} yielding the *Adams-Bashforth predictor*.

$$y_{n+1} = y_n + \frac{h}{24}\left[55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}\right] \tag{1}$$

The tentative computed value of y at x_{n+1} obtained from the predictor formula (1), together with values of y at x_n, x_{n-1}, x_{n-2} are used to construct a second cubic polynomial, which is then integrated over the interval x_n to x_{n+1}, yielding the *Adams-Moulton corrector:*

$$y_{n+1} = y_n + \frac{h}{24}[9f_{n+1} + 19f_n - 5f_{n-1} + f_{n-2}] \tag{2}$$

The step size h, should be decreased, when the difference between predicted and corrected values reaches or exceeds the accuracy criterion.

Worked Out Examples

Example 1: Solve $y' = x^2 - y$, $y(0) = 1$ on the interval [0, 0.5] with step size $h = 0.05$ and with initial values $y(0.05) = 0.95127058$, $y(0.10) = 0.90516258$, $y(0.15) = 0.86179202$. Compare with true (exact) solution.

Solution: Here $f(x, y) = x^2 - y$. $h = 0.05$ $x_0 = 0$, $y_0 = 1$, $f(x_0, y_0) = f_0 = 0^2 - 1 = -1$
$x_1 = 0.05$, $y_1 = 0.95127058$, $f(x, y) = f_1 = (0.05)^2 - 0.95127058$ so $f_1 = -0.94877058$.
$x_2 = 0.10$, $y_1 = 0.90516258$, so $f(x_2, y_2) = f_2 = (0.10)^2 - 0.90516258 = -0.89516258$
$x_3 = 0.15$, $y_3 = 0.86179202$ so $f(x_3, y_3) = f_3 = (0.15)^2 - 0.86179202 = -0.83929202$
Use Adams-Bashforth predictor ($n = 3$)

$$y_4 = y_3 + \frac{h}{24}[55f_3 - 59f_2 + 37f_1 - 9f_0]$$

substituting the above starting values

$$\begin{aligned} y_4 &= 0.86179202 + \frac{0.05}{24}[55(-0.8392920) \\ &\quad -59(-0.89516258) + 37(-0.94877058) \\ &\quad -9(-1)] \\ &= 0.86179202 + \frac{0.05}{24}[-19.45097924] \\ y_4 &= 0.86179202 - 0.040522873 = 0.821269146 \end{aligned}$$

Using this tentative predictor value, compute $f_4 = f(x_4, y_4) = f(0.20, 0.821269146)$

$$f_4 = (0.20)^2 - 0.821269146 = -0.781269146$$

Now we utilize this value of f_4 in the Adams-Moulton corrector ($n = 3$)

$$y_4 = y_3 + \frac{h}{24}[9f_4 + 19f_3 - 5f_2 + f_1]$$

substituting the above data, we get

$$\begin{aligned} y_4 &= 0.86179202 + \frac{0.05}{24}[9(-0.781269146) \\ &\quad +19(-0.83929202) - 5(-0.89516258) \\ &\quad +(-0.94877058)] \\ &= 0.86179202 + \frac{0.05}{24}[-19.45092837] \\ y_4 &= 0.86179202 - 0.040522767 = 0.821269252 \end{aligned}$$

The exact (true) solution is

$$y(x) = x^2 - 2x + 2 - e^{-x}$$

so $y(0.20) = 0.821269246$. Calculating further values with ABM method we get the follwing data:

n	x	Adams-Bashforth Moulton method $y(x)$	Exact solution $y(x)$	Error
4	0.20	0.821269252	0.821269246	6×10^{-9}
5	0.25	0.7836992	0.783699216	$1.69 \times 10^{-}8$
6	0.30	0.7491818	0.749181779	$+2.068 \times 10^{-}8$
7	0.40	0.6896800	0.689679954	$4.6036 \times 10^{-}8$
8	0.50	0.6434694	0.64346934	$5.9716 \times 10^{-}8$

Note: For $n = 4$, we have the following predictor corrector formulas.
Predictor: $y_5 = y_4 + \frac{h}{24}[55f_4 - 55f_3 + 37f_2 - 9f_1]$
Corrector: $y_5 = y_4 + \frac{h}{24}[9f_5 + 19f_4 - 5f_3 + f_2]$.
Here
$f_4 = f(0.2, 0.821269252) = (0.2)^2 - 0.821269252 = -0.781269252$

EXERCISE

ABM Method

Solve by ABM method using the given initial values. Compare with exact solution.

1. $y' = -2x - y$, $y(0) = -1$, find $y(0.4)$, $y(0.5)$
   $y(0) = -1$, $y(0.1) = -0.9145122$, $y(0.2) = -0.8561923$, $y(0.3) = -0.8224547$

Ans. At 0.4, Predictor value = -0.8109687
Corrector value : -0.8109652
Exact solution: $y(x) = -3e^{-x} - 2x + 2$
$y(0.4)$ = Exact value = -0.8109601
At 0.5 = P : -0.8195978, C : -0.819505, E: -0.8195920

2. $y' = (x - y)/2$, $y(0) = 1$, $h = 0.125$, find y on [0, 1]. Compare with exact solution. Initial values are : $y(0.125) = 0.94323919$, $y(0.250) = 0.89749071$, $y(0.375) = 0.86208736$.
Exact solution: $y(x) = 3e^{-x/2} - 2 + x$

Ans. $y(0.5) = 0.83640227$, $y(0.625) = 0.81984673$, $y(0.75) = 0.81186762$, $y(0.875) = 0.81194530$, $y(1.0) = 0.81959166$

3. $y' = 2xy^2$, $y(0) = 1$, $h = 0.05$, find y on [0, 0.5] Compare with exact solution.
Initial values are $y(0.05) = 1.0025063$, $y(0.10) = 1.0101010$, $y(0.15) = 1.0230179$.
Exact solution: $y(x) = 1/(1 - x^2)$

Ans. $y(0.2) = 1.0416675$ $y(0.25) = 1.0666688$
$y(0.3) = 1.0989052$, $y(0.4) = 1.1904878$,
$y(0.5) = 1.3333631$

4. $y' = y^2 \sin x$, $y(0) = 1$, find y on [0.05], $h = 0.05$.
Initial values: $y(0.05) = 1.0012513$, $y(0.10) = 1.0050209$, $y(0.15) = 1.0113564$.
Exact solution: $y(x) = \sec x$

Ans. $y(0.2) = 1.0203389$, $y(0.25) = 1.0320852$,
$y(0.3) = 1.0467519$, $y(0.4) = 1.0857051$
$y(0.5) = 1.1394953$.

5. $y' = -\frac{x}{y}$, $y(1) = 1$, find y on [1, 1.4]. $h = 0.05$
Initial values $y(1.05) = 0.94736477$, $y(1.10) = 0.88881944$, $y(1.15) = 0.82310388$
Exact solution: $y(x) = (2 - x^2)^{1/2}$

Ans. $y(1.2) = 0.7483205$, $y(1.25) = 0.6613998$,
$y(1.3) = 0.5566583$, $y(1.35) = 0.4208572$,
$y(1.4) = 0.1974740$.

6. $y' = x^3 + y^2$, $y(0) = 0$, $h = 0.1$. Find y on [0, 1.2]. Initial values are $y(0.2) = 0.0004$, $y(0.4) = 0.0064$, $y(0.6) = 0.0325$

Ans. $y(0.8) = 0.1035$, $y(0.9) = 0.1669$, $y(1.0) = 0.2574$, $y(1.1) = 0.3836$, $y(1.2) = 0.5581$

33.4 NUMERICAL SOLUTIONS TO PARTIAL DIFFERENTIAL EQUATIONS

Many physical phenomena can be modelled mathematically by partial differential equations, whose theory is quite difficult and numerical methods most difficult and needs extensive computation. Although analytical solutions can be obtained by separation of variable technique and integral transforms (Fourier

transforms), in many complicated cases one has to resort to numerical methods. Initial value problems or time dependent problems consists of two types: hyperbolic problems (containing both f_{xx} and f_{tt} term as in wave equation) and parabolic problems (which contain only f_{xx} and f_t but no f_{tt} term as in the heat equation). For these problems, the solution is known at an initial time and then propagates through space in time. The domain is open (goes to infinity) somewhere for example, in time. The other type of problems is the boundary value problem where the domain is closed and is entirely surrounded by boundary values. These problems are called elliptic (and are time independent as in the case of Laplace equation).

The dimension of a partial differential equation is the number of space variables x, y, z appearing in the equation as independent variables (apart from the independent variable time t). Thus a one-dimensional heat equation or one-dimensional wave equation involves only one space variable say x (and of course time t). Similarly two-dimensional Laplace equation contains two space variables say x and y.

The central idea in numerical methods to partial differential equations is to discretize the partial differential equation by replacing it, approximately, by a finite system of algebraic equations. One such important discretization method is finite differences, in which the derivatives are replaced by finite differences. The entire given domain is discretized by a discrete set of points, as follows:

Introducing $x_i = x_0 + ih,\ i = 0,\ \pm1,\ \pm2,\ \pm3,\ \ldots$ and $t_j = t_0 + jk,\ j = 0,\ \pm1,\ \pm2,\ \pm3,\ \ldots$, the (numerical) solution is obtained at the discrete points (x_i, t_j) known as grid or nodal or lattice or mesh or pivotal points of the computational grid. Here $h =$ constant is spacing (or size of interval) in x-direction while $k =$ constant is spacing in t-direction.

33.5 NUMERICAL SOLUTION TO ONE-DIMENSIONAL HEAT EQUATION

Consider the initial boundary value problem of the parabolic one-dimensional heat equation.

$$\text{P.D.E:} \qquad \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2},\ 0 \le x \le L, t \ge 0 \tag{1}$$

Two boundary conditions (B.C.'s):

$$u(0, t) = 0 \quad u(L, t) = 0 \tag{2}$$

One initial condition (I.C.)

$$u(x, 0) = f(x) \tag{3}$$

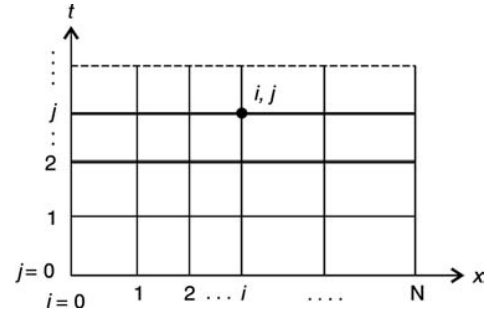


Fig. 33.3

Replacing u_t and u_{xx} by finite differences, equation (1) is approximated by a finite difference equation.

$$\frac{1}{k}\left(u_{i,j+1} - u_{ij}\right) = \frac{c^2}{h^2}\left(u_{i+1,j} - 2u_{ij} + u_{i-1,j}\right) \tag{4}$$

Here h is mesh size in x-direction i.e., $h = \frac{L}{N}$ and k is mesh size in t-direction. Rewriting (4),

$$u_{i,j+1} = \left(1 - 2r\right)u_{ij} + r(u_{i+1,j} + u_{i-1,j}) \tag{5}$$

where $r = \frac{c^2 k}{h^2}$. This method is known as *Schmidt explicit method* because the unknown $u_{i,j+1}$ is explicitly expressed in terms of the known (given) values $u_{i,j}; u_{i+1,j}; u_{i-1,j}$; whereas in the *implicit method* such as the Crank-Nicolson method the equations are implicit i.e., coupled and one has to solve a system of linear equations. Formula (5) provides a marching scheme to compute u at the mesh points. This method is convergent and stable if $r = \frac{c^2 k}{h^2} \le \frac{1}{2}$. In particular for $r = \frac{1}{2}$, (5) reduces to

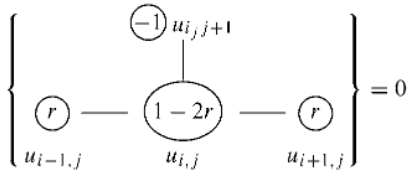
$$u_{i,j+1} = \frac{1}{2}(u_{i+1,j} + u_{i-1,j}) \tag{6}$$

which is known as *Bender-Schmidt recurrence relation*. So the values of h, k are so chosen (c is already given) that $r \le \frac{1}{2}$. For $r > \frac{1}{2}$ the method is divergent and unstable.

Stencil (pattern, molecule or star)

Finite difference approximations are often presented in terms of stencils or coefficient schemes which shows the pattern of connection in the difference equations. The entries in the stencil are coefficients in the difference equation.

For example the explicit method given by (5) is represented as a stencil shown below:



i.e., $(-1)u_{i,j+1} + r(u_{i-1,j}) \div (1-2r)u_{ij} + ru_{i+1,j} = 0$

WORKED OUT EXAMPLES

Example 1: Solve the initial boundary value problem:

$$\frac{\partial f}{\partial t} = 2\,\frac{\partial^2 f}{\partial x^2}$$

$$f(0,t) = 10 \qquad (1)$$

$$f(6,t) = 18 \qquad (2)$$

$$f(x,0) = \frac{x^2}{2} \qquad (3)$$

with $h = 1$ and $\tau = \frac{1}{8}$ by *explicit* method.

Solution: $m = \frac{a\tau}{h^2}$. Here $a = 2, h = 1, \tau = \frac{1}{8}$ so $m = \frac{1}{4}$. The explicit formula $f_{i,j+1} = (1-2m)\, f_{ij} + m(f_{i+1,j} + f_{i-1,j})$ with $m = \frac{1}{4}$ becomes

$$f_{i,j+1} = \frac{1}{2} f_{ij} + \frac{1}{4}(f_{i+1,j} + f_{i-1,j}) \qquad (4)$$

Here first row $t = 0$, consists of the initial values given by $IC(3) : f(x,0) = \frac{x^2}{2}$ at $x = 1, 2, 3, 4, 5$. The first column $x = 0$, gives the first boundary condition $f(0,t) = 10$. The last column $x = 6$, gives the second boundary condition $f(6,t) = 18$. The remaining values of f are calculated using (4). As $t \to \infty$, the process reaches steady state, so $f_{xx} = 0, f(x) = c_1 x + c_2, 10 = f(0) = 0 + c_2 \therefore c_2 = 10, 18 = f(6) = 6.c_1 + 10 \therefore c_1 = \frac{4}{3}$, analytical exact solution as $t \to \infty$ is $f(x) = \frac{4}{3}x + 10$. These are plotted for $t \to \infty$ in the last row of the Table 33.7.

Table 33.7

		B.C. (1)						B.C. (2)
	$t_j \backslash x_i$	0	1	2	3	4	5	6
I.C.(3)	$t = 0$	10	$\frac{1}{2}$	2	4.5	8	12.5	18
	$t = \frac{1}{8}$	10	3.25	2.25	4.75	8.25	12.75	18
	$t = \frac{1}{4}$	10	4.69	2.88	5.0	8.5	12.94	18
	$t = \frac{3}{8}$	10	5.56	3.86	5.34	8.73	13.09	18
	$t = \frac{1}{2}$	10	6.25	4.66	5.82	8.98	13.23	18
	$t = \frac{5}{8}$	10	6.79	5.35	6.32	9.25	13.36	18
	$t = \frac{3}{4}$	10	7.23	5.95	6.81	9.55	13.49	18
	$t = \frac{7}{8}$	10	7.60	6.49	7.28	9.85	13.63	18
	$t = 1$	10	7.92	6.97	7.73	10.15	13.78	18
	$t = 1\frac{1}{8}$	10	8.20	7.40	8.15	10.45	13.93	18
	$t = 1\frac{1}{4}$	10	8.45	7.79	8.54	10.75	14.08	18
	$t = 1\frac{3}{8}$	10	8.67	8.14	8.91	11.03	14.23	18
	$t = 1\frac{1}{2}$	10	8.87	8.47	9.25	11.30	14.37	18
	$t = \infty$	10	$11\frac{1}{3}$	$12\frac{2}{3}$	14	$15\frac{1}{2}$	$16\frac{2}{3}$	18

Example 2: Solve the above problem by Bender-Schmidt recurrence equation; with $h = 1$ and τ determined accordingly.

Solution: $m = \frac{a\tau}{h^2} = \frac{2\tau}{1}$. For Bender-Schmidt $m = \frac{1}{2} \therefore \quad \tau =$ time interval size $= \frac{1}{4}$. (See Table 33.8.)

Non-dimensional form

Example 3: Transform the heat equation

$$U_T = c^2 U_{XX}, \text{ where } 0 \le X \le L$$

to non-dimensional form.

Solution: Introducing

$$u = \frac{U}{u_0} \quad \text{and} \quad x = \frac{X}{L}, \quad t = \frac{c^2 T}{L^2}, \quad \text{we have}$$

$$\frac{\partial u}{\partial x} = \frac{\partial U}{\partial X} \cdot \frac{\partial X}{\partial x} = L\frac{\partial U}{\partial X}, \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial X}\left(L\frac{\partial U}{\partial X}\right)\frac{\partial X}{\partial x}$$

Table 33.8

		B.C.(1)						B.C.(2)
	$t_j \backslash x_i$	0	1	2	3	4	5	6
I.C.(3)	$t = 0$	10	$\frac{1}{2}$	2	4.5	8	12.5	18
	$t = \frac{1}{4}$	10	6	2.5	5.0	8.5	13	18
	$t = \frac{1}{2}$	10	6.25	5.5	5.5	9	13.25	18
	$t = \frac{3}{4}$	10	7.75	5.88	7.25	9.38	13.5	18
	$t = 1$	10	7.94	7.50	7.625	10.38	13.69	18
	$t = \frac{5}{4}$	10	8.75	7.78	8.94	10.66	14.19	18
	$t = \frac{3}{2}$	10	8.89	8.84	9.22	11.56	14.33	18
	$t = \frac{7}{4}$	10	9.42	9.05	10.20	11.77	14.78	18
	$t = 2$	10	9.53	9.81	10.41	12.49	14.89	18
	$t = \frac{9}{4}$	10	9.90	9.97	10.15	12.65	15.25	18
	$t = \frac{5}{2}$	10	9.99	10.53	11.31	13.20	15.33	18

$$\frac{\partial^2 u}{\partial x^2} = L\frac{\partial^2 U}{\partial X^2}\cdot L = L^2\frac{\partial^2 U}{\partial X^2}. \qquad \text{Similarly,}$$

$$\frac{\partial u}{\partial t} = \frac{\partial U}{\partial T}\frac{\partial T}{\partial t} = \frac{L^2}{c^2}\frac{\partial U}{\partial T}. \qquad \text{Then}$$

$$U_T = \frac{\partial U}{\partial T} = \frac{c^2}{L^2}\frac{\partial u}{\partial t} = c^2 U_{XX} = c^2\frac{\partial^2 U}{\partial X^2} = c^2\cdot\frac{1}{L^2}\frac{\partial^2 u}{\partial x^2}$$

Thus $\dfrac{\partial u}{\partial t} = \dfrac{\partial^2 u}{\partial x^2}$ i.e., $u_t = u_{xx}$

for $0 \le \frac{X}{L} \le \frac{L}{L}$ i.e., $0 \le x \le 1$. This form is known as non-dimensional standard form. Similarly, the wave equation

$$U_{TT} = a^2 U_{XX} \quad \text{where} \quad 0 \le X \le L$$

can be transformed to non-dimensional standard form

$$u_{tt} = u_{xx} \quad \text{where} \quad 0 \le x \le 1.$$

EXERCISE

Numerical solution to one-dimensional heat equation

1. Using Schmidt explicit method, solve $u_t = u_{xx}, 0 \le x \le 1, u(0,t) = 0, u(1,t) = 0, u(x,0) = 2x$ when $0 \le x \le 0.5$ and $u(x,0) = 2(1-x)$ when $0.5 \le x \le 1$ for $h = 1, k = 0.001$ at $t = 0.001, 0.002$.

Ans.

x :	0	.1	.2	.3	.4	.5	.6	.7	.8	.9	1.0
$u(x, 0.001)$:	0	.2	.4	.6	.8	.96	.8	.6	.4	.2	0
$u(x, 0.002)$:	0	.2	.4	.6	.796	.928	.796	.6	.4	.2	0

2. Use Schmidt explicit method to obtain numerical solution of $u_t = u_{xx}, u(0,t) = u(1,t) = 0, u(x,0) = \sin \pi x$, with $h = \frac{1}{3}, k = \frac{1}{36}$ for two levels.

Hint: Here $r = \frac{kc^2}{h^2} = \frac{1}{36}\cdot\frac{1}{\left(\frac{1}{3}\right)^2} = \frac{1}{4}$ so

$$u_{i,j+1} = \frac{1}{4}[u_{i-1,j} + 2u_{ij} + u_{i+1,j}]$$

Ans.

$x =$	0	$\sqrt{3}/2$	$\sqrt{3}/2$	0
$t = \frac{1}{36}$:	0	.65	.65	0
$t = \frac{2}{36}$:	0	.49	.49	0

3. Use Schmidt explicit method to solve $u_t = u_{xx}, u(0,t) = u(1,t) = 0, u(x,0) = \sin \pi x$; for $0 \le t \le 0.2$, with $r = 0.25$.

Hint: $r = k/h^2 = 0.25$, so $k = 0.01$, $u_{i,j+1} = 0.25\left(u_{i-1,j} + 2u_{ij} + u_{i+1,j}\right)$. Value at $x = 0.6$ is same as at $x = 0.4$; Value at $x = 0.8$ is same as at $x = 0.2$ due to symmetry.

Ans.

Table 33.9

	$x = 0.2$	$x = 0.4$
$t = 0$:	0.588	0.951
$t = 0.04$	0.393	0.637
$t = 0.08$	0.263	0.426
$t = 0.12$	0.176	0.285
$t = 0.16$	0.118	0.191
$t = 0.20$	0.079	0.128

4. Using Bender-Schmidt (explicit method with $r = \frac{1}{2}$) solve $u_{xx} - 2u_t = 0, u(0,t) = u(4,t) = 0, u(x,0) = x(4-x)$, with $h = 1, k = 1$.

Ans.

5. Solve $2u_{xx} - u_t = 0$, $u(x,0) = 50(4-x)$, $u(0,t) = 0, u(4,t) = 0$, with $h = 1, \tau = 0.25$.

Hint: use Bender-Schmidt scheme.

Ans. At $t = 1.50; u = 0, 6.25, 12.5, 6.25, 0$ (after six iterations)

Table 33.10

$t \setminus x$	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	1.5	2	1.5	0
3	0	1	1.5	1	0
4	0	.75	1	.75	0
5	0	.5	.75	.5	0

6. Compute $u(x, t)$ at $x = 0, 1, 2, \ldots 7$, $t = \frac{1}{8}j$, $j = 0,\ 1,\ 2,\ 3,\ 4,\ 5$ by solving $u_t = 4u_{xx}$, $u(0, t) = u(8, t) = 0$, $u(x, 0) = 4x - \frac{x^2}{2}$.

   Hint: $r = \frac{1}{2}$, use Bender-Schmidt recurrence relation.

Ans.

Table 33.11

$j \backslash x$	0	1	2	3	4	5	6	7	8
0	0	3.5	6	7.5	8	7.5	6	3.5	0
1	0	3	5.5	7	7.5	7	5.5	3	0
2	0	2.75	5	6.5	7	6.5	5		0
3	0	2.5	4.625	6	6.5	6	4.625		0
4	0	2.3	4.25	5.56	6	5.56	4.25		0
5	0	2.125	3.94	5.125	5.56	5.125	3.94	2.125	0

33.6 NUMERICAL SOLUTION TO ONE-DIMENSIONAL WAVE EQUATION

Consider the initial boundary value problem of hyperbolic one-dimensional wave equation

$$\text{P.D.E}: \quad \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, 0 \le x \le L, t \ge 0 \tag{1}$$

Two boundary conditions:

$$u(0, t) = u(L, t) = 0 \tag{2}$$

Two initial conditions: Prescribed initial displacement:

$$u(x, 0) = f(x) \tag{3}$$

Prescribed initial velocity:

$$u_t(x, 0) = g(x) \tag{4}$$

The finite difference approximation of (1) is

$$\frac{1}{k^2}\left(u_{i,j+1} - 2u_{ij} + u_{i,j-1}\right) = \frac{a^2}{h^2}\left(u_{i+1,j} - 2u_{ij} + u_{i-1,j}\right) \tag{5}$$

This explicit method converges and stable for $0 < r \le 1$. Here h is mesh size in x-direction and k is the mesh size in t-direction. For a given a^2, choose h and k such that

$$r = \frac{a^2k^2}{h^2} = 1.$$

Then Equation (5) simplifies to

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{ij-1} \tag{6}$$

The numerical solution is generally represented in the form of an open ended table, consisting of n columns, headed by $x_i = x_0 + ih$, where the mesh size $h = \frac{L-0}{n}$. Here $x_0 = 0, x_n = L$.

Step I All the entries in the first column are zeros, given by the first boundary condition $u(0, t) = 0$.

Step II Similarly, all the entries in last column are zeros, given by the second boundary condition.

$$u(L, t) = 0.$$

Step III Each row represents a time step. Entries in first row, are calculated using (3) the first initial condition $u(x, 0) = f(x)$ for $x_i = x_0 + ih, i = 1, 2, 3, \cdots, n$.

Step IV Observe that formula (6) involves 3 time steps namely the future $(j + 1)$ in terms of the present (j) and the past $j - 1$. At this stage, from step III, the first row gives only on time step $(j = 0)$. So use (4) the second initial condition which gives rise to another (second) row which will be the second time step $(j = 1)$.

Now from I.C. (4) in difference form is

$$\frac{1}{2k}(u_{i1} - u_{i,-1}) = g_i$$

or

$$u_{i,-1} = u_{i1} - 2\,k\,g_i \tag{7}$$

where $g_i = g(ih)$. For $t = 0$ (i.e., $j = 0$), we have from (6)

$$u_{i1} = u_{i-1,0} + u_{i+1,0} - u_{i,-1} \tag{8}$$

Substitute $u_{i,-1}$ from (7) into (8), we get

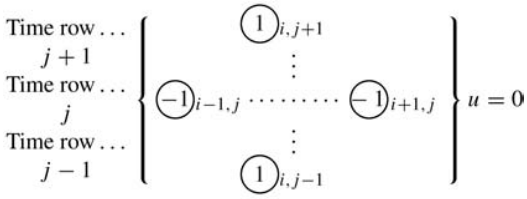
$$u_{i1} = u_{i-1,0} + u_{i+1,0} - (u_{i1} - 2k\,g_i) \quad \text{or}$$

$$u_{i1} = \frac{1}{2}(u_{i-1,0} + u_{i+1,0}) + kg_i \tag{9}$$

Equation (9) expresses u_{i1} in terms of the initial data. For $i = 0, 1, 2, \ldots, n$, Equation (9) gives the entries in the second row.

Step V Now that we have two rows, one from I.C.

(3) and one from I.C. (4), we can calculate third row entries by formula (6) using the first ($j = 0$) and second row ($j = 1$) entries. Continuing this process the 4th row is calculated by (6) using 2nd and 3rd row entries. Subsequent rows are obtained similarly.
Stencil for formula (6)



i.e., $u_{ij\,j+1} + u_{i,j-1} = u_{i-1,j} + u_{i+1,j}$

WORKED OUT EXAMPLES

Numerical solution to one-dimensional wave equation

Examples: Compute numerically the solution of the one-dimensional wave equation $25 f_{xx} - f_{tt} = 0$ with boundary conditions $f(0, t) = f(5, t) = 0$ and with initial conditions $f(x, 0) = 20x$, when $0 \le x \le 1$ and $f(x, 0) = 25\left(1 - \frac{x}{5}\right)$ when $1 \le x \le 5$; $\frac{\partial f}{\partial t}\big|_{t=0} = 0$.

Solution: Here $a^2 = 25$, choose $h = 1$ mesh size in x. Since

$$\frac{a^2 t^2}{h^2} = 1 \text{ so } \frac{25\tau^2}{1} = 1 \text{ or } \tau = \frac{1}{5}, \text{ mesh size in } t.$$

InItial values: At $t = 0$, (initial displacement).

$$f(x, 0) = \begin{cases} 20x, & 0 \le x \le 1 \\ 25\left(1 - \frac{x}{5}\right), & 1 \le x \le 5 \end{cases} \qquad \text{IC (1)}$$

For $x = 0$, $f_{00} = 0$, for $x = 1$, $f_{10} = 20$, for $x = 2$, $f_{20} = 15$, $f_{30} = 10$, $f_{40} = 5$, $f_{50} = 0$. These six values form the first row in the table.
Since at $\frac{\partial f}{\partial t}(x, 0) = 0$, I.C. (2)
the initial velocity is zero so

$$f_{i1} = \frac{1}{2}\left(f_{i-1,0} + f_{i+1,0}\right). \qquad \text{IC (2)}$$

For $i = 1$, $f_{11} = \frac{1}{2}(f_{00} + f_{20}) = \frac{1}{2}(0 + 15) = 7.5$,
For $i = 2$, $f_{21} = \frac{1}{2}(f_{10} + f_{30}) = \frac{1}{2}(20 + 10) = 15$
For $i = 3$, $f_{31} = \frac{1}{2}(f_{20} + f_{40}) = \frac{1}{2}(15 + 5) = 10$
For $i = 4$, $f_{41} = \frac{1}{2}(f_{30} + f_{50}) = \frac{1}{2}(10 + 0) = 5$
These six values form the second row in the table. The remaining displacements are calculated from

$$f_{i,j+1} = f_{i-1,j} + f_{i+1,j} - f_{i,j-1} \qquad (*)$$

which involves 2 time steps (at J and $J - 1$). Then the resulting (Table 33.12) is given below:

Table 33.12

		$f(0,t{=}0)$ B.C.(1)↓					$f(5,t){=}0$ ↓B.C.(2)
	$t \backslash x$	0	1	2	3	4	5
I.C.(1)	0	$f_{00} = 0$	$f_{10} = 20$	$f_{20} = 15$	$f_{30} = 10$	$f_{40} = 5$	$f_{50} = 0$
I.C.(2)	1	0	$f_{11} = 7.5$	$f_{21} = 15$	$f_{31} = 10$	$f_{41} = 5$	0
	2	0	−5	2.5	10	5	0
	3	0	−5	−10	−2.5	5	0
	4	0	−5	−10	−15	−7.5	0
	5	0	−5	−10	−15	−20	0
	6	0	−5	−10	−15	−7.5	0
	7	0	−5	−10	−2.5	5	0
	8	0	−5	2.5	10	5	0
	9	0	7.5	15	10	5	0
	10	0	20	15	10	5	0

The values in the 2nd, 3rd, 4th, . . ., 10th row are calculated using the formula (*). The displacement function $f(x, t)$ goes through a complete oscillation (cycle) in 10 steps, indicating a periodic behaviour with period $T = 10\tau = 10\left(\frac{1}{5}\right) = 2$.

EXERCISE

Numerical solution to one-dimensional wave equation

1. Evaluate the pivotal values of $16u_{xx} = u_{tt}$, $u(0, t) = u(5, t) = 0$, $u(x, 0) = x^2(5 - x)$, $u_t(x, 0) = 0$, with $h = 1$ and up to one half of the period of vibration.

   Hint: Period of vibration $= \frac{2L}{a} = \frac{2.5}{4}$ since $a^2 = 16$, $L = 5$. Period: $\frac{5}{2}$ secs or compute up to $t = \frac{5}{4}$ secs. From $u_t(x, 0) = 0$, $u_{i1} = u_{i0}$ so 1st and 2nd row are same.

Ans.

Table 33.13

$t\backslash x$	0	1	2	3	4	5
0	0	4	12	18	16	0
1	0	4	12	18	16	0
2	0	8	10	10	2	0
3	0	6	6	−6	−6	0
4	0	−2	−10	−10	−8	0
5	0	−16	−18	−12	−4	0

2. Determine the displacement $u(x, t)$ of a string if $u_{tt} = u_{xx}, u(x, 0) = \frac{x(10-x)}{100}, u(0, t) = 0, u(10, t) = 0, u_t(x, 0) = 0$ for $x = 0(1)10, t = 0(1)5$.

Ans.

x :	0	1	2	3	4	5
$t = 0$:	0	0.09	0.16	0.21	0.24	0.25
$t = 1$:	0	.08	.15	.20	.23	.24
$t = 4$:	0	.02	.04	.06	.08	.09
$t = 5$:	0	0	0	0	0	0

x :	6	7	8	9	10
$t = 0$:	.24	.21	.16	.9	0
$t = 1$:	.23	.20	.15	.08	0
$t = 4$:	.08	.06	.04	.02	0
$t = 5$:	0	0	0	0	0

3. Compute numerically the solution of $4u_{xx} - u_{tt} = 0, u(x, 0) = x(4 - x); u(0, t) = 0, u(4, t) = 0$, with $h = 1, \tau = \frac{1}{2}$.

Hint: Complete one oscillation in 8 steps; Period $= 8\frac{1}{2} = 4$.

Ans.

Table 33.14

$t\backslash x$	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	0	0	0	0
3	0	−2	−3	−2	0
4	0	−3	−4	−3	0
5	0	−2	−3	−2	0
6	0	0	0	0	0
7	0	2	3	2	0
8	0	3	4	3	0

4. Calculate the numerical solution with $h = k = 0.2$ by solving $u_{tt} = u_{xx}$, $u(x, 0) = \sin \pi x$, $u(0, t) = u(1, t) = 0, u_t(x, 0) = 0$.

Hint: Symmetric $u(x = 0.2, t) = u(x = 0.8, t)$ and $u(x = 0.4, t) = u(x = 0.6, t)$.

Ans.

Table 33.15

	$x = 0$	0.2	0.4
$t = 0$	0	.588	.951
$t = 0.2$	0	0.476	0.769
$t = 0.4$	0	0.182	0.294
$t = 0.6$	0	−0.182	−0.294
$t = 0.8$	0	−0.476	−0.769
$t = 1.0$	0	−0.588	−0.951

5. Evaluate the pivotal values for $\frac{1}{2}$ period of vibration by solving $25u_{xx} - u_{tt} = 0, u(0, t) = u(5, t) = 0, u(x, 0) = 2x, 0 \le x \le 2.5, u(x, 0) = 10 - 2x, 2.5 \le x \le 5$.

Hint: $h = 1, \tau = \frac{1}{5}$.

Ans.

Table 33.16

$j\backslash i$	0	1	2	3	4	5
0	0	2	4	4	2	0
1	0	2	3	3	2	0
2	0	1	1	1	1	0
3	0	−1	−1	−1	−1	0
4	0	−2	−3	−3	−2	0
5	0	−2	−4	−4	−2	0

33.7 NUMERICAL SOLUTION TO TWO-DIMENSIONAL LAPLACE EQUATION

Consider the boundary value problem of elliptic Laplace equation in two-dimensions defined in a rectangular domain with prescribed boundary conditions as follows:

$$\text{P.D.E}: \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial y^2} = 0, \quad \begin{matrix} 0 < x < a, \\ 0 < y < b \end{matrix} \tag{1}$$

Four B.C's

$$u(0, y) = p(y), u(x, 0) = q(x),$$
$$u(a, y) = r(y), u(x, b) = s(x) \tag{2}$$

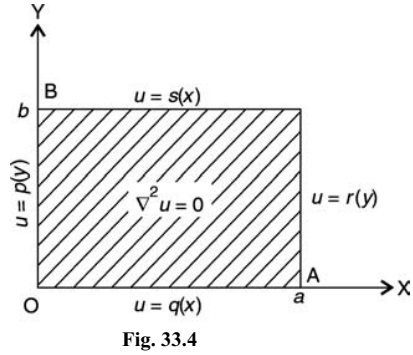


Fig. 33.4

The rectangular domain is discretized by dividing OA into M qual parts of size $\Delta x = \frac{a}{M} = h$, and dividing OB into N equal parts of size $\Delta y = \frac{b}{N} = k$. Thus the function $u(x, y)$ is to be determined at the nodal points $p_{ij} = (x_i, y_j) = (i\Delta x, j\Delta y) = (ih, jk)$ for $i = 0, 1, 2, \ldots M$ and $j = 0, 1, 2, \ldots N$. Replacing (1) by finite difference, we have

$$\frac{u_{i-1,j} - 2u_{ij} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} = 0$$

For a square domain, $h = k$. The above equation reduces to

$$u_{ij} = \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}\right] \quad (3)$$

(3) is known as the **standard five points formula** and represented by the stencil.

$$\left\{ \begin{array}{ccc} & (-1)_{i,j+1} & \\ (-1)_{i,j-1} & (-4)_{i,j} & (-1)_{i,j+1} \\ & (-1)_{i,j-1} & \end{array} \right\} u = 0$$

Since Laplace's equation is invariant under rotation of axes through 45°, we get a **five point diagonal formula**

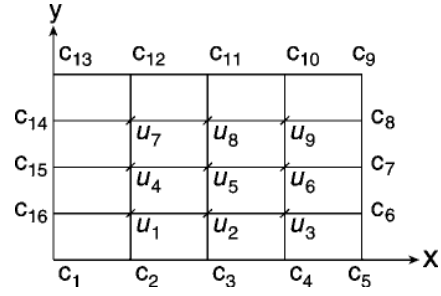
$$u_{ij} = \frac{1}{4}\left[u_{i-1,j-1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j+1}\right] \quad (4)$$

with stencil

$$\left\{ \begin{array}{ccc} (1)_{i-1,j-1} & & (1)_{i-1,j+1} \\ & (-4)_{i,j} & \\ (1)_{i+1,j-1} & & (1)_{i+1,j+1} \end{array} \right\} u = 0$$

Generally, standard five point formula is preferred over the diagonal formula.

Procedure: Suppose the square domain is divided into a grid as follows:



Here the boundary values $c_1, c_2, \cdots, c_{16}$ are given. To find the unknown function u at the 9 grid or mesh points 1, 2, 3, . . . , 9, first use the diagonal formula to compute first u_5 and then u_7, u_9, u_1, u_3 in that order. They are given by

$$u_5 = \frac{1}{4}[c_1 + c_5 + c_9 + c_3]$$

$$u_7 = \frac{1}{4}[c_{15} + u_5 + c_{11} + c_{13}]$$

$$u_9 = \frac{1}{4}[u_5 + c_7 + c_9 + c_{11}]$$

$$u_1 = \frac{1}{4}[c_1 + c_3 + u_5 + c_{15}]$$

$$u_3 = \frac{1}{4}[c_3 + c_5 + c_7 + u_5]$$

Now making use of these values u_5, u_7, u_9, u_1, u_3 compute the remaining values u_8, u_4, u_6, u_2 (in any order) by standard five points formula, as follows:

$$u_8 = \frac{1}{4}[u_5 + u_9 + c_{11} + c_7]$$

$$u_4 = \frac{1}{4}[u_1 + u_5 + u_7 + c_{15}]$$

$$u_6 = \frac{1}{4}[u_3 + c_7 + u_9 + u_5]$$

$$u_2 = \frac{1}{4}[c_3 + u_3 + u_5 + u_1]$$

The accuracy of these 9 values u_1 to u_9 can be improved by using iterative methods such as (i) **Jacobi's method** (ii) **Gauss-Seidel method** (iii) **Successive over relaxation method**. In the **Gauss-Seidel method (also known as Leibman's method)** a standard five point formula is used wherein the

latest (last available) values of u are utilized. If n denotes the iteration number then the value of u at the $(n+1)$th iteration is given by

$$u_{i,j}^{n+1} = \frac{1}{4}\left[u_{i-1,j}^{(n+1)} + u_{i+1,j}^{(n)} + u_{i,j-1}^{(n+1)} + u_{i,j+1}^{(n)}\right] \quad (5)$$

Observe that in R.H.S. of (5), the latest values

$$u_{i-1,j}^{n+1} \text{ and } u_{i,j-1}^{n+1} \quad \text{are utilized.}$$

Note: Solution of Laplace equation is known as **harmonic** function.

WORKED OUT EXAMPLES

Example 1: Compute a solution to the Laplace's equation $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$ at all mesh (grid) points of the following square in which the boundary values are indicated, using five point formula.



Solution: Let us index the mesh points as $f_1, f_2, f_3, \ldots, f_9$. First calculate f_5 using standard five point formula and then calculate f_1, f_3, f_7, f_9 using diagonal formula. Now the remaining values f_2, f_4, f_6, f_8 are again calculated by standard five point formula. Thus by standard formula

$$f_5 = \frac{1}{4}[0 + 17.0 + 21.9 + 12.1] = 12.525.$$

Now use diagonal formula

$$f_1 = \frac{1}{4}[0 + 0 + 17.0 + 12.525] = 7.38$$

$$f_3 = \frac{1}{4}[17.0 + 18.6 + 21.9 + 12.525] = 17.28$$

$$f_7 = \frac{1}{4}[12.1 + 0 + 0 + 12.525] = 6.16$$

$$f_9 = \frac{1}{4}[12.1 + 12.525 + 21.9 + 9] = 13.66$$

Resorting to standard formula

$$f_2 = \frac{1}{4}[7.38 + 17 + 17.28 + 12.525] = 13.55$$

$$f_4 = \frac{1}{4}[0 + 7.38 + 12.525 + 6.16] = 6.52$$

$$f_6 = \frac{1}{4}[12.525 + 17.28 + 21 + 13.66] = 16.12$$

$$f_8 = \frac{1}{4}[6.16 + 12.525 + 13.66 + 12.1] = 11.11.$$

Example 2: Solve the above problem, using the values of $f_1, f_2, \ldots, f_9$ as the initial approximation, by Gauss-Seidel (or Leibman's) method.

Solution:

$$f_{i,j}^{(n+1)} = \frac{1}{4}\left[f_{i-1,j}^{(n+1)} + f_{i+1,j}^{(n)} + f_{i+1,j}^{(n+1)} + f_{i,j+1}^{(n)}\right]$$

1st iteration:

$$f_1 = \frac{1}{4}[0 + 11.1 + 13.55 + 6.52] = 7.7925 \simeq 7.79$$

$$f_2 = \frac{1}{4}[7.79 + 17.0 + 17.28 + 12.525] = 13.65$$

$$f_3 = \frac{1}{4}[13.65 + 19.7 + 21.9 + 16.12] = 17.84$$

$$f_4 = \frac{1}{4}[0 + 7.79 + 12.525 + 6.16] = 6.62$$

$$f_5 = \frac{1}{4}[16.12 + 11.11 + 13.65 + 6.62] = 11.875$$

$$f_6 = \frac{1}{4}[21.0 + 13.66 + 17.84 + 11.875] = 16.09$$

$$f_7 = \frac{1}{4}[11.11 + 8.7 + 0 + 6.62] = 6.607$$

$$f_8 = \frac{1}{4}[13.66 + 12.1 + 11.875 + 6.607] = 11.06$$

$$f_9 = \frac{1}{4}[17.0 + 12.8 + 16.09 + 11.06] = 14.238$$

2nd iteration

$$f_1 = \frac{1}{4}[0 + 13.65 + 6.62 + 11.1] = 7.84$$

$$f_2 = \frac{1}{4}[7.84 + 17.84 + 11.87 + 17] = 13.639 \simeq 13.64$$

$$f_3 = \frac{1}{4}[13.64 + 16.09 + 19.7 + 21.9] = 17.83$$

$$f_4 = \frac{1}{4}[0 + 7.84 + 6.607 + 11.873] = 6.58$$

$$f_5 = \frac{1}{4}[6.58 + 13.64 + 11.01 + 16.09] = 11.843$$

$$f_6 = \frac{1}{4}[11.843 + 17.833 + 14.238 + 21] = 16.2285$$

$$f_7 = \frac{1}{4}[6.58 + 11.059 + 8.7 + 0] = 6.58$$

$$f_8 = \frac{1}{4}[12.11 + 6.587 + 14.238 + 11.843] = 11.19$$

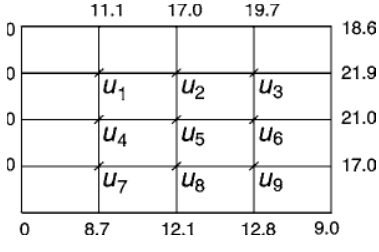
$$f_9 = \frac{1}{4}[11.19 + 16.25 + 17 + 12.8] = 14.31.$$

Note that in every calculation, the latest (last available) values of f has been taken into the right hand side of the formula.

Ans: 7.8, 13.6, 17.8, 6.6, 11.9, 16.2, 6.6, 11.2, 14.3.

EXERCISE

1 (a) Determine the values of the interior lattice points of a square region of the harmonic function u with boundary conditions prescribed as shown in the figure.



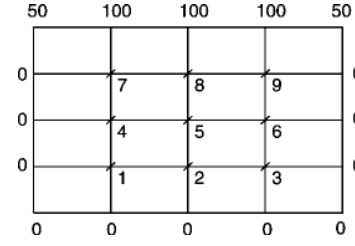
(b) Iterate by Gauss-Seidel (Leibman's) method.

Ans. (a) $u_1 = 7.4, u_2 = 13.6, u_3 = 17.3, u_4 = 6.5, u_5 = 12.5, u_6 = 16.1, u_7 = 6.2, u_8 = 11.1, u_9 = 13.7$

(b) $u_1 = 7.9, u_2 = 13.7, u_3 = 17.9, u_4 = 6.6, u_5 = 11.9, u_6 = 16.3, u_7 = 6.6, u_8 = 11.2, u_9 = 14.3$ (after 3 iterations)

2 (a) Compute solution of Laplace's equation in the following square grid with prescribed boundary conditions.

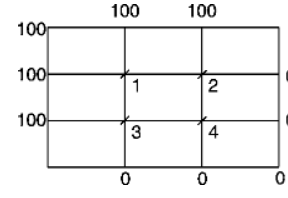
(b) Iterate the values by Gauss-Seidel method.



Ans. (a) $u_1 = 6.25, u_2 = 9.38, u_3 = 6.25, u_4 = 18.75, u_5 = 25, u_6 = 18.75, u_7 = 43.75, u_8 = 53.12, u_9 = 43.75$

(b) 7.17, 9.86, 7.16, 18.78, 25.04, 18.77, 42.88, 52.70, 42.87 (after 3 iterations)

3. Compute solution of Laplace's equation in the square grid.



Hint: Solve the system of equations

$$u_2 + u_3 - 4u_1 + 100 + 100 = 0$$

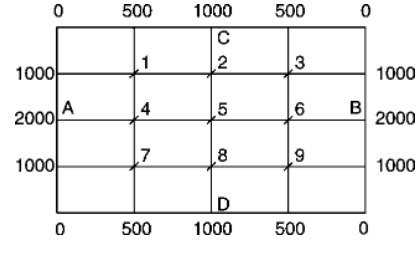
$$u_1 + u_4 - 4u_2 + 100 + 0 = 0$$

$$u_1 + u_4 - 4u_3 + 100 + 0 = 0$$

$$u_2 + u_3 - 4u_4 + 0 + 0 = 0$$

Ans. $u_1 = 75, u_2 = 50, u_3 = 50, u_4 = 25$

4 (a) Determine $u_1, u_2, u_3, \ldots, u_9$ if u satisfies Laplace's equation in the grid with given boundary condition.



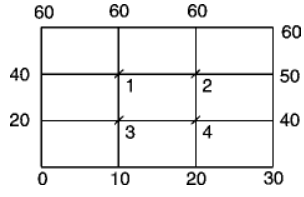
(b) Iterate by Gauss-Seidel method.

Hint: Use symmetry about AB and CD.

Ans. (a) $u_1 = u_3 = u_7 = u_9 = 1125, u_2 = u_8 = 1188, u_5 = 1500, u_6 = u_4 = 1438$

(b) $u_1 = 939, u_2 = 1001, u_4 = 1251, u_5 = 1126$ (after 12 iterations)

5. Solve Laplace's equation.



Hint: Solve the four equations

$$40 + 60 + u_2 + u_3 - 4u_1 = 0$$

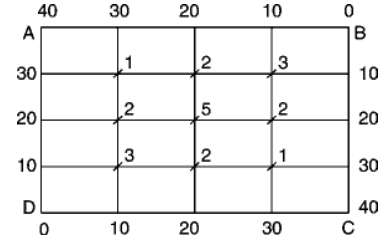
$$60 + 50 + u_1 + u_4 - 4u_2 = 0$$

$$20 + 10 + u_1 + u_4 - 4u_3 = 0$$

$$20 + 40 + u_3 + u_2 - 4u_4 = 0$$

Ans. $u_1 = 43.33, u_2 = 46.66, u_3 = 26.66, u_4 = 33.33$

6. If $\nabla^2 u = 0$, find u at the mesh points.



Hint: Square is symmetric about both the diagonaly AC and BD.

Ans. $u_1 = 25, u_3 = 15, u_2 = 20, u_5 = 20$